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Organizational Learning and Value Co-creation in the Context of Digital Service Innovation: a study on software development teams

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José Antônio Afonso Nonato

Organizational Learning and Value Co-creation in the Context of Digital Service Innovation: a study on software development teams

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“Do not conform to the pattern of this world, but be transformed by the renewing of your mind. Then you will be able to test and approve what God’s will is – his good, pleasing and perfect will”.

Romans, 12: 2 (NIV).

ABSTRACT

With this study, we investigated how software development teams experience organizational learning and value co-creation in the context of digital service innovation. Using qualitative research and analysis based on Grounded Theory and the constructivist paradigm, the study explores the intricate interplay between learning processes, collaborative relationships, and innovation practices. The findings culminate in a substantive theory centered on Building Digital Service Innovation as the core process that integrates organizational learning, value co-creation, and service innovation. This theory posits that innovation emerges from the dynamic synthesis of learning, trust-building, and iterative collaboration, ultimately balancing technological, problem-oriented, and product-driven strategies. Organizational learning is proposed as a cornerstone of service innovation, driven by experimentation, problem-solving, and collaborative knowledge-sharing. Trust, cultivated through openness and vulnerability, may act as a critical enabler of value co-creation, fostering more profound engagement with clients and stakeholders. Notably, the study highlights the counterintuitive finding that vulnerability – expressed through openly sharing limitations, uncertainties, or imperfect prototypes – can transform relationships by fostering authenticity, mutual respect, and a shared sense of purpose. Design thinking and iterative approaches are proposed as effective frameworks for integrating empathy, experimentation, and client feedback into innovation processes. However, challenges such as resource constraints, operational pressures, and communication misalignment appear to underscore the need for strong leadership, adaptive cultures, and team empowerment to navigate these barriers. The study contributes to the literature by proposing a conceptual framework that bridges organizational learning, trust-building, and service innovation through the lens of Building Digital Service Innovation as a core enabler. Practical recommendations include adopting design thinking, leveraging trust as a foundation for collaboration, embedding experimentation into workflows, and fostering organizational resilience to sustain innovation under uncertainty. By integrating theoretical and practical perspectives, this research equips organizations with actionable strategies and offers a platform for future studies about service innovation and value co-creation across diverse contexts.

Keywords: Service Innovation, Building Innovation, value co-creation, organizational learning, Grounded Theory.

RESUMO

Com este estudo investigou-se como equipes de desenvolvimento de software experienciam a aprendizagem organizacional e a cocriação de valor no contexto da inovação em serviços digitais. Utilizando pesquisa de natureza qualitativa e análise usando teoria fundamentada nos dados (*Grounded Theory*) e paradigma construtivista, a pesquisa explora a intrincada interação entre processos de aprendizagem, relacionamentos colaborativos e práticas de inovação. Os achados culminam em uma teoria substantiva centrada na Construção da Inovação de Serviços Digitais como o processo principal que integra aprendizagem organizacional, cocriação de valor e inovação em serviços. Essa teoria postula que a inovação emerge da síntese dinâmica de aprendizagem, construção de confiança e colaboração iterativa, equilibrando estratégias tecnológicas, orientadas por problemas e direcionadas por produtos. A aprendizagem organizacional é proposta como um alicerce da inovação em serviços, impulsionada pela experimentação, resolução de problemas e compartilhamento colaborativo de conhecimento. A confiança, cultivada por meio de abertura e vulnerabilidade, pode atuar como um elemento crítico para a cocriação de valor, promovendo um engajamento mais profundo com clientes e partes interessadas. Notavelmente, o estudo destaca o achado contraintuitivo de que a vulnerabilidade – expressa ao compartilhar abertamente limitações, incertezas ou protótipos imperfeitos – pode transformar relacionamentos ao fomentar autenticidade, respeito mútuo e um senso compartilhado de propósito. Design thinking e abordagens iterativas são propostos como estruturas eficazes para integrar empatia, experimentação e feedback dos clientes nos processos de inovação. Contudo, desafios como restrições de recursos, pressões operacionais e desalinhamentos na comunicação parecem enfatizar a necessidade de uma liderança forte, culturas adaptativas e empoderamento das equipes para superar essas barreiras. O estudo contribui para a literatura ao propor uma estrutura conceitual que conecta aprendizagem organizacional, construção de confiança e inovação em serviços por meio da lente da Construção da Inovação de Serviços Digitais como habilitador central. Recomendações práticas incluem a adoção de design thinking, o uso da confiança como base para colaboração, a integração da experimentação nos fluxos de trabalho e a resiliência organizacional como sustento da inovação em cenários de incerteza. Ao integrar perspectivas teóricas e práticas, a pesquisa fornece estratégias acionáveis e uma plataforma para estudos futuros sobre inovação em serviços e cocriação de valor em contextos diversos.

Palavras-chave: Inovação em Serviços, Construindo Inovação, Cocriação de Valor, Aprendizado Organizacional, Teoria Fundamentada nos Dados.

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LIST OF ABBREVIATIONS AND ACRONYMS

CAPES	Coordernação de Aperfeiçoamento de Pessoal de Nível Superior
CGT	Constructivist Grounded Theory
GT	Grounded Theory
OL	Organizational Learning
S-D	Service-Dominant Logic
SDT	Service Development Teams
SI	Service Innovation
VCC	Value Co-Creation

SUMMARY

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1. INTRODUCTION

The unprecedented societal and economic disruptions triggered by the pandemic compelled organizations to rethink the workplace and how value is created. In an environment marked by increasing uncertainty and the rapid evolution of business operations, disruption has become the norm rather than the exception (Heinonen & Strandvik, 2021). To navigate this reality, service organizations are under relentless pressure to enhance their flexibility, agility, and capacity for learning and innovation (Bessant et al., 2005; Carbonell et al., 2009; Xie et al., 2021).

Powered by unprecedented creativity, service companies must rely on resources outside their boundaries and capacities (Wang, 2021) to cope with disruptive changes, anticipate innovation trends, and appropriate emerging ideas before competitors recognize them (Heinonen & Strandvik, 2021).

Furthermore, companies that can address customers' opportunities and pain points not in a prescriptive manner but in a tailored, contextualized, and consistent way will thrive (Xie et al., 2021). Therefore, understanding and internalizing customer needs are paramount in digital services companies where customers access value instead of owning it (Akbar & Hoffmann, 2020), such as subscription-based value propositions.

As the global economy becomes more service-oriented (Xie et al., 2021), the creation of innovative services is considered a differentiation factor and an essential determinant of company growth (Helkkula et al., 2018), especially in turbulent times or crises (Heinonen & Strandvik, 2021) and under intense competitive pressure from technology and globalization (Patrício et al., 2018). Nevertheless, service innovation is not constrained by technologies, places where things can be done, or actors (Chandler & Lusch, 2015). It is a broad and often vague construct, lacking a common understanding regarding its meaning (Witell et al., 2016).

To address this lack of conceptual clarity, it is necessary to establish a working definition of service innovation within this study. It can be understood as a new process or service offering that is put into practice by an organization, adopted by, and creates value with/for one or more actors in an ecosystem (Helkkula et al., 2018; Witell et al., 2016).

Besides, to enhance the understanding and provide a theoretical background of how value is created among parties, this study drew from the value co-creation literature. Given the conceptual relevance of this perspective, the study adopted the classical definition proposed by Prahalad and Ramaswamy (2004), where value co-creation is a joint venture between providers and customers, from creating value propositions to implementing those expectations.

Notably, despite calls for a multidimensional view of service innovation, such studies remain rare (Helkkula et al., 2018). Most existing work has been conceptual and narrow in empirical scope (Heinonen & Strandvik, 2021). Besides, inconsistent findings are revealed in research streams about the participation of customers in the value creation of services (Akbar & Hoffmann, 2020; Desmarchelier et al., 2020; Xie et al., 2021). Thus, we argue that further empirical research can augment the existing understanding of the role of value co-creation for digital service innovation and increase awareness of how team learning can contribute to the inception of new service offerings.

Besides, few studies addressed the difficulty of integrating efforts from multiple actors in service ecosystems (Akbar & Hoffmann, 2020; Wang, 2021; Xie et al., 2021). Others, like Tuunanen et al. (2023), suggest a holistic approach combined with micro-level mechanisms such as social use, customer orientation, service experience, context, and customer values and goals.

Regarding integration efforts, we understand it is more than "just" listening to the customer/partner but promoting active participation where actors depart from their context to innovate (Patrício et al., 2018). Considering the relevance of service innovation and the advent of an access-based economy, understanding how customers can foster innovation becomes relevant (Akbar & Hoffmann, 2020), and, even more importantly, understanding if the customer involvement is perceived by the ones that create the value proposition as beneficial for innovation performance (Xie et al., 2021).

In that context, software development teams become particularly relevant for understanding service innovation and value co-creation since they are the primary conduits for integrating individual knowledge in organizations (Dixon, 2017). Software development teams can also be seen as the locus of learning, a sort of group mind as described by Weick and Roberts (1993), and with whom organizations heavily rely upon the implementation and ideation of innovative digital services (Duc et al., 2020; Mehta et al., 2022). However, in many organizations, these development teams are traditionally distant from the customers (internal and external). Teams are incentivized to make a few mistakes while simultaneously requesting to increase the speed of innovation generation and falling short in time for learning (Hoda et al., 2013). Besides the apparent paradox, the need to "productify" digital services moves those teams away from the customer perspective and perception (Xin et al., 2021).

In contrast, Desmarchelier et al. (2020) pointed out an intriguing paradox. These authors depart from an organizational learning framework to test a computational value co-creation model. Although more customer involvement can amplify value co-creation during offering

elaboration, when it deepens in co-productions (value configuration and capture), simulations show that contradictions between consumers' aspirations and the practices and values of their organizations engender ambiguity and create "viscosity" in the learning cycle. The term "viscosity" comes from the organizational change literature and refers to the resistance produced in the organizational structure that delays decision-making and favors suboptimal alternatives (Hannan et al., 2003). Desmarchelier et al. (2020) argued the result of increased levels of ambiguity is less use-value potential.

Notwithstanding, contrary to what common sense may suggest, the critical role of learning in general, and team learning in particular, in conjunction with service innovation and customer involvement in co-creation, has received marginal attention in the literature (Xie et al., 2021).

The above elaboration reinforces the opportunity to use the theoretical lens of organizational learning to understand the phenomenon of value co-creation in the context of digital services innovation (Gustafsson et al., 2020). These authors argued that organizational learning can be used to develop the service innovation construct further. Drawing from the definition proposed by Takahashi (2015), organizational learning is the result of a continuous movement of reassessing meaning and establishing a new order from change and continuity.

Considering the very nature of innovations, embedded in ambiguity, uncertainty, and prone to failure, and the need for mutual learning and action in the face of value co-creation, we argued that organizational learning could contribute to the comprehension of how these constructs related to each other in the context of software development teams.

To do so, this study relied on Grounded Theory because of its emphasis on generating propositions from emerging concepts and relationships in the context of social settings that are connected to data collected (Urquhart & Fernández, 2013). We argued that existing theoretical frameworks, although invaluable, do not fully explain how the participation of customers in software development teams co-creates value and promotes innovations.

1.1 Research Question

The following research problem guided this study: **How do software development teams experience organizational learning and value co-creation in the context of digital service innovation?**

1.2 Research Objectives (General and Specific)

According to Creswell (2010), the statement of objectives of a study defines what is intended to be achieved and must communicate this intention to the reader. In this way, this research aimed to **understand the perceptions and learning experiences of members of software development teams when they participate in innovation processes with value co-creation.**

Additionally, Blaikie (2009) argues that once the research question and the general objective are laid out, the researcher must clarify the types of knowledge that the study produces. In this sense, the specific objectives limit the scope of the study, and the researcher must account for these at the end of his work (Creswell, 2010). Based on the general objective, the following specific objectives were proposed:

1) Understand the key elements of the phenomenon of organizational learning and value co-creation in software development teams.
2) Understand the relationships of these elements in the context of innovation in digital services.
3) Offer explanations for the behavior of individuals in development teams working in innovation processes with co-creation of value.
4) Develop a conceptual scheme capable of relating organizational learning, value co-creation, and digital service innovation.

1.3 Justification for the Study

The increasing importance of service innovation in today's competitive, service-oriented economy has placed immense pressure on organizations to enhance their flexibility, adaptability, and capacity for learning (Agazu & Kero, 2024). Service companies, particularly those delivering digital services, must continuously innovate to address customer needs, create tailored solutions, and maintain operational stability while driving new revenue streams (Xie et al., 2021).

While the concepts of organizational learning, value co-creation, and service innovation are well-established, studies that integrate these three constructs in a cohesive, empirical framework are notably rare (Bosisio, 2024). Existing research tends to examine these phenomena in isolation or through a narrow lens, leaving gaps in understanding their dynamic interrelationships (Helkkula et al., 2018; Heinonen & Strandvik, 2021). This study aimed to

respond to these gaps by investigating the interplay between learning processes, collaborative relationships, and innovation practices within the context of software development teams, which serve as critical conduits for knowledge integration and value creation in service organizations (Dixon, 2017; Duc et al., 2020).

Besides, software development teams occupy a central role in service innovation. They are responsible for transforming client needs, ideas, and feedback into tangible solutions that deliver value-in-use. However, these teams face considerable challenges, such as limited proximity to customers, misaligned communication, and the paradox of being expected to minimize errors while accelerating innovation cycles (Hoda et al., 2013; Xin et al., 2021). Additionally, the pressure to scale digital services often distances teams from the user-centric principles that underpin value co-creation, making it difficult to foster collaborative learning and experimentation.

At the same time, emotions and psychological enablers, hedonic aspects, remain overlooked in the literature on innovation and value co-creation (Tuunanen et al., 2023). Trust, for instance, plays a pivotal role in enabling collaboration, yet few studies have examined how mechanisms such as vulnerability, transparency, and shared purpose influence trust-building in teams (Wilhelm et al., 2019).

Further complicating these dynamics is the paradox noted by Desmarchelier et al. (2020), where increased customer involvement can amplify value co-creation in the early stages but may create contradictions and resistance. This phenomenon underscores the need for organizations to balance the benefits of client participation with the structural challenges it may pose, an issue that remains underexplored in empirical research.

Given the complexity of these interdependent constructs, this study employs Grounded Theory as its methodological foundation. Grounded Theory offers a rigorous, inductive approach to uncovering emerging patterns and relationships directly from data (Urquhart & Fernández, 2013). By capturing the lived experiences of software development teams and their role in organizational learning and value co-creation, this study developed a conceptual framework that aims to explain how innovation emerges as a dynamic, iterative process. This aligns with the use of Constructivist Grounded Theory to uncover context-bound substantive explanations for social and information system fields (Hewitt et al., 2022).

From both theoretical and practical perspectives, this study is significant. It bridges existing gaps in the literature by offering an integrated understanding of how learning, trust-building, and value co-creation enable service innovation. Software development teams, positioned as learning vehicles, emerge as critical actors in this process, where experimentation,

client collaboration, and trust serve as foundational mechanisms for driving innovation.

Practically, the findings may provide actionable insights for organizations striving to innovate in increasingly competitive markets. By uncovering strategies that foster collaborative learning, build trust through vulnerability, and enable iterative co-creation, this study can equip practitioners with tools to overcome barriers to innovation.

In conclusion, this research aimed to advance the understanding of service innovation by integrating organizational learning and value co-creation into a unified framework. It highlighted the unique role of software development teams as engines of innovation, shedding light on the psychological, relational, and structural mechanisms that drive their success. This study can contribute not only to academic discourse but also to the practical realities of organizations navigating the complexities of digital service innovation.

The remainder of this study is structured as follows. First, the theoretical framework is laid out and presents the concepts used to enlighten the relevant gaps within service innovation, value co-creation, and organizational learning. Besides, a short section was dedicated to explaining why software development teams were chosen as the context for addressing the research question. Second, the methodological procedures are described, starting from the paradigm selection, method, procedures, and instruments. Next, findings and results are discussed by means of a theoretical framework, followed by final remarks. Finally, the anchorage matrix, the initial interview guide, and GenAI prompts used during the analysis are outlined in the appendixes.

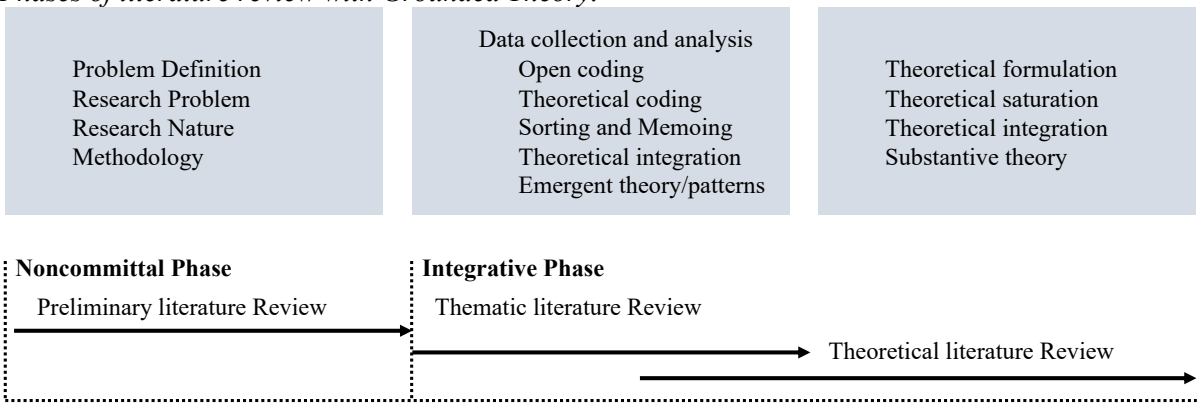
2. THEORETICAL FRAMEWORK

This section is formed by the theories supporting the concepts or areas of interest where this study will unfold. It will provide the context and capture the ongoing dialog of scholars about key definitions and relevant gaps and opportunities for the proposed research framework. The motivation for this brief literature review is to develop theoretical sensitivity before conducting the empirical work so that, as researchers, we understand the nature of the phenomenon under scrutiny. This is done without losing sight of the recommendation made by Urquhart and Fernández (2013, p. 230), "the generated grounded theory will determine the relevance of the literature, never the converse".

Therefore, the concepts presented in the following sections are part of the initial and noncommittal phase to find a broader research problem/scope and learn about extant domain literature (Urquhart & Fernández, 2013). Figure 1 depicts the different phases and scope of literature review on grounded theory projects:

Figure 1

Phases of literature review with Grounded Theory.



Source: Urquhart and Fernández (2013, p. 229).

As we move along the research process, the initial review gives place to a second phase of integrative nature, thematic in scope, which will drive the comparison between the emergent theory and the extant body of knowledge to formulate a substantive theory, during that stage the researcher will determine whether further theoretical sampling is required to progress towards saturation. Finally, the theoretical transposes the substantive theory into a formal theory (Urquhart & Fernández, 2013). The subsequent sections will describe the selected concepts used during the noncommittal phase.

2.1 Innovation

It is broadly accepted that Joseph Schumpeter is the founder of innovation research with his works in *Theory of Economic Development* and *Capitalism, Socialism and Democracy* back in 1934 and 1942 respectively (Lundvall, 2010). However, according to Lundvall, it was in the 1980s that innovation research flourished under the assumption that it was an interactive and manageable process.

In the last decades, innovation has become pervasive in our society, and organizations must secure a space in the hearts and minds of customers through innovative products and processes (Messmann & Mulder, 2020). Additionally, customers have been exposed to a broader range of information, creating a growing demand for newer products and greater quality, all while fierce competition leaves less room for attractive margins (Al-Omari et al., 2019).

Furthermore, innovation has also been the focus of an increasing number of scientific studies over the last few years dedicated to understanding how innovation activities, processes, and outputs affect our economies and general development for business and society (Cancino et al., 2017; M. Gupta, 2018). Nevertheless, being "innovative" is easier said than done. Despite the growing interest of practitioners and researchers, except by good practices and design principles (Bessant et al., 2005), precise prescriptions on how to implement innovation and the unquestionable determinants that engender it are still some sort of "holy grail" (M. Gupta, 2018).

However, innovation also shares a complex nature with other concepts in social science, embodying social and psychological dimensions that make it hard to manage and measure (M. Gupta, 2018). This combination of relevance and impact drew the attention of a plethora of researchers, and the following sections will highlight common definitions and typology.

2.1.1 Concepts of Innovation

Although innovation, as a concept, has been extensively studied from various disciplines and distinct perspectives, there seems to be no consensus or authoritative definition of innovation, leading to confusion and ambiguity (Baregheh et al., 2009). Nonetheless, prominent authors converge on its importance for value creation and organizational sustainability (Zahra & Covin, 1994), where the role of innovation is to renew companies' capabilities and offerings (Bessant et al., 2005) to cope with an ever-changing environment (Damanpour, 1991), through

learning processes (Dosi, 1990).

Indeed, the perspective that innovation is the process of filling the gap between unstructured beliefs, opportunities, and a new market offer or process to achieve competitive advantage is shared by Dosi (1990), who emphasizes the need to discover and learn from the resulting output or process.

According to Damanpour (1991), innovation is defined as adoption and encompasses the conception, development, and implementation of new ideas and behaviors. It can be adopting something created internally or acquired externally, whether a technology, product, service, policy, or process.

Zahra and Covin (1994) also approached the definition of innovation as a multidimensional concept where the two main dimensions are type and source, where type refers to the focus and source refers to the pattern followed by the organization for developing or acquiring innovations. For these authors, the different types are product, administrative, and process. The sources are incubative, acquisitive, and imitative (Zahra & Covin, 1994). However, their work did not offer a definition as such.

To this point, definitions converge for a view where innovation does not act in isolation or a single discrete event but rather a multi-stage process with distinct social actors that determine how ideas are transformed into new products or services (Baregheh et al., 2009). On the other hand, Lundvall (2010) posits that the processing of adaptation and competence building is accomplished by learning, and understanding the learning process is the most fundamental theoretical foundation for comprehending innovation. In that sense, Lundvall argued that innovation is the outcome of combining knowledge from different sites (sources) in a specialized innovation system.

However, the approach proposed by Lundvall fits for explaining how incremental innovation works, in which the outcome is an extension of an existing offer or the recombination of knowledge but falls short when dealing with disruptive innovation (Christensen, 2013).

Christensen originally proposed disruptive innovation in 1997 in his book “The Innovator’s Dilemma” (Si & Chen, 2020). These latter authors proposed an integrative definition of disruptive innovation as a process where the output, whether technology, product, or services, is initially inferior to existing offers in the attributes perceived by mainstream users but can satisfy other consumers with advantages in performance attributes. Learning would still be key, but as a process, metrics of good/bad performance are distinct, and less optimal alternatives sometimes are the right thing to pursue. Table 1 summarizes the distinct definitions

and perspectives found during the literature review of the innovation concept.

Table 1
Concepts of Innovation

Authors	Perspective	Concept
Dosi (1990).	Defined as process	Innovation concerns processes of learning and discovery about new products, new production, processes, and forms of economic organization
Damanpour (1991); Zahra and Covin (1994)	Defined in terms of type and source	Innovation is defined as the adoption of technology, policy, plan, program, process, product, or service that is new to the adopting organization
Baregheh et al. (2009).	Multi-stage	Innovation is the multi-stage process whereby organizations transform ideas into new/improved products, services, or processes to advance, compete, and differentiate themselves successfully in their marketplace
Toivonen and Tuominen (2009)	Schumpeterian	Innovation is carried into practice, provides benefits to its developer, and is reproducible.
Lundvall (2010)	Knowledge and learning-dependent	Innovation is the outcome of combining knowledge located at different sites in a specialized innovation system
Si and Chen (2020).	Disruptive Innovation	Innovation is a process where the initial output is initially inferior to mainstream offers but can still satisfy new markets with distinctive performance attributes.

Source: Prepared based on authors researched.

In this study, we draw the definition of innovation from Baregheh et al. (2009). Using content analysis, these authors analyzed a corpus of 60 cross-disciplinary definitions of innovations. They assert that "innovation is the multi-stage process whereby organizations transform ideas into new/improved products, services or processes, in order to advance, compete and differentiate themselves successfully in their marketplace" (Baregheh et al., 2009, p. 1334). It is worth observing that innovation is not limited to the creative act or the generation of ideas. This definition of innovation emphasizes transforming ideas into something and the subjective nature of its relationship with success (Hartley & Knell, 2021).

On the other hand, Baregheh et al. (2009) omitted in their definition the "how" and "where" this innovation happens. However, they allude to the "how" and "where" in their analysis when they organize the 60 definitions in 6 dimensions that consider stages, from creation and generation to implementation and adoption, and the locus, or social spheres where it happens: organizational, among firms, with customers, social systems, employees, and developers.

On the other hand, Hartley and Knell (2021) argued that innovation has an inherent

propensity for failure, where risk and unpredictability are intertwined with the process and the outcome. From their perspective, the renewed interest in innovation is not followed with the same enthusiasm for innovation failure. They asserted that active experimentation to learn and improve is central to innovation.

The following section will present the different types of innovation before discussing service innovation.

2.1.2 Types of Innovation

As a multidimensional concept, innovation can be described from the perspective of its different types (Hartley & Knell, 2021). Where "type" refers to the focus of a given innovation, whether a product, engendering an external focus, or a process that is an internal focus (Zahra & Covin, 1994). According to Gupta (2018), innovation can be categorized into two major types: product and process. Product innovation is about outputs and consists of goods and services created to attain profits or to address market needs. Process innovation aims to improve organizational processes by utilizing technical and managerial changes.

Damanpour (1991)argued that innovation is predominately about changing an organization in response to external influences. Therefore, outputs and processes are intertwined; in that context, technological innovations encompass new process technologies, while managerial innovations are a subtype of process innovation and relate to innovations in organizational structure, administrative systems, and routines (M. Gupta, 2018).

Baregheh et al. (2009) use a similar approach to categorization, with the caveat that types of innovation are related to the kind of output resulting from innovation, whether product or service. In contrast, the managerial and technical are referred to as means of innovation, that is, resources that need to be placed for innovation to happen. Damanpour (1991) contrasts managerial or administrative with technical innovation. The former relates to social structures, whereas the latter is concerned with products, services, and production technology.

Regarding the degree of novelty or the radicalness of innovation (Damanpour, 1991), innovations can be classified as radical or incremental. Damanpour explains that radical innovations result from drastic changes in the activities that an organization performs to produce value. In comparison, incremental innovations are extensions of existing practices with little departure from what is currently done.

Besides the dichotomy of product and service innovations, other authors highlighted the purpose of the innovation. Service innovations that tackle social, economic, and environmental

challenges are defined as social innovations (Windrum et al., 2016). According to these authors, social innovation involves muti-agents and value co-creation is guided by "social" outcomes and processes. Table 2 summarizes the different types of innovations found in the literature review.

Table 2
Types of Innovation

Authors	Perspective	Concept
Damanpour (1991); Baregheh et al. (2009)	Managerial vs. Technical Radicalness of Innovation	Managerial is about social structures; technical innovation concerns with products, services, and production technologies. Radical innovations produce fundamental changes and a clear departure from existing practices; Incremental innovations result in a little departure from existing practices.
Zahra and Covin, (1994); Gupta (2018),	Product vs. Process	Product: external focus and about outputs to address market needs and environment changes; Process: internal focus and to address technical and managerial changes
Windrum et al. (2016)	Social Innovation	Social innovators seek to develop new services that improve the quality of life of individuals

Source: Prepared based on authors researched.

The following section deepens the distinctive nature of service innovation and summarizes the theoretical perspective of the Service-Dominant Logic.

2.2 Service Innovation (SI)

It is relevant to start this section by reviewing the distinction between products and services in the innovation literature. Although Gallouj and Weinstein (1997) contrast the intangible nature of services with the inherent exteriority of products and their physical existence, they also argued that this dichotomy is hardly problematic and the boundaries of that distinction are fuzzy and can even be unnecessary for studying innovation (Gallouj & Weinstein, 1997; M. Gupta, 2018; Toivonen & Tuominen, 2009).

The initial idea behind service innovation relied on technological development as the main driver (Barras, 1986). The model proposed by Barras is considered the pioneer in the field, and for the first time, the relationship between innovation theory and services was established (Kowalkowski & Witell, 2020; Toivonen & Tuominen, 2009).

Drawing from Schumpeterian and evolutionary perspectives, Sundbo (1997) defines service innovation by conciliating the model of radical, abrupt changes with a model of

continual learning and asserts that service innovation is about creating an entirely new solution, or a new combination of old elements, to address a customer's problem in a repeatable manner. He argued that service innovation and organizational learning are just different theoretical lenses over the same phenomenon. From his perspective, "the innovation process in service firms is a search-and-learning process" (Sundbo, 1997, p. 450). Years back, Brown and Duguid (1991, p. 40) asserted that "working, learning, and innovating are closely related forms of human activity.".

In addition, Sundbo (1997) also emphasized the need for repeatable or reproducible service products and argued that this implies standardization of the service production and delivery processes. Therefore, although the customer problem is the point of departure for service innovation, the need for control of the production process and costs, and the fear of imitation tend to drive how the innovation comes to market and limit customer involvement during the development.

Also aligned with the perspective of a customer-centric focus on service innovation, Toivonen and Tuominen (2009, p.887) asserted that service innovations "emerge in the process of service provision on the basis of client's needs and are recognized as innovations only a posteriori.". According to these authors, the emergence of service innovation does not mean it occurs by chance or serendipity but instead as the deliberate enterprise to improve customer satisfaction and quality.

The service innovation literature portrays diverse perspectives for defining service innovation (Witell et al., 2016), where some authors categorize service innovations in terms of underlying assumptions and epistemologies (Heinonen & Strandvik, 2021), others emphasize that its conceptualization remains vague and lacks common meaning among researchers (Mele et al., 2019). For example, Coombs and Miles (2000) argued that different perspectives to define service innovations can be grouped into three main types: assimilation, demarcation, and synthesis.

According to the assimilation perspective, service innovation activities are generally the same as manufacturing activities and focus on new technologies as the main driver (Toivonen & Tuominen, 2009). Therefore, product and service innovation are similar (Coombs & Miles, 2000). The demarcation perspective emphasizes services' unique, intangible nature and dynamic characteristics (Witell et al., 2016). It is worth mentioning that in both approaches, the locus of innovation is within the firm boundaries and places customers as interested just in the output. Nonetheless, whether dealing with goods or services, the innovation research that falls into these perspectives tends to privilege firm-centric production activities and offerings

(Coombs & Miles, 2000).

On the other hand, the synthesis perspective emphasizes the value. It characterizes service innovation as a multidimensional construct with a compelling need for customer integration and the contribution of organizational knowledge and non-technological elements, such as extensive customer contact and experience (Hipp & Grupp, 2005). This broader view of the scope of innovation that spawns beyond firm-centered production activities and offerings has, in turn, generated new knowledge and practical solutions that are both collaborative and integrative (Lusch & Nambisan, 2015; Vargo et al., 2015). Besides, service research has moved beyond traditional output- and process-based archetypes to a more experiential (phenomenological) and systemic understanding of value creation (Karpen et al., 2012; Prahalad & Ramaswamy, 2003).

Nevertheless, Helkkula et al. (2018) departed from the three perspectives proposed by Coombs and Miles (2000) to suggest a typology of four theoretical archetypes of service innovation: output-based, process-based, experiential, and systemic. The authors argued that output-based and process-based archetypes occur most frequently among the four. At the same time, experiential and systemic accounts can be characterized as "emerging" archetypes of service innovation where social aspects are considered, and the service scope is beyond firm-centered activities (Kowalkowski & Witell, 2020). Those authors went further and anticipated that these archetypes would prevail as predominant perspectives for research.

When service innovation is viewed as an output, the research focuses on attributes as critical success factors or performance indicators. In that sense, customers would consume the outputs without participating in its conception (Helkkula et al., 2018). Conversely, when service innovation is viewed as a process-based archetype, emphasis is placed on architectural elements and the succession of activities related to service experience rather than as output (Toivonen & Tuominen, 2009). However, no other process is considered (Chan et al., 2010). Thus, we may argue that learning processes are overseen or ignored in both the output- and process-based approaches.

According to the experiential archetype, service innovation can be characterized as the subjective or individual experience that happens in a social context. Therefore, the experience of people using the service is socially constructed, the outcome of human interaction (Rubalcaba, 2018). The technology or process is not central but serves merely as a distribution mechanism for service provision, creating no value per se (Helkkula et al., 2018). As a convergent proposal to previous archetypes, the systemic archetype broadens the scope of service innovation by emphasizing multiple items, such as partners, customers, other actors in

the market, regulations, norms, and rules – in seeking a holistic view to understand service innovation and its environment as part of an ecosystem (Chandler & Lusch, 2015; Rubalcaba et al., 2012). Table 3 summarizes the distinct definitions and perspectives found during the literature review of service innovation.

Table 3
Concepts of Service Innovation

Authors	Perspective	Concept
Sundbo (1997)	Schumpeterian	Service innovation is about creating an entirely new solution, or a new combination of old elements, to address a customer's problem in a repeatable manner.
Toivonen and Tuominen (2009); Coombs and Miles (2000);	Assimilation	Service innovation activities are generally the same as manufacturing activities and focus on new technologies as the main driver. Therefore, product and service innovations are similar.
Witell et al. (2016)	Demarcation	It emphasizes the unique, intangible nature and dynamic characteristics of services. As in the assimilation perspective, the locus of innovation is within the firm boundaries and places customers as interested in the output.
Hipp and Grupp (2005); Chandler and Lusch (2015); Rubalcaba et al. (2012); Rubalcaba (2018),	Synthesis	It is defined as a multidimensional construct with a compelling need for customer integration and the contribution of organizational knowledge and non-technological elements, such as extensive customer contact and experience. Service scope goes beyond the boundaries of the organization. Service innovation is about co-creating new experiences.

Source: Prepared based on authors researched.

This study draws from a combined view because it enables researchers and managers to analyze service innovation from multiple angles. It defines strategies for value co-creation and integrates the resulting knowledge as fuel for a continuous service innovation process. In that sense, an option was made toward the systemic archetype once it considers a holistic view and aspects such as value creation in an ecosystem, the importance of value co-creation in the customer context, and that comprehends both the output and the process perspective (Helkkula et al., 2018).

Therefore, service innovation can be understood as a new process or service offering that is put into practice by an organization and is adopted by and creates value with/for one or more actors in an ecosystem (Helkkula et al., 2018; Witell et al., 2016). In comparison, actors can be customers, organizations, and other participants in the value-creation process (Patrício et al., 2018).

In sum, service innovation requires new forms of organizing the complex and interdependent relationships between actors, promoting alignment and integration of learning (Wang, 2021). Consequently, innovation is not restrained to the firm's boundaries but needs new working arrangements and mechanisms to thrive. As we describe the latter concept in the next section, the convergence of service innovation and organizational learning becomes apparent.

2.3 Organizational Learning (OL)

Although organizational learning is a relevant topic across the business and administration literature, convergence for a widespread theory or model has not yet been achieved (Blanco-Fernandez et al., 2022; Ricciardi et al., 2021). Decades ago, Fiol and Lyles (1985) pointed out a dilemma: the wide acceptance of the notion of organizational learning has engendered more confusion than explanation. The authors departed from the perspective that organizational learning is related to new insights or knowledge (Hedberg, 1981), assuming that in this process of improving action by knowledge, future performance will be promoted (Elbawab, 2024). Therefore, learning would be about aligning and adapting the organizational capabilities to environmental changes (Lundvall, 2010) and works as a response to reduce uncertainty (Schönherr et al., 2023).

Organizational learning can be defined as the change in the organizational knowledge stock promoted by practical experiences (Argote & Miron-Spektor, 2011). The change can be intentionally or unintentionally provoked (Huber, 1991), however, OL is related to the process through organizations learn (Domínguez-Escrig et al., 2023).

Fiol and Lyles (1985) also distinguish between individual and organizational, where the latter is not the summation of the former, but the result of cognitive systems and memories developed and shared by individuals and preserved as behaviors, norms, and values. In other words, OL occurs through individuals, but it is retained and embedded through the company’s strategic orientation, structure, culture, and interactions with the environment. In that sense, organizational learning is about the alternance of change and stability (Hedberg, 1981). Said differently, individual learning is a necessary but not sufficient condition for organizational learning (Fiol & Lyles, 1985).

This study departs from the perspective that learning in the organizational setting is a process and a social endeavor that happens on different levels, from the individual and group to the organizational sphere (Pérez López et al., 2005; Takahashi, 2015). These authors argued

that as an interdependent process, organizational learning requires coordination of the underlined levels and a shared purpose that gives sense and direction to activities.

Moreover, as a social endeavor, learning takes place in the presence of social relations, where emotions and perceptions can foster or stifle learning (Watzek et al., 2019; Wilhelm et al., 2019). Dixon (2017, p. 138) argued that "teams are the unit of learning within organizations"; therefore, working in teams is a combined endeavor of learning behaviors, interleaving action, and reflection to reach the outcomes.

Consistent with that perspective, Bontis et al. (2002) argued that organizational learning is a dynamic process where learning occurs over time, within and across levels. Learning within a level is related to what the authors called stock and the one that occurs between levels as a flow (feed-forward and feed-back). The authors assert that individuals use their intuition and interpretation to make sense of changes and opportunities as the ignition of this dynamic process.

According to Van Grinsven & Visser (2011), organizational learning is the "detection and correction of error". Where error is the difference between the aspirations or expectations and the actual outcome achieved. Drawing from an analogy, consider the physical concept of voltage. The difference in potential between the two points establishes the flow of charges (current). In the organizational setting, the difference between the expected and the actual result of an action or experiment triggers learning. The immediate social context is the conduit that can facilitate or even block learning from happening.

Nevertheless, authors like Putz et al. (2013) argued that learning is an unnegotiable prerequisite for companies competing in todays and future landscape. Consequently, structure, incentives, and organizational climate shall be aligned to promote employees' willingness to learn, experience, and share what was learned, what worked, and what did not work as well. For these authors, learning is inexorably bound to an experience-based process, where new information is compared and recombined with what individuals know, creating expectations and changes in behaviors (Argyris & Schon, 1978), and once it is experienced, can be gauged against the results.

This view is also shared by Weiss (1990), who argued that the term 'learning' is commonly defined as an experience-based process of a relatively permanent change in knowledge or skills. Likewise, Takahashi (2015) argued that organizational learning results from a continuous movement of reassessing meaning and establishing a new order from change and continuity. It thrives in the presence of trust and strong relationships where individuals share common values. Table 4 summarizes the distinct definitions and perspectives found

during the literature review of the concept of Organizational Learning.

Table 4
Concepts of Organizational Learning

Authors	Perspective	Concept
Hedberg (1981)	New insights bound to performance	Learning is a new insight or knowledge that will improve future performance. The focus is on the content of learning or the cognitive aspect.
Pérez López et al. (2005); Bontis et al. (2002)	Learning as a process	Organizational learning is a dynamic process where learning occurs over time, within and across levels
Van Grinsven and Visser (2011)	Trial and Error	Organizational learning is about the "detection and correction of error.".
Weiss (1990); Putz et al. (2013)	Learning as an experience-based process	Learning is an experience-based process where new information is compared and recombined with what individuals know, creating expectations and behavioral changes.
Watzek et al. (2019); Wilhelm et al. (2019)	Learning is embedded in social relations and emotions	Learning in organizations takes place in the presence of social relations, where emotions and perceptions can foster or stifle learning

Source: Prepared based on authors researched.

Regarding team learning, not all teams learn at the same pace, and the perception of social bounds and psychological safety affect the team's behaviors and performance (Edmondson, 1999). Individuals working in teams constantly monitor the environment to determine whether it facilitates or inhibits learning (Onça et al., 2018). Therefore, learning is not a spontaneous response to a mismatch between output and expectation but an induced process, a contingent response to the group learning climate where external and internal feedback is crucial (Bido et al., 2010).

Extending the idea of team learning to involve other stakeholders, such as customers and partners, for the co-creation of value is the subject of the next section in this theoretical framework.

2.4 Value Co-creation (VCC)

Before discussing the concept of value co-creation, it is worth conveying the definition of value in the context of the service literature. The meaning of value in the service literature is embedded in a broader discussion on value that traces back to Aristotle, who argued that different people assign different meanings to value (Akbar & Hoffmann, 2020). However, Adam Smith was the first to distinguish between value in exchange, typical of an owning or

product-centric economy, and value-in-use (Akbar & Hoffmann, 2020). The latter is related to satisfaction, with cognitive and emotional dimensions (Vargo & Lusch, 2008; Yu & Sangiorgi, 2018).

Value in services is commonly associated with value-in-use. It is the result of the consumer's cognitive evaluation or perception of the service experience (Akbar & Hoffmann, 2020). Consequently, value is influenced by the customer's functional, emotional, sociocultural, and relational experiences over multiple interactions in time (Yu & Sangiorgi, 2018). In the context of services, those interactions are known as service encounters (Rubalcaba, 2018), like the online reservation of a hotel, the review of a telecommunication service bill on the provider's portal, reading a book online, etc.

The service-dominant (S-D) logic literature also emphasizes the "phenomenological" nature of value, meaning that it is uniquely determined, interactionally, by the beneficiary, and the customer is always a co-creator of value (Vargo & Lusch, 2008). Therefore, value is an experiential concept and can only be created and determined by the user (Lusch & Vargo, 2006). Conversely, the enterprise providing the service cannot deliver value but promises or propositions of value (Vargo & Lusch, 2008).

In that context, value co-creation can be understood as a process where customers participate jointly in the value creation, starting from the problem definition and exploration of ideas and perspectives to the creation of expectations of experiences (Prahalad & Ramaswamy, 2004). That definition is consistent with the S-D logic. Nevertheless, in their updated version, Vargo and Lusch (2008) explained that co-production activities are optional for value co-creation.

Conversely, according to Desmarchelier et al. (2020), creating value-in-use propositions does not imply co-production. That is, the involvement in formulating or evaluating an offer may not be followed by the actual production/delivery of the value proposition. In that sense, the value proposition is no more than a promise whose realization occurs later when the customer buys, experiences, and compares the result against the expectation (Vargo et al., 2008).

Besides, according to Prahalad and Ramaswamy (2003, p. 14), value is co-created through experience, where value is defined by "the experience of a specific consumer, at a specific point in time and location.". Accordingly, engaging customers in their context to foment the innovation process is potentially more effective but would require an engagement platform for value co-creation (Patrício et al., 2018). As experience matters and customers are seen as the true experts in some domains, the participation of actors would be more beneficial

for incremental innovations. In contrast, radical innovations by default are unrelated to previous services (Patrício et al., 2018).

Creating value using an external focus is necessary for firms engaged in exploring market opportunities and their capacity to explore asymmetries with competitors (Brandenburger & Stuart, 1996). According to these scholars, "value creation is oriented towards the buyers they serve and the suppliers they rely upon" (Brandenburger & Stuart, 1996, p. 6). Authors like Desmarchelier et al. (2020) argued that filling the gaps created by asymmetries between customers' aspirations and the company offers can trigger learning processes as an adaptive response to the environmental misfit.

Furthermore, to realize the gap and to make efficient use of the process for value creation, companies must intentionally create a common stock of knowledge that facilitates the transfer of knowledge within groups (Kogut & Zander, 1992). This stock of knowledge also enhances the ability to evaluate, assimilate, and utilize external knowledge (W. M. Cohen & Levinthal, 1990). Table 5 summarizes the distinct definitions and perspectives found during the literature review of value co-creation.

Table 5
Concepts of Value Co-creation

Authors	Perspective	Concept
Brandenburger and Stuart (1996)	Value is Buyer-centric	Value creation is oriented towards the buyers they serve and the suppliers they rely upon
Prahalad and Ramaswamy (2004); Prahalad and Ramaswamy (2003, p. 14)	Co-creation of experiences	A joint venture where customers participate in the value creation, starting from the problem definition and exploration of ideas and perspectives to the creation of expectations of experiences. "The personal meaning derived from the co-creation experience is what determines the value to the individual."
Patrício et al. (2018).	Customer/partner journey	Value is co-created through multiple touch points of interactions known as customer/partner journeys. When experience matters, customers can contribute to incremental innovations as they are true experts in some domains.
Desmarchelier et al. (2020),	Value-in-use propositions	The involvement of customers in the formulation and evaluation of an offer.

Source: Prepared based on authors researched.

The classical definition proposed by Prahalad and Ramaswamy (2004) was used for this study, where value co-creation is a joint venture between providers and customers, from creating value propositions to implementing those expectations. As the interactions or encounters are the heart of service companies, understanding how the different actors such as

customers, partners, and others get aware of and experience services becomes crucial for opening new opportunities for innovation (Patrício et al., 2018). The multiple interactions are known as the customer/partner journey and form the perceived experience of what happened in the different touchpoints (Patrício et al., 2018). Similarly, the interactions of one specific actor or experience may also affect how other actors perceive the service. In this networked context, understanding how the relationship of multiple actors frames the value co-creation in service ecosystems offers a great research opportunity (Patrício et al., 2018).

The following section will discuss the locus of innovation for digital services. Software development teams will be presented as the key context where customers, partners, and employees can learn from each other and co-create service innovation.

2.5 Software Development Teams (SDT)

Before the discussion of software development teams as a concept, it is necessary to align what a team is in the organizational setting. From the multiple definitions found in the literature, the one proposed by S. G. Cohen and Bailey (1997) based on the work of Hackman (1987) will endorse the upcoming discussion. According to these authors, a team is a collection of individuals who are interdependent in their tasks, who have a shared purpose, who see themselves and are seen by others as a unit embedded in one or more large social systems, and who manage their relationships across organizational boundaries. Therefore, teams are interdependent social units embedded in organizational systems with common goals (Watzek et al., 2019).

Teams are seen as “vehicles” of collaboration and are important to organizations involved in innovation that must satisfy unceasing pressure for new products (Hoegl & Parboteeah, 2006). Teams are also a key source of learning and accomplishing in organizations (Watzek et al., 2019), in other words, “teams are the place where the rubber meets the road” where organizational learning happens (Santa, 2015). Also, the inherent characteristics of innovations reflect ambiguity, complexity, uncertainty, and high task variety upon the work of those teams making them different from ones assigned to routine projects (Sicotte & Langley, 2000). Teams can be seen as learning units and key drivers for innovation (Decuyper et al., 2010).

In that sense, teams are also equipped with innovation capabilities to generate novel ideas, turn those ideas into practice, and claim the benefits of the outcome (Duc et al., 2020). They argued that departing from prior knowledge about the service, teams can re-evaluate

evolving customer needs and intentionally promote, develop, and test new ideas that have the potential to update previous knowledge.

S. G. Cohen and Bailey (1997) suggested there are four types of teams in organizations: (1) work teams, (2) parallel teams, (3) project teams, and (4) management teams. The authors explained that work teams are a stable form of work units and are responsible for producing outcomes (goods and services). Parallel teams are formed by pulling people from the formal structure for problem-solving or improvement-related activities, having limited authority and mandate. Project teams are time-bounded arrangements where one-time outcomes are produced. In general project teams work on non-repetitive tasks and rely on resources from different parts of the organization to increment or create new value. The last type is the Management Team. It provides coordination and direction for the members of sub-units.

On the other hand, Devine et al. (1999) proposed a taxonomy of organizational teams with four types: (1) ad hoc project teams, (2) ongoing project teams, (3) ad hoc production teams, and (4) ongoing production teams. Ad hoc and ongoing project teams are similar in the sense they aim to solve problems, make plans or decisions, and interact with customers. They differ from each other in the time dimension, where ad hoc teams exist for a finite period, and project teams are more stable. In the same way, ad hoc and ongoing production teams are output-oriented and aim to build, construct, or assemble products, differing from each other in the temporary nature of the former versus the regular and more recurrent of the latter.

Software development teams (SDT) nowadays overlap with more than one type of team in the definition proposed by S. G. Cohen and Bailey (1997) and Devine et al. (1999). Software development teams usually follow a life cycle that goes from formation, storming, norming, and performing (Koc, 2007). According to Koc, an SDT is a multipurpose arrangement of human resources responsible for delivering, maintaining, and enhancing a software product over time. They may start as project teams with enough autonomy to coordinate their work with other groups and transition to a more stable, production-oriented structure to support the evolutive aspect of software requirements.

Learning is at the heart of the software development activity. According to Koc (2007), the work of software development teams consists of acquiring domain knowledge about a specific problem (internally and externally), understanding user requirements and business opportunities, sharing findings and interpretations (user stories and specifications), and finally integrating this knowledge into software artifacts. To this author, cross-functional cooperation and the involvement of key stakeholders are key to effective teamwork.

Software development teams become particularly relevant for understanding service

innovation and value co-creation since they are the primary conduits for integrating individual knowledge in organizations (Dixon, 2017). They can be understood as the locus of learning, experimentation, and socialization of knowledge (Mehta et al., 2022). According to Sicotte and Langley (2000), a distinct mix of skills is often necessary to fuel innovation processes and perform complex activities needed to develop new products. Therefore, knowledge integration capabilities, an experimentation mindset, and a psychologically safe environment where trust and collaboration exist are antecedents to enact learning and innovation (Dixon, 2017; Sicotte & Langley, 2000).

2.6 Learning, Innovation, and Value Co-creation

Innovation is inherently uncertain, so lacking answers to well-defined questions triggers a learning process (M. Gupta, 2018; Leoncini, 2016). Besides, customers using digital services also learn through the experience with the new offer and create use-value in that process. By leveraging customer participation as part of the development team, organizations could potentially lower the uncertainty while providing a means for knowledge exchange (Weiss, 1990) and opportunities for developing consensus about problem definition and expectations (Sicotte & Langley, 2000).

However, some may argue that bringing the customer to participate in the value-creation process is like inviting restaurant clients to visit the kitchen; it is a double-edged sword (Chan et al., 2010). On one side, higher levels of satisfaction may be produced (Helkkula et al., 2018), on the other side, it engenders increasing levels of anxiety and stress (Chan et al., 2010) and may expose suboptimal processes and technologies that otherwise would not be seen.

Prior discussion may lead the reader to a conclusion that is not corroborated by empirical findings. Bringing customers to participate in the innovation process does not ensure more effectiveness and use-value as some may expect (Desmarchelier et al., 2020). According to these authors, despite more customer involvement can amplify value co-creation during offering elaboration when it deepens in co-productions (value configuration and capture), simulations show that contradictions between consumers' aspirations and the practices and values of their organizations engender ambiguity and create viscosity in the learning cycle, resulting in less use-value potential.

2.6.1 Learning, Innovation, and Value Co-creation in Software Development Teams

The intriguing relationship between teamwork and effectiveness is not new. Hackman (2002) argued that common sense suggests that bringing more resources and diverse perspectives could create better outcomes. However, he referred to what empirical evidence shows as a puzzle. He argued that "research evidence about team performance shows that teams usually do less well—not better—than the sum of their members' individual contributions" (Hackman, 2002, p. 246), that is, teams consistently underperform despite the extra resources.

The same problems found in groups that don't work as a team (Hackman, 2002) can affect the outcomes of software development teams where customers participate in the innovation process: conflict of authority (who determines the direction to follow? What is the level of autonomy of the team? What the team is supposed to be doing?); constrained support from the organization; a clear group structure – tasks, composition, and norms; the assumption that everybody knows how to work as a team, even when new members (customers) are invited to participate. Hackman then suggested a few questions that can help to determine how the enterprise operates to install the necessary conditions to foster effectiveness in teamwork, not only in business-as-usual situations but also under circumstances where innovation is the outcome:

- Who decides? Who decides what ideas and problems are important to pursue?
- Who is responsible? What is the demarcation of responsibility and accountability?
- Who gains? What are the incentives allocated to the different individuals?
- Who learns? How do team members appropriate learning, growth, and career opportunities?

According to Decuyper et al. (2010), team learning is simultaneously a process and output where acquisition, participation, and creation of knowledge promote a collaborative adaptation or improvement. That is, the result is co-created, but it is not free of conflict, which is expected and key for innovation and transformation. Therefore, team learning goes beyond sharing ideas and requires an integration process, an ongoing dialog, where ideas are built on each other's contributions, but also involves questioning, confrontation, and contradictory opinions as a constructive conflict mechanism for refining, building on, or modifying an

original offer. That is, team learning is at the center of innovation processes where co-creation occurs.

Therefore, considering that knowledge is the main operant capable of fueling the transformation of value propositions into value created through interaction with customers (Vargo & Lusch, 2008), software development teams would have the potential to drive the alignment of value propositions to customer-determined benefits through learning and innovation.

The emphasis on co-construction or constructive conflict processes is intertwined and can be related to more exploratory innovation or more fundamental transformation respectively (Decuyper et al., 2010). On the other hand, Hackman (2002) suggested that to achieve better performance, teams that used to work together need a deviant who can challenge homogeneity and instill creativity and learning.

The theoretical discussion has shown that learning is at the heart of service innovation and teamwork and that customer involvement can bring the diversity necessary for constructive conflict that engenders new ideas. Also, service innovation is a customer-centric process, and it is about how value is experienced (use-value) by the end consumer. Besides, a paradox appears when introducing an external member to a software development team; it creates "viscosities" related to conflicting interests, authority, and expectations that may lead to less use-value instead of more (Desmarchelier et al., 2020). Table 6 summarizes the perspectives of learning, innovation, and value co-creation in software development teams.

Table 6

Learning, innovation, and value co-creation in software development teams

Authors	Perspective	Concept
Hackman (2002)	Team performance is about balancing homogeneity and diversity.	Teams that used to work together need a deviant to instill creativity and learning.
Decuyper et al. (2010).	Team learning is about the co-creation of knowledge	Team learning requires co-construction and constructive conflict.
Desmarchelier et al. (2020).	Value co-creation may lead to learning viscosity	Customer involvement can amplify the creation of value propositions, not necessarily more use-value.

Source: Prepared based on authors' research.

Likewise, those service innovations can respond to environmental changes or unfulfilled customer needs. Still, in one way or another, software development teams are ultimately responsible for bringing forth digital service innovations. The next chapter will detail the research procedures that will guide the empirical work considering the study's objectives.

3. METHODOLOGICAL PROCEDURES

This section describes the research procedures adopted in this study, starting from the paradigm, followed by a brief discussion of the research method, participants, and the data collection technique.

Before detailing the procedures followed in the empirical work, it is important to establish what paradigms mean and how they affect how the research will be performed. According to Kuhn (1989), paradigms are models or standards expressed in the form of accepted assumptions that determine what the world is like (ontology) and how knowledge about it can be created (epistemology). Additionally, they represent the commitments made by the researcher and determine how the phenomena should be observed. Khun further argues that paradigms can direct research even in the absence of rules.

Ryan (2018) classifies the distinct paradigms as Comte positivism, Logical positivism, Critical positivism, Critical theory, Interpretivism, Post-modernism, Institutionalism, and Symbolic interactionism.

In general terms, for positivism, “reality” consists of sensitive impressions, and empirical knowledge is the only way of knowing (Giddens, 1998). From this author's perspective, the advancement of knowledge in positivism occurs through the principle of verification or refutation, where statements are considered significant only if there are ways to test them.

Unlike the positivist view, the interpretivism paradigm considers that the reality resulting from the interaction between subject and object is not objective but that meanings are imposed on objects (Mingers, 2003) and, therefore, if known, can be used to explain other phenomena in different arrangements/contexts. The unit of analysis in interpretivism research is meaning. Thus, the researcher, using methods of reliability and authenticity, can generate knowledge and understand the phenomenon in a controlled way without “disturbing” its essence.

Likewise, from the perspective of symbolic interactionism, reality is intersubjective, created between social authors, as being the product of interactions that also never occur in an apolitical way (Harvey & Katovich, 1992). Refuting structural functionalism, interactionism highlights that humans are simultaneously subjects and objects of the changes that they engender or by which they are affected.

According to Harvey and Katovich (1992), symbolic interactionism originated in the need to explain the changes engendered in the United States in the last decades of the 19th

century with the end of agrarian society. With the end of the First World War, American society became an excellent incubator for the development of social psychology: an urban, modern, inventive, and heterogeneous society. The term symbolic interactionism was initially coined by Helber Blumer in his writings back in 1937 (Bryman, 2012), and it is rooted in three premises (Blumer, 1980):

- Human beings act towards things based on the meanings that these things have for them.
- The meaning of such things is derived from, and arises from, the social interaction one has with one's fellows.
- These meanings are handled and modified through an interpretive process used by the person in dealing with the thing he encounters.

According to Bryman (2012), in symbolic interactionism, the subject interprets his environment and acts based on the meanings he assigns. For this author, social life is a process that unfolds actions and meanings associated with experiences. Therefore, sensing-making and experience are at the heart of symbolic interactionism as a paradigmatic perspective, which makes this an appropriate lens for studying the phenomena object of this study.

This study draws from a symbolic interactionism (SI) conception aiming to understand the phenomenon of organizational learning and value co-creation in the context of service innovations of software development teams. As a theoretical lens, SI values interaction and strives to understand the meaning people assign to things and other entities as they experience social relations. Therefore, SI focuses on comprehending human actions and behaviors in a group where reality is said to be socially constructed (Blumer, 1980).

The following characteristics of symbolic interactionism, extracted from the work of Harvey and Katovich (1992), corroborate the choice of paradigmatic approach to this study:

- Empathy: in symbolic interactionism, the "I" is part of a process of fantasy and posture, as well as empathizing and reflecting. The researcher is not apathetic and a mere observer; he participates in the process of interpreting reality.
- Holistic: society is inseparable. Individuals, when in groups, reconstitute their environment by adjusting their behavior in response to that of others.
- Dual causality: Human nature is seen as a system of beliefs, desires, and purposes, and it presupposes continuous "interaction" between the individual and the environment. Individuals are subjects and objects of social change.

- Evolutionary: constant adaptation to the new, to the emerging, acting concomitantly with the actions of others, influencing, and being influenced, making 1+1>2.
- Manipulation: social actors creatively modify social reality through manipulations of perspectives and identities.
- Shared expectations: they are the basis of social control and ethics. It's a way of anticipating the response of others and adjusting your perspective accordingly.

Many of the above characteristics are also common properties of the concepts outlined in the theoretical framework. The interpretation that software development teams associate with value, learning, and innovation is not dissociated from shared expectations about accepted behaviors and habits. By adopting the lens of interactionism, the researcher can understand the predispositions and automatic modes of response of individuals.

3.1 Grounded Theory (GT)

The last few decades have seen a growing corpus of studies using grounded theory, where authors used this qualitative method as an inductive process to systematically gather and analyze data to discover a theory grounded in participants' accounts of a phenomenon (Urquhart & Fernández, 2013). According to these authors, a distinctive aspect of GT, compared to other qualitative research methods, is the continuous interplay between data collection and analysis. Although the development of new theory is often associated with the primary objective of grounded theory, other contributions such as models, justifications, rich descriptions, boundary conditions, and explanations are common in studies that rely on this method (Wiesche et al., 2017). Before describing the procedures followed by this study, this section introduces the origins and distinct types of grounded theory as a research method.

Grounded Theory was initially developed in the School of Nursing at the University of California in San Francisco by sociologists Glasser and Strauss in a work about the Awareness of Dying (Glaser & Strauss, 1967). According to these authors, Grounded Theory (GT) is “the discovery of theory from data systematically obtained from social research.” (Glaser & Strauss, 1967, p. 2). Influenced by symbolic interactionism, GT departs from the understanding that human beings act based on the meanings they associate with things and that those meanings are embedded in social interactions that influence the interpretative process (Blumer, 1980).

Glasser and Strauss wrote a book conceived to assist scholars in generating new theories based on systematically working out the collected data during the research (Glaser & Strauss, 1967). The central idea was to assist researchers in discovering their way from problematic situations through reflexive propositions and richness of descriptions to theories that emerge from them (Lacerda & Santos, 2019). Discovery in grounded theory is closely related to the principle that we are creatures of habits so that behavior patterns can be noticed (Giddens & Turner, 2005).

Grounded theory strives to provide rich descriptions as narratives based on empirical observations (Langley, 1999). These narratives include objects, people, systems, activities, and events, illuminating a phenomenon, providing practical relevance, and potentializing the development of the new theory (Wiesche et al., 2017). Therefore, using GT as a research method in information systems has gained renewed interest and offers a systematic way to explore a social phenomenon experienced by participants, along with process and context-embedded problems (Urquhart & Fernández, 2013).

In that sense, Grounded theory can be seen as an inductive process that interplays, as a pendulum does, a constant movement between data and theory. Therefore, the theory is conceptualized within the study by working systematically with data rather than verifying previous theoretical assumptions (Charmaz, 2014; Farragher & Coogan, 2020; Glaser & Strauss, 1967). Besides, the researcher does not approach reality as a tabula rasa (Glaser & Strauss, 1967). The extant theory helps challenge emergent findings, driving areas of interest, suggesting ideas for new data collection, and enabling better conceptualization of emergent patterns (Levina & Lazarsfelt, 2021). The recommendation from Glaser and Strauss is a warning to researchers to set aside misconceptions and prevent the extant literature from dictating the research findings and hypothesis (Urquhart & Fernández, 2013).

In fact, since the original work was published, Glaser and Strauss followed different paths in developing Grounded Theory (Bryman, 2012). The GT, as developed by Glaser, is the classical version and considers that the research should avoid preconceptions and conjectures. According to this perspective, the researcher should enter the field without a research question and be open to emerging themes (Dey, 1999). GT does not give permission to ignore the existing literature, instead, it should be recruited as warranted by the evidence from the fieldwork (Dey, 1999).

On the other hand, the Straussian perspective is consistent with a pos-positivist paradigm. This alignment explains why this GT version is more structured and prescriptive in analyzing and coding data (Dey, 1999). For Straussian practitioners, the development of

explanatory theories is a process that is simultaneously inductive and deductive, where the researcher approaches the field with a broad research question and gradually refines it as the research unfolds.

The Straussian perspective relies on a paradigmatic model where, by the end of the research, the author should be able to assert under which specific conditions people assign meaning and what actions are expected with what results in such context (Lacerda & Santos, 2019). Urquhart and Fernández (2013) recounted the tension between the different perspectives of GT and pointed out the main disagreement in terms of the coding process and the use of the "conditional matrix", in summary, of emergence (Glasian) vs. forcing the theory (Straussian).

Charmaz, a student of Glaser, proposed that the theory is not discovered; instead, it is built from the meaning and expressions of the participants (Dey, 1999). For Charmaz (2014), the Grounded Theory has a pragmatic ontology with a relativist epistemology, meaning it is constructed, and participants refer to constructs when they explain events or describe a phenomenon. Therefore, for Charmaz, constructive GT emphasizes the mutual creation of knowledge resulting from the interaction between researcher and participant.

Although the three main strands of Grounded Theory are regularly discussed, it is worth mentioning that there are "probably as many versions of grounded theory as there were grounded theorists" (Dey, 1999, p. 2). However, the variations usually gravitate around one of the three main perspectives in some of the dimensions depicted in Table 7.

Table 7

The different perspectives of Grounded Theory

Classification Criteria	Classical GT	Straussian GT	Constructivist GT (CGT)
Main Authors	Glaser and Strauss (1967)	Strauss and Corbin (1990)	Charmaz (2014)
About Paradigms	Glaser follows a positivist ontological approach. But Glaser himself refuses that.	It allows for an abductive reasoning influence rather than purely inductive.	Charmaz proclaims CGT has a pragmatist ontology with a relativist epistemology.
The role of the researcher	Glaser maintained empiricism; thus, the experiment or fieldwork drove the research, not the preconceived theory.	The researcher and participants collaborate in a sensing-making exercise.	The theory is not discovered. It is instead built from the meaning and expressions of the interaction between researchers and participants
About Prior Knowledge	Considers that prior knowledge lends itself to bias, the same way the researcher's background should be put aside. Neutrality is necessary	Asserts that researchers should be aware of the influence and not be blind to prior literature.	CGT's previous knowledge can strengthen a research project, provided it does not define it. CGT researchers are to carefully navigate and

Classification Criteria	Classical GT	Straussian GT	Constructivist GT (CGT)
			control their perspectives rather than attempt to erase or forget their knowledge.
The research question	Glaser suggests that the researcher starts the field work without a research question, but with a general area of interest.	The research question needs to be broad, open, and not focused. But it will be gradually refined as the research process unfolds. Answering this question determines what is pertinent or not.	Participants are experts in matters relating to them, therefore the research question starts broad, and it is flexible to be shaped by the researcher interaction with participants.

Source: Prepared based on authors researched.

It is also evident that Grounded Theory is not free of criticism. According to Dey (1999), it is a paradox that a methodology based on "interpretation" proved hard to interpret, given the multiplicity of perspectives it has unfolded. By using a vivid metaphor, the author argued that "meaning is like a bubble floating before our (researcher) eyes: we see it plainly enough, but as soon as we try to grasp it, the bubble bursts and meaning evaporates" (Dey, 1999, p.23).

Urquhart and Fernández (2013) highlighted that studies using GT have also been criticized for having low levels of theory development and for being used merely as a coding method instead of a method for generating theory. However, the authors argued that the antidote for such criticism relies in great measure on a systematic approach or the use of a framework as a guideline. Urquhart and Fernández (2013) pointed out four distinctive characteristics of the systematic use of grounded theory:

- Its primary purpose is to theory building. Therefore, the researcher needs to be capable of theoretical sensitivity, that is, understanding the field of investigation and the general context of the phenomenon.
- Prior knowledge can lead the research into biased preconceptions and hypotheses. Dey (1999) suggested having an open mind is not the same as having an empty head; thus, the researcher can be familiar with the field's literature but should be aware of his preformulated ideas.
- Constant comparison of roots analysis and data collection are the core processes of theory discovery/building. Data collection, analysis, coding, and analysis are done simultaneously and engender the emergence of theory from the data.
- Theoretical sampling is the selection mechanism that decides where to sample from next. Concepts, categories, and ideas from the initial data will guide further data collection.

In sum, Grounded Theory (GT) is a qualitative research method (Chamberlain-Salaun et al., 2013), and while scholars differ in their views on its epistemological assumptions, it is essential for researchers to explicitly articulate their own ontological and epistemological stance (Urquhart et al., 2010). This clarity is crucial because these positions influence the coding and analysis processes. According to these scholars, any assertion that GT is inherently positivist, interpretive, or constructivist lacks robust support from the existing GT literature.

In this research, we emphasized the reflexivity inherent to symbolic interactionism (Hewitt et al., 2022) Besides, we adopted Grounded Theory as the methodology to construct interpretations (Charmaz, 2014) with the expectation to promote the researcher and participants' awareness of the different bubbles of meaning, why they were formed, and by whom, to advance the current knowledge about services innovations in software development teams. By following a systematic framework depicted in the following section, we aimed to firmly ground the research's empirical work and provide the required rigor for this scientific endeavor.

Therefore, as a disclaimer to contribute to the research rigor, the reader should be aware that my professional background and experience inevitably shape my analytical perspective as a researcher. Having worked extensively in the Information Technology Industry for over 30 years, I brought practical insights and a solid theoretical foundation to the field under study. However, as Dey (1999) suggested, familiarity with the literature and industry practices is not inherently a source of bias but rather a resource for informed interpretation. Therefore, my prior knowledge allowed me to engage with participants more effectively, formulate meaningful questions, and navigate the complexities of the field. At the same time, I remained critically aware of any preconceptions, ensuring that data, rather than assumptions, guided the emerging insights. This reflexive approach aligns with best practices in qualitative research, where the researcher is both an observer and an instrument of analysis, requiring ongoing self-awareness and methodological rigor.

3.1.1 Theorizing in Grounded Theory Studies

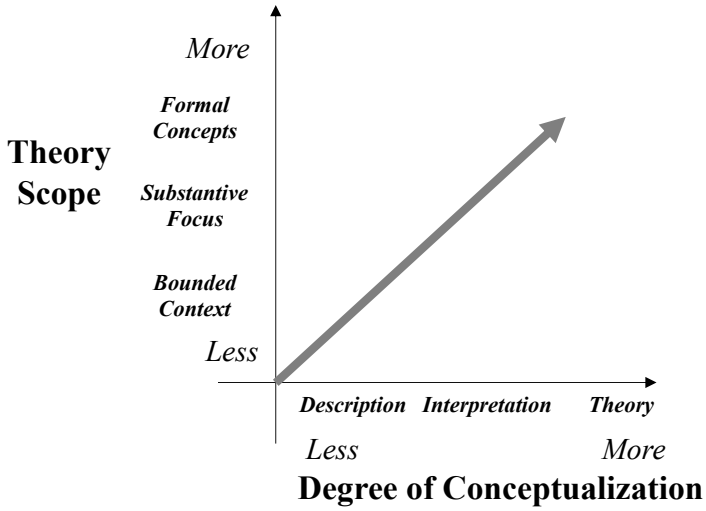
Considering the primary outcome of using GT as the research method is theory building, it is necessary to understand the different degrees to which this objective can be achieved. With this intention, Urquhart et al. (2010) highlighted two relevant aspects of theorizing: conceptualization and theory scope. Conceptualization corresponds to the degree of data analysis; the greater the depth of the analysis, the better the use of GT is (Glaser & Strauss, 1967). Theory scope relates to the coverage of phenomena explained by the proposed theory

and the primary objective of GT is to develop theories with a greater scope (Dey, 1999). Urquhart and Fernández (2013) capture this framework and position these dimensions along two axes, as depicted in Figure 2.

According to Glaser and Strauss (1967), the degree of conceptualization varies from the most basic conceptual stage, Description, passing through interpretation, and finally, theoretical coding. “Description” is about identifying categories and properties. Interpretation can help to explain the interaction and focus on understanding and explaining areas of interest. The last stage, theoretical coding, aims to create inferential or predictive statements (Urquhart et al., 2010).

Figure 2

A framework for analyzing grounded theory studies.



Source: Urquhart et al. (2010, p. 366).

The second dimension, y axis, theory scope, aims to move from a bounded context, passing through substantive focus, to formal concepts (Glaser & Strauss, 1967). According to Urquhart et al. (2010), the bounded context is the narrowest scope, and it is no more than the researcher’s hunches from the context with little support of empirical evidence. Substantive focus has strong empirical support and extends the theory's predictive and explanatory power to the specific phenomena in which it was developed. The formal concept is the greater scope of a theory developed using GT. A formal theoretical construct applies to a family of substantive areas.

3.1.2 Constructivist Grounded Theory (CGT)

Therefore, this study relied on constructivist grounded theory (Charmaz, 2014). The choice for the constructivist approach is consistent with the nature of the phenomena under scrutiny, where people working in software development teams interpret learning and value co-creation according to their social setting and expectations. According to Farragher and Coogan (2020), the emphasis CGT places on reflexivity and the position of researcher/subjectivity makes its choice ideal for exploring integral social relationships and the behavior of groups where the researcher co-produces knowledge in the interaction with subjects. For Charmaz (2014), the process of building theory out of data consists in:

- Simultaneous collection and analysis of data.
- The creation of analytic codes and categories is developed from data and not by pre-existing conceptualizations (theoretical sensitivity).
- Discovery of basic social processes in the data.
- Inductive construction of abstract categories.
- Theoretical sampling to refine categories.
- Writing analytical memos is the stage between coding and writing.
- The integration of categories into a theoretical framework.

Urquhart and Fernández (2013)argued that before starting the research, one must familiarize oneself with an area of inquiry and rely on seed concepts as ideation sources. This initial level of abstraction informs research gaps or opportunities and starts the process of theory building.

3.2 Research Procedures

As a research method, Grounded Theory required a formalized approach to its application. This formalization involved outlining how the analysis was systematically conducted (Levina & Lazarsfelt, 2021), including identifying interesting questions within the data, defining a sampling strategy, and carrying out systematic data comparisons to identify insightful patterns and verify their consistency (Dey, 1999).

3.3 Data Collection Instrument

The data collection process relied on intensive interviews to explore participants' experiences, meanings, and perceptions in-depth. The interviews were guided by open-ended questions aligned with the grounded theory method. The interview approach was flexible enough to accommodate emerging themes and incidents, and how participants solved those concerns. Specifically, Dey (1999) suggested that at the beginning of the study, the data collection process resembles more of a conversation. In contrast, more focused interactions will be conducted on specific topics as the study evolves. This author envisaged the sequence (Dey, 1999, p. 6):

- Use qualitative methods for data collection.
- Look for a variety of data sources.
- Start with unstructured methods of collecting data.
- Develop a dialog between data and analysis.
- As theory emerges, use more focused methods of observation.

According to Charmaz (2014), the intensive nature of grounded theory interviews aims at eliciting relevant experiences and meanings participants associate with those and how they portray these experiences. The adoption of intensive interviews using open-ended questions facilitates the exploration of an area in which the participant understands or perceives how a phenomenon unfolds and may offer an explanation, concern, or constraint to what is happening in their world.

An initial interview guide was developed based on the matrix presented in Appendix A – Anchorage Matrix. The initial questions are detailed in Appendix B – Interview Guide. This guide was designed as a flexible tool to outline the questions to be addressed during the interviews, providing direction and structure to the conversations.

The interview guide aimed to facilitate a dialogic exploration of participant perceptions and the identification of relevant issues. Furthermore, it was designed to be inherently aligned with the central research question. The guide served as a criterion for determining the relevance of the data collected (Dey, 1999).

To uphold the principle of transparency in scientific research, the steps and decisions made during the data analysis process were thoroughly detailed in the corresponding section (Shenton, 2004). All interviews were conducted between November 2023 and September 2024.

The interviews were conducted via Microsoft Teams, which provided an initial automated transcription. This transcription served as a starting point for the researcher's rigorous manual transcription process. By personally transcribing the interviews, the researcher ensured accuracy and maintained a deep engagement with the data, aligning with the grounded theory's emphasis on the researcher's close interaction with the material.

It is important to note that two initial interviews were conducted to validate the interview guide and familiarize the researcher with the field. The same guide, with necessary adjustments, was utilized for subsequent interviews. All interviews were recorded and transcribed in their entirety by the author, ensuring a consistent and uniform transcription process.

The first interview provided valuable insights that significantly shaped the refinement of the interview guide. Initially, the opening question about the participant's experience proved too broad, leading to lengthy responses and consuming more time than anticipated. Additionally, one question assumed the participant had an active role in co-creation processes, which was not necessarily true for all cases, highlighting the need for more neutral phrasing. Besides, while well-intentioned, the alignment of the questions with research objectives resulted in wording that felt somewhat detached from the participant's day-to-day reality. I often found it necessary to complement the questions with additional explanations to address this, disrupting the interview flow. These observations prompted revisions to ensure the guide was accessible and adaptable to participants' diverse experiences.

3.4 Research Participants

This section provides an overview of the participants involved in this study. Understanding the characteristics and backgrounds of the participants is essential for contextualizing the findings and ensuring the credibility of the research. The participants were selected using purposive/convenience sampling followed by theoretical sampling (Bryman, 2012). In this approach, key individuals or relevant cases were identified as the starting point for a successive refinement process in which the data collection aimed at maximizing the opportunity to find variances, develop concepts, and identify relationships between emerging concepts (Dey, 1999).

Participants with first-hand experience in software development and service innovation were selected within the same team and among teams of different organizations. In total, 12 participants were recruited, from nine distinct companies, representing diverse demographics and professional experiences that enrich the data collected. The recruitment process involved

direct contact with participants through WhatsApp or e-mail, which proved time-consuming and ineffective.

After the sixth interview, the process was changed to ask previous participants to indicate new names or to forward the "Invitation E-mail" (Appendix C – E-mail Invitation for Interview) to contacts on their network. The prospective participants, in turn, could schedule time on a shared calendar directly (Appendix D – Online Appointment Schedule Tool) and this process turned out to be more effective. Within 15 calendar days, six interviews were scheduled and performed. Table 8 describes the participants in this study. With no exceptions, all participants work for information technology companies and all of them reported their gender as male.

Table 8
Research Participants

Participant	Job/Role	Company Classification	Educational Level	Country Region	Team Size
P1	Software Developer Director (R&D)	Traditional software company (Org. 1)	Bachelor's degree / MBA	Brazil South	15 persons
P2	Senior Manager (Cross Product Integration)	Traditional software company (Org. 1)	Bachelor's degree / MBA	Brazil South	18 persons
P3	Software Engineer (R&D)	Traditional software company (Org. 1)	Bachelor's degree	Brazil South	4 persons
P4	Software Architect (Cross Product Integration)	Traditional software company (Org. 1)	Bachelor's degree	Brazil South	5 persons
P5	Entrepreneur and Back-end developer	Software House (Org. 2)	Bachelor's degree	Brazil North	5 persons
P6	Freelancer and Product Owner	Software House (Org. 3)	Bachelor's degree	Brazil Southeast	5-10 persons (project related)
P7	Product Manager	Traditional software company (Org. 4)	Bachelor's degree	Brazil North	58 persons
P8	Entrepreneur	Startup (Org. 5)	Bachelor's degree	Brazil Southeast	22 persons
P9	Entrepreneur and Tech Leader	Software House (Org. 6)	High school	Brazil Southeast	7 persons
P10	Entrepreneur	Startup (Org. 7)	Bachelor's degree	Brazil South	21 persons
P11	Entrepreneur	Startup (Org. 8)	PhD. Computer Science	Brazil North	4 persons
P12	Entrepreneur and Product Manager	Startup / Traditional software company (Org. 9)	Bachelor's degree	Brazil North	Not informed

Source: Research Data (2024).

The organizations the participants work for were categorized as described in column "Company Classification" using the criterion below. In total, the participants represented 9

distinct organizations (also indicated in the column “Company Classification” by the Org. #number):

- **Traditional Software Companies**: Typically refer to established companies that primarily engage in software development and focus on serving established markets. They often operate as product-oriented companies, follow a business plan, and tend to emphasize scalability and innovative product development (Blank, 2013).
- **Software Houses**: refer to companies that primarily engage in software development and provide services such as custom software solutions, system integration, and maintenance for clients. They often operate on a project basis and may not focus on scalability or innovative product development (Blank, 2013).
- **Startups**: Generally defined as newly created companies that seek to develop a product or service highly focused on solving customers' pain points or fulfilling their needs (Kemell et al., 2020). They often aim to scale quickly starting from a niche market and operating under conditions of uncertainty (Blank, 2013). They are characterized by innovation, high risk, and a focus on disrupting existing markets (Giardino et al., 2014).

All interviews conducted as part of this qualitative study were recorded with the explicit consent of the participants. The recordings were subsequently transcribed verbatim to ensure accuracy and fidelity in data representation. This data collection method was employed to facilitate a comprehensive analysis of the participants' perspectives, experiences, and insights related to the research objectives. All recordings and transcripts will be handled with the utmost confidentiality and following ethical research practices.

Table 9
Interview Schedule and Duration

Participant	Date	Duration	Schedule Type
P1	28/Nov/2023	45:44 min	Direct contact
P2	27/Mar/2024	55:16 min	Direct contact
P3	16/May/2024	48:00 min	Direct contact
P4	17/May/2024	1:01:01 hr	Direct contact
P5	28/Jun/2024	40:10 min	Direct contact
P6	09/Jul/2024	50:53 min	Direct contact

Participant	Date	Duration	Schedule Type
P7	18/Set/2024	44:33 min	Calendly Booking
P8	20/Set/2024	41:15 min	Calendly Booking
P9	23/Set/2024	37:19 min	Calendly Booking
P10	24/Set/2024	30:26 min	Calendly Booking
P11	24/Set/2024	31:28 min	Calendly Booking
P12	26/Set/2024	54:20 min	Calendly Booking

Source: Research Data (2024).

Participants were informed about the purpose of the study, how the data would be used, and their right to withdraw from the study at any time without any negative consequences. Anonymity will be maintained in all published findings.

3.5 Data Analysis

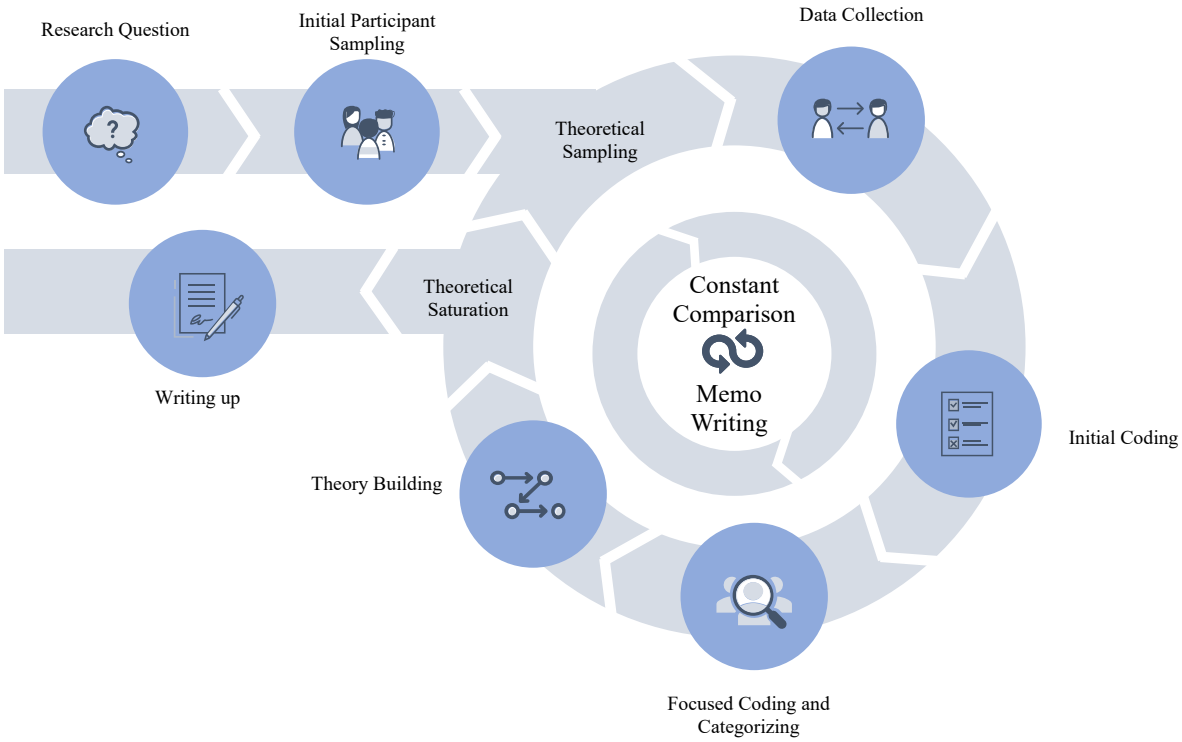
Data analysis using grounded theory as a research method is an iterative and systematic process in which data collection and analysis occur simultaneously (Langley, 1999). This approach employs constant comparison, with the analysis "anchored" in the research question (Urquhart & Fernández, 2013). However, it also remains open to serendipity and unexpected insights the data reveals (Bryman, 2012). As a result, "the analytical guidelines offered by GTM cannot be applied mechanically" (Urquhart & Fernández, 2013, p. 225) but instead should be used with flexibility to adapt to emerging findings (Charmaz, 2014).

In that way, Figure 3 presents a graphical representation of the constructivist grounded theory process adopted in this research. The existing literature was reviewed to define the research scope and formulate the research question. An initial sample of participants was then selected to explore the fieldwork and refine the interview guide. The interview guide (refer to Appendix B – Interview Guide) was adjusted after the fourth interview to facilitate more open-ended conversations and allow emergent themes to surface without compromising the study's objectives. Please refer to Table 8 for a brief description of the participants and the sequence of interviews.

Next, a cyclical data collection, coding, and analysis process was done for each additional interview. Along that process, we made sense of the participant experience, captured as memo-writing. Also, incoming data was compared against initial codes while the process ensured that emergent themes kept the analysis creative and fresh.

Figure 3

Visual representation of grounded theory



Source: Adapted from Charmaz (2014).

In addition, a purpose-specific tool, CAQDAS, assisted the data analysis process. The term CAQDAS refers to Computer Aided Qualitative Data Analysis Software. In this research, the option was made for the use of Atlas.TI (2024), as the CAQDAS tool. The use of tools to support data analysis presents clear benefits in processing large volumes of data and risks associated with possible bias in the analytical process (Lage & Godoy, 2008). However, the option for a software-based tool does not imply automatic coding or the creation of meanings. As Bryman (2012) highlighted, coding is only part of the analysis process, and it is up to the researcher to add their interpretation and identify the meanings that participants associate with their experiences and opinions, just as they would do in a manual process.

Besides, according to (Flick, 2009), using tools to support qualitative analysis has the following advantages: speed of handling, researching, and presenting data and contributing to improving the quality of demonstrating results.

As previously explained, theoretical sampling was employed to obtain selective data (both convergent and divergent), as required by the analysis. The intended stopping criterion for this cyclical process was theoretical saturation—the point at which no new properties or categories emerge from additional data collection. While valuable insights were consistently identified throughout the process, the study concluded after 12 interviews due to time

constraints, rather than an explicit confirmation of data saturation. However, the richness and depth of the collected data sufficiently supported the analytical process and theoretical development.

4. FINDINGS

This section presents the study's findings, which emerged through an iterative data collection and analysis process using constructivist grounded theory (Charmaz, 2014). The findings are structured around the initial codes, focused codes, and categories developed through constant comparison and analytical coding. These categories represent key processes, experiences, and phenomena relevant to the research question.

The analysis began with open coding to identify initial themes, followed by focused coding to refine and organize the data into meaningful patterns. Finally, theoretical coding was applied to integrate these patterns into a coherent explanatory framework. The findings are presented in a narrative form, supported by representative quotes from participants to illustrate the emergent concepts and ensure the grounding of the analysis in the data. Theoretical sensitivity and reflexivity were maintained throughout the process to ensure rigor and trustworthiness. The following sections detail this process and describe the key codes, categories, and interrelationships, providing insights into the phenomenon under investigation.

4.1 Initial Coding

The initial coding phase represents the initial step of grounded theory analysis, where the data is systematically examined to identify and label discrete units of meaning (Charmaz, 2014). During this phase, the focus was on breaking the data into smaller segments and assigning codes capturing the essence of participants' experiences and perspectives regarding actions, ideas, or phenomena. This process involved line-by-line coding to ensure a detailed and comprehensive data exploration, allowing early patterns and themes to emerge without being constrained by preconceived theoretical frameworks.

The initial coding was conducted entirely using Atlas.TI (2024). After loading the interview transcripts and thoroughly reading the text, the coding process was performed manually without relying on automated or generative AI tools. This approach aligns with Charmaz (2014), who recommends that researchers should "appropriate" the data, immersing themselves in the material to fully understand and engage with the participants' experiences.

Initial codes were developed through an iterative process and divided into distinct moments. First, each of the six initial interviews (from P1 to P6) was coded using constant comparison to refine and adjust the coding as new data was analyzed. This first pass of coding ended with the creation of 143 individual codes derived from the constant comparison of data

across six interview transcripts. Each transcript was analyzed line-by-line, ensuring the codes captured actions, ideas, or phenomena directly grounded in the participants' narratives. This step allowed for a detailed and open data exploration, as recommended in grounded theory methodology (Charmaz, 2014).

Following this, a second round of constant comparison was conducted, focusing not on the interview transcripts but on the relationships and overlaps between the initial codes. Through this iterative process, similar codes were identified and consolidated, and more precise and representative codes were developed to refine the analytical framework without losing sight of the participant's meaning. As a result, the initial 143 codes were streamlined into 99 individual codes, providing a clearer and more cohesive foundation for subsequent focused coding and categorization.

For the remaining interviews (P7 to P12), each was coded individually, followed by a constant comparison process conducted after an initial coding schema was applied. This approach ensured that the codes remained closely connected to the raw data, reflecting the participants' language and context (Charmaz, 2014; Urquhart et al., 2010). Where appropriate, participants' own words were used to maintain authenticity and avoid premature abstraction. Through repeated constant comparison, the final coding emerged organically from the data. This process resulted in the creation of 220 unique codes. Table 10 presents the number of extracts coded for each participant.

Table 10

Initial Codes per Participant

Participant	#Extracsts Coded
P1	58
P2	70
P3	63
P4	70
P5	50
P6	50
P7	62
P8	66
P9	42
P10	43
P11	56
P12	78

Source: Research Data (2024).

Table 11 presents a subset of codes and the corresponding groundedness. The concept

of "*groundedness*" refers to the degree to which data within the qualitative dataset support a code. It represents a quantitative indicator of how frequently a specific code has been applied to data segments during the coding process. Yet, Charmaz (2014) and other grounded theorists caution against over-relying on quantitative measures like groundedness. Charmaz emphasized that qualitative insights, context, and relationships among codes are equally critical, but *groundedness* could be used to: a) report a subset of the codes created, b) be used as a starting point to identify prominent patterns and assess the density of specific concepts.

Table 11
Initial Codes and corresponding groundedness.*

Code	Groundedness
Cod.195 - Empowering the team	20
Cod.197 - Integrating teams (Diversity and Inclusion)	17
Cod.117 - Understanding real problems	16
Cod.13 - Understanding the business vision	14
Cod.23 - Addressing individual needs without scaling	13
Cod.98 - Balancing Innovation and Sustainability (Agile)	13
Cod.113 - Overcoming technical debt	13
Cod.7 - Seeking external knowledge	11
Cod.66 - Shooting in all directions - Lack of focus	11
Cod.173 - Innovating incrementally	11
Cod.19 - Translating the demand	10
Cod.97 - Accommodating changes	10

Source: Research Data (2024).

Note: *Only codes with groundedness greater than or equal to 10 were represented. Where groundedness refers to the number of code occurrences (Charmaz, 2014).*

4.2 Focused Coding

The focused coding phase represents a critical step in grounded theory analysis, where the most significant and frequently occurring codes from the initial open coding are refined and synthesized into broader concepts based on their similarities, relationships, or relevance (Charmaz, 2014). This stage shifts the analysis from exploring the data's breadth to identifying patterns that advance theoretical development. In line with Charmaz's constructivist grounded theory approach, focused coding is not a rigid or linear process but a reflective and iterative engagement with the data, emphasizing flexibility and responsiveness to emerging insights.

Furthermore, Urquhart and Fernández (2013) similarly highlighted the importance of coding in connecting higher-level concepts to theoretical insights, noting that this phase

involves interpreting and categorizing the data into meaningful units while remaining closely anchored to the research question. By continuously comparing codes and refining their conceptual boundaries, this stage ensures that the analysis captures the complexity and coherence of the phenomenon under investigation.

Table 12 presents an extract of the derived Focused Codes and their corresponding groundedness. From the 220 initial codes, 91 focused codes were created. However, to enhance clarity and maintain the focus on the most relevant findings, only the most significant items were selected for display in this text. This approach ensures that the discussion remains concise and emphasizes the key insights without overwhelming the reader with excessive detail. The complete table will be available upon request for those who wish to explore the full dataset.

Table 12

Focused Codes and corresponding groundedness.

Focused Code	Initial Codes Related	Groundedness
FC.5 - Understanding the Business	Cod.13, Cod.14, and Cod.16	25
FC.47 – Empowering the Team	Cod.138, and Cod.195	23
FC.48 - Integrating Teams - Diversity and Inclusion	Cod.164, Cod.190, Cod.191, Cod.194, and Cod.197	23
FC.4 - Combining Knowledge	Cod.1, Cod.2, Cod.5, Cod.7, Cod.8, and Cod.9	22
FC.75 - Acting as an Organizational Therapist	Cod.117, Cod.119, and Cod.132	20
FC.30 - Maintaining Focus Amid Complexity	Cod.65, Cod.66, and Cod.76	17
FC.37 - Adopting Agile Practices for Implementation	Cod.136, Cod.137, Cod.139, Cod.143, Cod.144, and Cod.145	17
FC.64 - Balancing Innovation and Sustenance - Agile	Cod.98 and Cod.101	17
FC.15 - Finding Product-Market Fit	Cod.89 and Cod.93	16
FC.55 - Losing Sight of the Customer	Cod.24, Cod.67, Cod.68, Cod.71, Cod.75, and Cod.141	16
FC.79 - Developing Tools, Not Solutions	Cod.122, Cod.123, Cod.124, Cod.129, Cod.133, and Cod.134	15
FC.8 - Participating in Value Definition	Cod.47, Cod.48, Cod.50, Cod.52, Cod.54, and Cod.55	15
FC.18 - Creating Quick and Unembellished Value	Cod.44, Cod.102, Cod.206, and Cod.207	14
FC.22 - Balancing Scalability and Customization	Cod.23 and Cod.28	14
FC.76 - Prioritizing Processes Over Tools	Cod.112, Cod.115, Cod.121, Cod.140, and Cod.168	14
FC.74 - Overcoming Technical Debt	Cod.113	13
FC.3 - Learning from Problems	Cod.4 and Cod.11	12
FC.31 - Prioritizing Scalability	Cod.78, Cod.79, and Cod.80	12
FC.21 - Navigating Feedback and Alignment Issues	Cod.208, Cod.210, and Cod.213	11
FC.88 - Driving Efficiency Through Innovation	Cod.159, Cod.160, Cod.166, and Cod.167	11

Focused Code	Initial Codes Related	Groundedness
FC.39 - Creating Competitive Advantage	Cod.158, Cod.161, Cod.162, and Cod.165	11
FC.41 - Creating Incentives	Cod.146, Cod.148, Cod.154, and Cod.179	11
FC.69 - Innovating in Increments	Cod.173	11
FC.80 - Prioritizing Support	Cod.127, Cod.128, Cod.131, and Cod.135	11
FC.14 - Understanding Need	Cod.91, Cod.94, and Cod.169	10
FC.17 - Validating Hypotheses	Cod.106, Cod.107, and Cod.108	10
FC.57 - Resource and Time Constraints	Cod.22, Cod.25, and Cod.130	10
FC.66 - Accommodating Changes	Cod.97	10
FC.7 - Internalizing Knowledge	Cod.19	10

Source: Research Data (2024).

Focused Coding is the first step during the grounded theory analysis in which the researcher decides and selects which initial codes are worth pursuing and categorizing (Charmaz, 2014). They advance the abstraction level and are the basis for categories and subcategories.

4.3 Categorization

In the constructivist grounded theory approach outlined by Charmaz (2014), categories serve as pivotal analytic constructs that bridge the emergent codes developed during focused coding and the development of a cohesive theoretical framework. Categories are more abstract than focused codes, representing patterns or key themes that synthesize and explain larger data portions. They reflect the researcher's analytic interpretation of what the data conveys, moving beyond descriptive levels to conceptual ones.

Moreover, Charmaz (2014) emphasized that categories are not static; instead, they are constructed through an iterative process of comparing data, codes, and developing concepts, ensuring they remain closely grounded in the data while offering broader explanatory potential. This dynamic process positions categories as the scaffolding for building theory, allowing researchers to refine their analytic lens and draw connections between emerging ideas and the overarching phenomena under study. Table 13 describes the relationship between the 25 Categories and the 91 Focused Codes found in the study.

Table 13

Relationship between Categories and Focused Codes.

Category	Focused Code	Comments about
CT.1 - Team-Based Learning	FC.1 - Learning from Mistakes FC.2 - Learning through Experimentation FC.3 - Learning from Problems FC.4 - Combining Knowledge	The category captures the dynamics of how teams learn and grow through collaboration, problem-solving, and shared experiences.
CT.2 - Learning from and with the Client	FC.5 - Understanding the Business FC.6 - Learning Together FC.7 - Internalizing Knowledge	This category explores how organizations learn in partnership with their clients, emphasizing mutual understanding and knowledge sharing.
CT.3 - Internal Challenges to Co-creation	FC.21 - Navigating Feedback and Alignment Issues FC.22 - Balancing Scalability and Customization FC.23 - Managing Control and Ownership FC.24 - Overreliance on Prescriptive Practices FC.25 - Materializing Value Co-creation	This category reflects difficulties within the organization or team related to mindset, processes, and organizational practices that hinder effective co-creation.
CT.4 - Client-Driven Challenges in Co-creation	FC.26 - Outcome-Driven Focus FC.27 - Trust and Knowledge Asymmetry	The category reflects difficulties stemming from client attitudes, behaviors, or fears that impact the co-creation process.
CT.5 - Leveraging Relationships for Effective Co-creation	FC.29 - Maintaining and Strengthening Relationships FC.84 - Fostering New Connections FC.85 - Expanding Points of Interaction	The category focuses on strategically using relationships to foster effective value co-creation, emphasizing building, maintaining, and expanding collaborative connections.
CT.6 - Co-Defining and Shaping Value	FC.8 - Participating in Value Definition FC.10 - Participating in Value Configuration	This category highlights the active involvement of teams and stakeholders in defining, articulating, and delivering value through co-creation.
CT.7 - Identifying Innovation Inputs	FC.61 - Starting from Market Gaps FC.62 - Starting from 'Pain Points' FC.63 - Starting from Reflection	This category focuses on the sources and starting points for innovation, highlighting how gaps, pain points, and reflective practices inspire new ideas.
CT.8 - Executing Innovation	CT.8 - Executing Innovation FC.30 - Maintaining Focus Amid Complexity FC.31 - Prioritizing Scalability FC.32 - Innovating with Legacy Systems FC.33 - Prioritizing Utilitarianism FC.34 - Prioritizing Outcomes FC.35 - Finding Product-Market Fit FC.37 - Adopting Agile Practices for Implementation FC.86 - Funding Innovation	This category emphasizes the practical aspects of implementing innovation, including focus, scalability, resources, and prioritization.
CT.9 - Communicating and Scaling Value	FC.9 - Articulating the Value Proposition FC.11 - Generalizing the Solution	This category emphasizes the need to articulate the value proposition and to generalize the development to address other customers.
CT.10 - Balancing Innovation and Sustainability in Organizations	FC.64 - Balancing Innovation and Sustenance - Agile FC.65 - Building Sustainable Solutions FC.66 - Accommodating Changes FC.67 - Segregating Profiles	It reflects the challenge of balancing innovation with sustainability and points to practices and tools that drive innovation.

Category	Focused Code	Comments about
CT.11 - Creating Feedback Loops for Continuous improvement	FC.13 - Creating Feedback Flow	The category reflects the importance of iterative feedback cycles for refining value creation.
CT.12 - Focusing on the Technology	FC.73 - Viewing Technology as a Driver of Solutions FC.74 - Overcoming Technical Debt	This category emphasizes the role of technology in innovation, addressing both its use as a solution and the challenges of managing technical debt.
CT.13 - Focusing on the Problem	FC.75 - Acting as an Organizational Therapist FC.76 - Prioritizing Processes Over Tools FC.77 - Being Essential to Business Operations FC.78 - Starting with Service, Not Product	This category highlights the importance of addressing core organizational issues and aligning processes before implementing technological solutions.
CT.14 - Focusing on the Product	FC.79 - Developing Tools, Not Solutions FC.80 - Prioritizing Support FC.81 - Diversifying Products	This category focuses on the challenges of product-centric innovation, emphasizing tool development, support prioritization, and diversification.
CT.15 - Leadership Practices for Driving Innovation	FC.41 - Creating Incentives FC.42 - Promoting Diversity FC.43 - Managing by Values FC.44 - Establishing Governance FC.45 - Tolerating Honest Mistakes FC.46 - Sharing the Vision	This category captures the role of leadership in driving and fostering an innovative environment through vision, governance, values, and incentives.
CT.16 - Defining Innovation for Competitive and Customer-Centric Goals	FC.39 - Creating Competitive Advantage FC.40 - Enhancing the Customer Experience FC.88 - Driving Efficiency Through Innovation	This category explores innovation's foundational goals and impacts, including efficiency, competitive advantage, and customer experience.
CT.17 - Tools and Practices for Organizational Innovation	FC.68 - Ritualizing Innovation FC.69 - Innovating in Increments FC.70 - Creating Innovation Spaces FC.71 - Creating Incentives for Innovation FC.72 - Focusing on Internal Processes	This category centers on the methodologies, rituals, and incentives that support the innovation process, focusing on incremental and organizational improvements.
CT.18 - Organizational Culture as a Catalyst for Innovation	FC.36 - Aligning Values FC.38 - Committing to Your Own Values FC.87 - Focusing on Learning FC.89 - Promoting Detachment FC.91 - Leading by Culture	This category explores how cultural elements serve as the bedrock for sustaining and enabling innovation.
CT.19 - Team Dynamics as a Driver of Innovation	FC.47 - Empowering the Team FC.48 - Integrating Teams - Diversity and Inclusion FC.49 - Building Strong Teams	This category highlights the importance of team composition, empowerment, and inclusivity in fostering an innovative environment.
CT.20 - Organizational Structures: Balancing Constraints and Opportunities for Innovation	FC.50 - Bridging Gaps Across Structures FC.51 - Startup - Operating with Limited Resources FC.52 - Traditional - Prioritizing Consistency FC.53 - Startup - Customer Orientation	This category examines how different organizational structures and resource constraints influence innovation practices.
CT.21 - Empowering Individuals for Innovation Success	FC.54 - Understanding the Whys FC.90 - Taking Ownership of Self-Development	This category focuses on individual ownership and understanding as critical elements of the innovation process.
CT.22 - Adapting for Sustained Innovation	FC.55 - Losing Sight of the Customer FC.56 - Overcoming Resistance to Change FC.58 - Ensuring Innovation Sustainability	This category explores the challenges and adjustments necessary for organizations to

Category	Focused Code	Comments about
		remain innovative, focusing on adaptability, vision, and sustainability.
CT.23 - Building Trust in Collaborative Relationships	FC.12 - Transforming Vulnerability into Trust FC.82 - Balancing Transparency and Protection FC.83 - Demonstrating Commitment to Relationships	This category revolves around the process of establishing trust in co-creation contexts, focusing on transparency, vulnerability, and commitment as foundational elements.
CT.24 - Overcoming Structural and Resource Constraints in Innovation	FC.17 - Validating Hypotheses FC.18 - Creating Quick and Unembellished Value FC.19 - Managing Experimentation - Governance FC.20 - Failing Fast, Learning Smart	This category focuses on using experimentation to test ideas, create value efficiently, and refine approaches through iterative learning.
CT.25 - Managing Conflicts in Innovation Through Leadership	FC.59 - Resolving Conflicts Through Authority	This category explores the role of authority and power in resolving conflicts that arise in innovation contexts, focusing on leadership styles and decision-making processes.

Source: Research Data (2024).

Throughout the process, memo-writing was used to support the creation of focused coding by documenting initial observations, questions, and potential connections between emerging concepts. According to Wiesche et al. (2017), it is a critical tool in GT to document theoretical insights and process reflections throughout the study. Examples of theoretical and process memos can be seen in Gasson & Waters (2013). Those examples were used as best practices. The following paragraphs provide samples of how memos were used during the analysis to connect emerging ideas with key categories. For instance, during the interview with Participant 3, when questioned about how he created room for innovation, he highlighted:

> **[P3]:** It is our responsibility to take the time and do this. We have the freedom to do so, but in practice, it hasn't been happening because of the sheer volume of tasks in our queue, and we end up not focusing on it at the moment.
>
> […] Some products were developed for years without involving the client, while we had several clients come to us seeking a solution that wasn't yet a product. We had to try to turn that demand into a product while the plane was already in flight. In these cases, we built the plane while it was moving, and it was quite challenging.

Later on, when prompted about how new team members are trained, he commented:

> **[P3]:** When someone first joins, we have material for at least three weeks to help them

> get acclimated, understand the company, how we work, and our rituals (daily meetings), etc. After that, we gradually introduce problems, starting from lower complexity to higher complexity, and this is essentially problem-oriented learning.

These extracts suggest that, initially, new team members bring fresh perspectives and may have some space to review processes and experiment with new ideas. However, once these individuals are fully integrated into the team, they are absorbed by the operational demands of product development. Their focus shifts almost entirely to the day-to-day tasks for which they were hired. This dynamic often diminishes the capacity for experimentation and process refinement over time. Table 14 presents a memo written after the interview with Participant 3:

Table 14
Memo 1.

The Role of Team Dynamics in Sustaining Innovation
Related Categories
CT.19 - Team Dynamics as a Driver of Innovation
There is a recurring challenge in innovation-driven organizations: the balance between operational efficiency and sustained creativity. While homogeneity within teams may foster smooth collaboration and alignment, it often comes at the expense of diversity in thought and the ability to challenge the status quo. New team members initially have the opportunity to act as agents of change, bringing fresh perspectives that can lead to process improvements and experimentation. However, as they become "normalized" within the team's established routines and expectations, this innovation potential is often lost. The organizational pull toward operational demands and existing norms creates a tension between immediate deliverables and the long term benefits of process innovation. Association with CT.19 - Team Dynamics as a Driver of Innovation The participant emphasizes the interplay between team composition, integration, and the potential for innovation. There is a missed opportunity to empower team members, particularly new hires, to continuously challenge processes and drive innovation. True empowerment would involve creating an environment where all team members, regardless of tenure, have the autonomy to think critically and innovate beyond their immediate tasks (FC.47 - Empowering the Team). The importance assigned to homogeneity to deal with problems within teams highlights the challenges in maintaining diversity of thought and perspective. While new members bring initial diversity, their eventual assimilation into established norms limits their ability to contribute unique ideas in the long-term (FC.48 - Integrating Teams - Diversity and Inclusion). Effective team-building involves not just integrating new members but also sustaining their ability to challenge and improve processes. Strong teams balance operational alignment with the capacity for ongoing innovation (FC.49 - Building Strong Teams). Implications for Team Dynamics and Innovation Team dynamics play a critical role in sustaining innovation. While operational alignment and homogeneity can enhance short-term efficiency, they risk undermining the long-term benefits of diverse perspectives and process improvement. By associating this memo with CT.19 - Team Dynamics as a Driver of Innovation, we emphasize the importance of intentional team-building practices that balance routine tasks with sustained opportunities for reflection, creativity, and innovation.

Source: Research Data (2024).

Another recurring theme during the interview with Participant 3 is related to the tension between the need for scalability in software development and the importance of maintaining a customer-centric perspective. In his perspective, the participation of customers in the process of defining value is fundamental, but he recognizes the challenge of maintaining this throughout the software lifecycle:

> **[P3]:** We don't have the process that I would consider ideal, where you start with a simple but functional application and then continue making incremental improvements in direct collaboration with the client, shaping the process together based on their needs. Often, the client themselves don't know how it should be systematized, but they discover how their process should become a system as they understand and operate the tool.

When questioned about the main challenges to make his team more innovative, Participant 3 stated:

> **[P3]:** Today, time. Our team is spending a lot of time dealing with legacy systems, which ends up taking up much of our capacity. If there were a support layer for customizations that allowed us to focus more on development and innovation, we could move forward and develop new technologies. At the moment, it would mean having a separation between the support team and the team working on the product.
>
> […] If we averaged the past year to assess how much time I could dedicate to the product, I would say about 30% of my time. Around 70% is spent solving various issues, whether with new clients or legacy clients. There's also time spent on documentation, which is part of the product but not the product itself. We had to invest a lot of time creating documents, and I also spent a significant amount of time helping the team and teaching them how to solve problems. Still, I hope that this year we can turn that 30% into more than 50%.

The participant expressed a sense of frustration, noting that while solutions are initially crafted with the client as a key source of knowledge about their business and processes, this focus shifts as the software evolves into a standardized product intended for commercialization where the primary metric (value) is the number of problems solved in time (Service Level Agreement). Table 15 presents a memo written after the interview with Participant 3.

Table 15
Memo 2.

The Tension Between Scalability and Customer-Centric Problem Definition
Related Categories
CT.13 - Focusing on the Problem
In the early stages of development, the customer's voice plays a central role in defining problems and shaping the solution. Their insights inform the design of functionalities that address specific business challenges. However, as the product matures into a scalable solution meant to serve broader markets, the development process shifts. Listening to the customer becomes deprioritized, and existing functional capabilities are "shielded," leaving little room for further adaptation or input from end-users. This shift often results in a partial understanding of the problem, as the product team begins to prioritize internal goals—such as reducing bugs, optimizing functionality, and achieving market fit—over deep engagement with how the software is actually used by customers. This shift away from customer engagement during productization reflects a critical challenge in maintaining a problem-centered approach to innovation. By ceasing to listen to customers actively, the development process risks losing sight of how users interact with the solution, the pain points they encounter, and the evolving needs of their business. The product may still function effectively, but it may fail to solve the right problems or align with the customer's experience in meaningful ways. This insight fits directly within the category "CT.13 - Focusing on the Problem", particularly in the following ways: - FC.75 - Acting as an Organizational Therapist: The disconnect between scaling the product and understanding the customer highlights the need for organizations to diagnose and address the root causes of this misalignment. Teams must maintain a deep understanding of the customer's context and evolving problems, even during commercialization. - FC.76 - Defining the Process Before the Software: P3's observation underscores the importance of continuing to define and refine processes based on customer feedback. As products scale, organizations must avoid reducing the development process to software maintenance alone, ensuring the customer's voice remains integral to the lifecycle. - FC.77 - Being Part of an Essential Process: the initial collaboration with customers reflects their central role in defining value. However, by excluding them during the scaling phase, the company risks sidelining a process essential to innovation: co-defining and iteratively addressing the customer's core needs. The extracts illustrate how the focus on scalability can dilute a problem-centered approach to software development. By anchoring development processes in problem definition, organizations can ensure that scaling efforts do not come at the expense of customer-centric innovation. This insight highlights the need for constant engagement with the customer's context to avoid creating solutions that are functional but fail to meet real needs.

Source: Research Data (2024).

Likewise, Participant 4 reflected on how the innovation forum of his company assigns priorities and resources to new experiments:

> **[P4]:** Usually, when we're working on continuous improvement, it's easier to reach an agreement because we can come together and say, "Look, we have this problem, and this and that is happening." Various people then start giving their input.
>
> […] But when we reach a "good enough" point—where it won't cause headaches and we're not over-polishing or nitpicking unnecessarily—it's because we have another ton of things to handle, right? So, once we get that other workload running smoothly and up to a good enough standard, we can revisit this one, or perhaps another where "good enough" is no longer acceptable, and we need to improve it further. […] So, when it comes to process improvement, in my view, it's easier…

Participant 4's comment provides valuable insight into the trade-offs organizations face when balancing incremental improvements and radical innovation. Process improvements often offer a straightforward path to increasing efficiency, reducing waste, and enhancing performance. In contrast, creating entirely new solutions, products, or services involves greater complexity, risk, and uncertainty, making it a more demanding form of innovation. Table 16 presents a memo written after the interview with Participant 4:

Table 16
Memo 3.

Balancing Process Improvement and New Creation in Innovation
Related Categories
CT.10 - Balancing Innovation and Sustainability in Organizations
The participant's comment on how innovating through process improvement is simpler than creating new things provides valuable insight into the trade-offs organizations face when committing resources for innovation. The quotation is directly linked to CT.10 - Balancing Innovation and Sustenance, as it reflects the tension between sustaining operational excellence through incremental process improvements and pursuing transformative change through new creations. Specifically, the following subcategories are relevant: - FC.64 - Balancing Innovation and Sustenance: process improvement represents an approach to sustaining innovation—delivering measurable benefits with relatively low risk. However, balancing this with the pursuit of new ideas is critical to ensure the organization remains competitive and adaptable. - FC.65 - Building Sustainable Solutions: although process improvements lead to operational sustainability, organizations aim to have solutions that are easy to maintain, with low error levels, and that can fit most customer needs with few modifications. Indeed, those are often conflicting interests that are difficult to reconcile. - FC.66 - Accommodating Changes: The comment indirectly highlights the need for organizations to accommodate changes and be nimble to promote improvements while sustaining service levels and keeping existing customers unaffected by existing functionalities. Furthermore, process improvements provide a low-barrier entry point to innovation, leveraging existing knowledge, workflows, and resources. They are particularly valuable for: 1. Enhancing efficiency: Small, incremental changes can streamline operations, reduce costs, and improve overall performance. 2. Risk management: Since these innovations typically operate within known parameters, they carry fewer risks compared to the uncertainties of creating new solutions. 3. Building momentum: Successful process improvements can foster a culture of continuous improvement, creating the conditions for larger-scale innovation initiatives. The tension between process improvement and new creation is not an either-or scenario but rather a balancing act. Organizations must strategically decide when to focus on incremental enhancements and when to invest in transformational change. Possible alternatives to address that balance could be: 1. Adopt a dual innovation strategy where separate but complementary tracks for process improvements and radical innovations could ensure both approaches are resourced and supported. 2. Prioritize based on strategic goals. By aligning the choice of innovation type with the organization's current challenges and future aspirations, the company may decide whether process improvements are prioritized for immediate gains or whether new creations are needed for long-term opportunities. 3. Encourage experimentation. Even in process improvement efforts, foster a culture of experimentation to gradually build toward larger-scale innovations.

Source: Research Data (2024).

4.4 Theory Building

The following sections detail the relationship of categories that emerged during this phase, illustrating the initial foundation of the analytical process and highlighting participants' voices in shaping the direction of the research. Representative quotes demonstrate the breadth and depth of the identified categories.

Before working on the details, it is necessary to converge on using theoretical codes as an alternative to express the relationship of categories and their applicability to grounded theory. According to Charmaz (2014), theoretical coding is the most abstract level of coding, where relationships between categories are identified to develop an integrated theoretical framework. The objective of using theoretical coding is to build a coherent theory that explains the central process or phenomena in the study. Table 17 illustrates the differences between the abstraction levels used for producing a theory grounded in data.

Table 17
Different Abstraction Levels for Grounded Theory

Aspect	Initial Coding	Focused Coding	Categories	Theoretical Coding
Abstraction	Low (close to data)	Moderate	High	Highest
Purpose	Explore the data	Synthesize and prioritize	Conceptualize patterns	Integrate into a theory
Scope	Fragmentary	Condensed and refined	Broad and conceptual	Holistic and integrative
Relationship to Data	Closely tied	It is still tied but more conceptual	Abstracted from codes	Focused on relationships between categories
At this Study	220 Initial Codes	91 Focused Codes	25 Categories	9 Theoretical Code

Source: Adapted From Charmaz (2014).

According to Glaser & Strauss (1967), the elements of a theory are categories and propositions or generalized relations among the categories and their properties. At this stage, called the integrative phase by Urquhart & Fernández (2013), the researcher returns to the extant literature to help develop the emerging concepts. This time, constant comparison is performed between emerging patterns against the literature, and theoretical memos are key resources that support the emergence of theoretical insights (Glaser & Strauss, 1967). Table 18 presents the theoretical codes created by establishing a relationship between categories and the comparison against extant literature.

Table 18

Theoretical Codes

Theoretical Code	Related Categories	Comments
TC.1 - Organizational Learning Across Teams and Client Partnerships	CT.1 - Team-Based Learning CT.2 - Learning from and with the Client	It emphasizes the interconnectedness of internal and external learning processes as antecedents for innovation and value creation.
TC.2 - Building Collaborative Relationships and Trust for Co-creation	CT.5 - Leveraging Relationships for Effective Co-creation CT.23 - Building Trust in Collaborative Relationships	It emphasizes the interconnectedness of leveraging relationships and fostering trust as complementary processes in successful co-creation.
TC.3 - Navigating the Challenges of Value Co-creation	CT.3 - Internal Challenges to Co-creation CT.4 - Client-Driven Challenges in Co-creation	The code reflects an active process of identifying and addressing the obstacles in co-creating value, emphasizing awareness and action.
TC.4 - Integrating Client Understanding, Experimentation, and Active Value Co-creation	CT.6 - Co-Defining and Shaping Value CT.9 - Communicating and Scaling Value CT.11 - Creating Feedback Loops for Continuous improvement CT.22 - Empathizing with Clients to Drive Value Creation CT.24 - Driving Value Creation Through Experimentation	Value co-creation is a multifaceted process combining client understanding, deliberated experimentation, and communication to achieve meaningful outcomes.
TC.5 - Enabling Innovation: Leadership, Culture, and Empowerment	CT.15 - Leadership Practices for Driving Innovation CT.18 - Organizational Culture as a Catalyst for Innovation CT.19 - Team Dynamics as a Driver of Innovation CT.20 - Organizational Structures: Balancing Constraints and Opportunities for Innovation CT.21 - Empowering Individuals for Innovation Success	It refers to how organizations integrate leadership, culture, team dynamics, organizational structure, and individual empowerment into a comprehensive perspective on innovation.
TC.6 - Adapting, Resolving, and Overcoming Barriers to Innovation	CT.22 - Adapting for Sustained Innovation CT.24 - Overcoming Structural and Resource Constraints in Innovation CT.25 - Managing Conflicts in Innovation Through Leadership	It captures processes and strategies to overcome obstacles that threaten innovation.
TC.7 - Building Digital Service Innovation: From Inputs to Tools and Sustainability	CT.7 - Identifying Innovation Inputs CT.10 - Balancing Innovation and Sustainability in Organizations CT.17 - Tools and Practices for Organizational Innovation	This overarching code highlights the innovation lifecycle, from identifying sources of inspiration to balancing sustainability and implementing structured practices.
TC.8 - Innovation Focus: Technology, Problems, and Product Strategies	CT.12 - Focusing on the Technology CT.13 - Focusing on the Problem CT.14 - Focusing on the Product	It encapsulates the interplay between focusing on technology, addressing core problems, and refining product strategies to foster innovation.
TC.9 - Implementing Service Innovation: Execution and Impact	CT.8 - Executing Innovation CT.16 - Defining Innovation for Competitive and Customer-Centric Goals	It highlights the balance of strategies execution and conceptual goals for implementing innovation effectively in services.

Source: Research Data (2024)

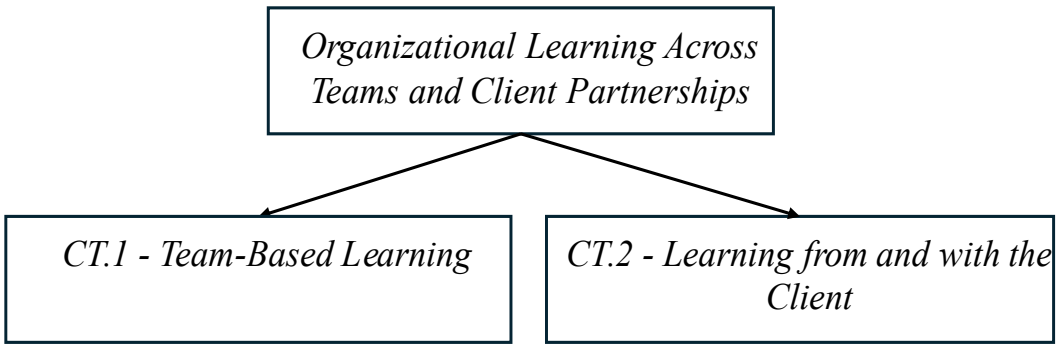
Naming theoretical codes effectively bridges the categories into an integrated, coherent theory that resonates with the data and the broader academic or practical context (Charmaz, 2014). The approach used for naming the theoretical ensured groundedness so that the concept reflects the patterns and ideas from the data and connection by capturing the relationships and interactions between categories. The following sections will present the integration of theoretical codes and the extant literature, when applicable, to explain the central processes or phenomena in the study.

4.4.1 TC.1: Organizational Learning Across Teams and Client Partnerships

This theoretical code emphasizes the interconnectedness of internal and external learning processes as key drivers for value creation and innovation, allowing organizations to adapt to challenges and promote competitive advantage. The code explores how organizations learn internally, through team collaboration and problem-solving, and externally, by engaging with clients and non-clients to co-create knowledge. The interplay between these two categories creates a learning ecosystem that enhances the organization's ability to adapt, grow, and deliver value in dynamic environments. The code comprises two categories, "CT.1 - Team-Based Learning" and "CT.2 - Learning from and with the Client", as depicted in Figure 4.

Figure 4

Categories related to TC.1: Organizational Learning Across Teams and Client Partnerships



Source: Research Data (2024).

Within organizations, teams act as central hubs of learning (CT.1 - Team-Based Learning), or using the metaphor of Hoegl & Parboteeah (2006), teams can be seen as "vehicles" for collaboration and sharing of resources. They engage in iterative cycles of

problem-solving, where learning from mistakes, experimenting with new approaches, and addressing challenges collectively contribute to their growth. The underlined focused codes are aligned with team learning in general and, more specifically, with organizational learning from errors. The latter refers to the intentional, manageable, and risky process in which individuals use trial and error to take ownership of what works best for an intended (Dahlin et al., 2018).

Additionally, integrating diverse expertise within and across teams creates opportunities for knowledge synergies, enabling innovative solutions that draw on the strengths of individual members. This internal dynamic not only strengthens the team's capabilities but also positions them as key contributors to the organization's broader learning ecosystem (Edmondson, 2012).

Beyond internal learning, organizations derive significant value from engaging with clients as knowledge partners (CT.2 - Learning from and with the Client). By understanding the client's business, organizations gain deep insights into their operations, objectives, and pain points. Collaborative learning dynamics further enhance this relationship, creating opportunities for both parties to co-create knowledge. This external knowledge is then internalized within the organization, where it informs strategies, improves processes, and drives innovation. Such partnerships enable organizations to align more closely with client needs while enhancing their capabilities.

Nevertheless, it is easier said than done. Learning from mistakes and fostering a culture of experimentation are counterintuitive in organizations where errors or failures are often associated with deviant behavior, non-conforming results, losses, and negative consequences (Leoncini, 2016; Zhao & Olivera, 2016). If understood as a social endeavor, organizational learning takes place in the presence of social relations, where emotions and perceptions can foster or stifle learning (Watzek et al., 2019; Wilhelm et al., 2019). Therefore, learning within teams and with partners requires "a shared belief that the team is safe for interpersonal risk-taking" (Edmondson, 1999, p. 352).

The interplay between internal and external learning, as well as individual and team learning, is consistent with Bontis et al. (2002), where organizational learning is seen as a process occurring over time, within and across levels, where individuals use their intuition and interpretation to make sense of changes and opportunities as ignition of this dynamic process.

Besides, the ability to recognize external knowledge requires prior knowledge and context so that it is possible to make sense of the new information (Todorova & Durisin, 2007). Furthermore, according to W. M. Cohen & Levinthal (1990), before learning new knowledge from external sources, that is, absorbing knowledge, organizations need the capacity to recognize the value of the new knowledge and drive it toward a commercial end. Therefore, the

concept of Absorptive Capacity (ACAP) seems suitable for explaining that just listening to customers is not the same as learning from them; it can be used to explore collaborative strategies between different organizations (Dyer & Singh, 1998).

In this context, ACAP refers to the organization's ability to identify, access, incorporate, transform, and apply ideas and external knowledge into the organization's processes (W. M. Cohen & Levinthal, 1990). Yet, despite a critical mass of research that draws upon absorptive capacity, a few studies performed a comprehensive assessment of the role of this construct as a mediator of value co-creation in the digital service ecosystem (Xie et al., 2021). Moreover, prior studies are inherently biased toward the exploitation of existing knowledge (Roberts et al., 2012), whereas this study proposes that ACAP also facilitates the exploration of new ideas and experiences created conjointly with partners and customers.

In contrast, there seems to be a drawback when learning from new employees is entirely based on problem-solving. Employees, upon being onboarded into the company, undergo an initial process of knowledge normalization, during which they become familiar with the company's rituals, practices, and products. See the extract from Participant 3:

> **[P3]:** I usually introduce the problem and guide the paths to follow. I ask them to proceed and then we schedule another meeting. We don't recommend specific reading material, but we provide guidance based on the problems […] For me, it is the easiest and most effective way to expose the person to the situation and enable them to have a faster learning curve for what we need.

Tools and processes are also introduced during this phase. Subsequently, the way of working is taught through problem-solving, especially in niche companies. That is, customer tickets are assigned to new employees so they can address nonconformities. This may suggest that these employees gain a partial view of the solution, often from the software's perspective and rarely from the business or customer perspective. Participant 3 acknowledges that fact during a subsequent extract:

> **[P3]:** The real challenge lies in understanding what it actually does. I believe this can only be achieved when the person understands the problem and tries to reverse-engineer it.

The interconnectedness between team-based learning and client-collaborative learning forms a dynamic learning ecosystem within organizations. Insights gained from client partnerships often serve as catalysts for refining internal processes, enabling teams to address

real-world challenges more effectively. Similarly, the expertise and adaptability of teams enhance the quality of client collaborations, ensuring that both parties benefit from the shared learning experience. This reciprocal relationship amplifies the organization's ability to innovate and sustain competitive advantage. Table 19 captures the memorandum that consolidates the information corresponding to the associated categories.

Table 19
Memo 4.

Organizational Learning Across Teams and Client Partnerships
Related Categories CT.1 - Team-Based Learning CT.2 - Learning from and with the Client
Organizational learning emerges as a dynamic interplay between internal team development and external client collaboration. Teams thrive on problem-oriented learning, where challenges are tackled incrementally, fostering skill-building and accelerating adaptation. Leadership plays a crucial role in guiding teams through this learning journey, emphasizing cross-skilling and cultivating a culture of experimentation. Mistakes are reframed as opportunities, enabling continuous improvement while balancing operational demands with the pursuit of innovation. However, operational pressures often limit time for learning, highlighting the need for intentional resource allocation to sustain both exploration and exploitation. Externally, clients serve as key collaborators, providing valuable insights that shape product development and refine solutions. Effective communication channels, such as direct feedback loops, help bridge gaps and align solutions with client needs. Contextual understanding enhances customization, particularly in niche markets like healthcare, where incremental changes can generate significant impact. Despite these efforts, organizations face challenges in prioritizing opportunities, scaling solutions, and maintaining agility. A supportive culture that embraces risk-taking, fosters collaboration, and seeks external perspectives ensures that organizations remain adaptable and innovative in dynamic environments.

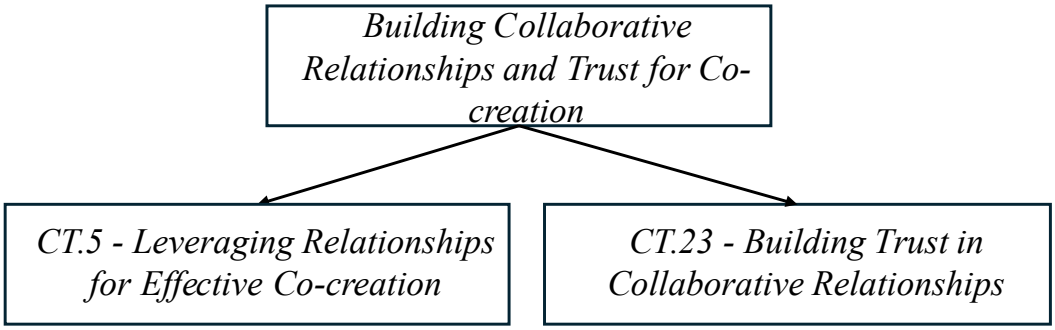
Source: Research Data (2024).

4.4.2 TC.2: Building Collaborative Relationships and Trust for Co-creation

In the context of co-creation, relationships and trust are the foundational pillars that enable meaningful collaboration (Neghina, 2015). This theoretical code, "Building Collaborative Relationships and Trust for Co-creation" explores how organizations leverage relationships and establish trust as enablers of value co-creation. It delves into the seemingly counter-intuitive yet powerful idea of fostering trust through vulnerability, where openness about limitations or uncertainties can drive deeper engagement, more authentic partnerships, and accelerated learning. According to Rubalcaba (2018), service co-production is impossible without trust. The theoretical code comprises two categories, "CT.5 - Leveraging Relationships for Effective Co-creation" and "CT.23 - Building Trust in Collaborative Relationships", as depicted in Figure 5.

Figure 5

Categories related to TC.2: Building Collaborative Relationships and Trust for Co-creation



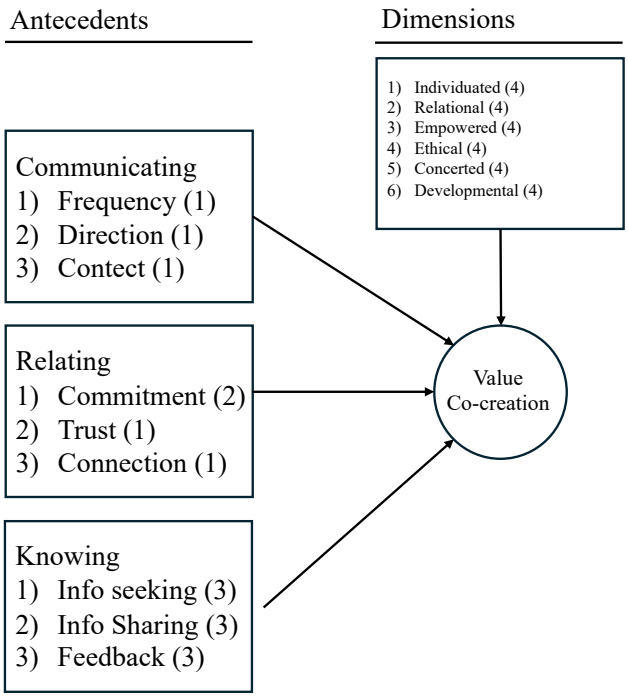
Source: Research Data (2024).

Effective value co-creation begins with cultivating and sustaining relationships that foster collaboration and innovation. According to Neghina (2015), value co-creation can be seen as planned coordination driven by outcomes, where the endeavor or joining activity requires three elements: (1) a plan for the desired value, (2) the roles actors are playing, and (3) the awareness that value can only be created with the support of another actor. Therefore, it involves maintaining existing partnerships, building new connections, and diversifying interaction points. Maintaining relationships ensures continuity in collaboration by nurturing established partnerships and keeping communication channels open. At the same time, building relationships focuses on initiating new collaborations that expand the network of contributors and create opportunities for fresh perspectives. By diversifying points of contact, organizations deepen engagement by utilizing multiple communication methods, such as direct feedback loops or informal channels, to facilitate authentic exchanges.

Moreover, leveraging or creating the type of relationships that are useful for value co-creation requires more than shared intention. Neghina (2015) argued that trust, commitment, and connection are key components to forging and feeding relations necessary for value co-creation. However, the author did not explain the mechanisms used by organizations to promote those dimensions. Turning to the research data, we found that vulnerability may play a key role in the efforts to build trust and commitment in addition to what Neghina (2015) describes in his conceptual value co-creation framework, as depicted in Figure 6. By openly inviting collaboration through early prototypes or acknowledging areas for improvement, organizations signal trustworthiness and encourage genuine input from their partners or customers.

Figure 6

Visual representation of grounded theory



Source: Adapted from Neghina (2015).

Likewise, the research data suggests that trust is the cornerstone of effective value co-creation. It is built on transparency, vulnerability, and consistent commitment. Transforming vulnerability into trust highlights how openly sharing challenges, experiment results, or admitting uncertainties creates space for collaboration and strengthens relational bonds. For example, presenting an incomplete or "ugly" product accelerates feedback and builds a sense of shared purpose. However, vulnerability must be balanced with transparency – strategically managing openness to protect sensitive information while maintaining credibility. Finally, by demonstrating commitment, the organization reinforces trust through consistent follow-through on feedback, showing that vulnerability is paired with action. Together, these practices create a culture of trust that empowers partners to engage deeply and confidently in the co-creation process.

Therefore, the synergy between leveraging relationships and building trust is what makes co-creation promising. Relationships provide the platform for collaboration (Neghina, 2015), while trust—fostered through vulnerability—deepens these connections and ensures their longevity. Vulnerability acts as a bridge between the two: by openly admitting imperfections or seeking input, organizations not only strengthen their relationships but also build trust rooted in authenticity and mutual respect. When relationships are diverse, and trust is strong, collaborators feel empowered to take risks, share ideas, and co-create solutions that

are innovative and aligned with shared goals. This interplay ensures that co-creation is not just a transactional process but a transformative partnership driven by trust and meaningful collaboration. Table 20 captures the memorandum that consolidates the information corresponding to the associated categories.

Table 20
Memo 5.

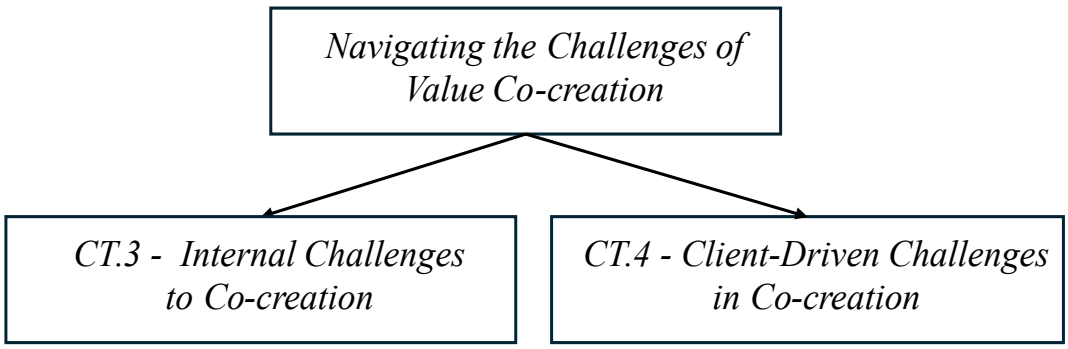
Building Collaborative Relationships and Trust for Co-creation
Related Categories CT.5 - Leveraging Relationships for Effective Co-creation CT.23 - Building Trust in Collaborative Relationships
The interviews underscore the critical importance of **relationships** and **trust** in driving effective co-creation. Participants highlighted that while existing channels like Product Owners or Customer Solution teams provide valuable insights, they are often insufficient. Building closer, more transparent partnerships with clients creates opportunities for deeper engagement, particularly when customers are invited into the development process as collaborators. Regular, open-ended discussions with clients about their daily challenges and workflows strengthen these bonds, enabling organizations to align solutions more closely with real needs. Trust, however, does not emerge overnight. It requires consistent effort, demonstrated commitment, and a willingness to embrace **vulnerability**. Participants emphasized that sharing the "imperfect" state of products or internal challenges with clients fosters trust and accelerates feedback. Yet, this must be balanced with protecting sensitive or strategic information to avoid misunderstandings. Developing trust often involves navigating initial skepticism and investing significant time to prove sincerity and reliability. Over time, this approach not only builds strong partnerships but also transforms clients into long-term collaborators who co-create value and provide crucial direction for product and organizational growth.

Source: Research Data (2024).

4.4.3 TC.3: Navigating the Challenges of Value Co-creation

Value co-creation is an intricate process that demands collaboration between organizations and their clients to create value for both (Payne et al., 2008). However, it is not without its challenges, which can be broadly categorized as internal obstacles and client-driven complexities. This theoretical code, "Navigating the Challenges of Value Co-creation," explores these two dimensions and highlights how overcoming them requires intentional actions to foster alignment, trust, and flexibility. The theoretical code comprises two categories: "CT.3 - Internal Challenges to Co-creation" and "CT.4 - Client-Driven Challenges in Co-creation", as depicted in Figure 7.

Figure 7
Categories related to TC.3: Navigating the Challenges of Value Co-creation



Source: Research Data (2024).

Internal Challenges to Co-creation reflect the barriers within organizations that hinder the effective realization of co-creation initiatives. Going beyond traditional buyer-seller dynamics requires suppliers and customers to collaborate to generate mutual value while leveraging unique and complementary resources (Sheth, 2020). The author suggested that internal capabilities, such as institutionalizing co-creation practices like establishing cross-functional teams and shared performance metrics, are necessary to deal with external factors.

Additionally, our research goes further and sheds light on specific internal challenges. For instance, teams often struggle to integrate client feedback into their workflows, resulting in misaligned priorities and slower progress. During his response about how team members feel about customer participation, Participant 1 mentioned his team struggles to "feel comfortable about the business" because there are layers of translations between the customer and the developer and mainly because the feedback about what was developed takes a long time to get to his team:

> **[P1]:** To give you an idea, a project we undertook this year to provide insight to the engineering group involved surveying clients and identifying which version they are using. As a result, we discovered that newer clients are currently seeing work that was completed two years ago.
>
> […] When receiving feedback 2 or 3 years later, people have often already changed departments or even companies, and as a result, the knowledge is lost. So, one of the objectives we need to work on is creating a way to gather this feedback as soon as possible, as quickly as possible.

Participant 12 brings a similar discussion but emphasizes the lack of tools and processes to get feedback from the final user and not the customer who demands the solution:

> **[P12]:** Currently, this is the challenge we face. We still don't have tools here to measure the level of service and satisfaction that the client achieved during their interaction using the tool.

Avila-Garzon & Bacca-Acosta (2024)corroborated that ensuring effective communication using openness and transparency among stakeholders is essential. However, differences in language, culture, and organizational jargon can also create barriers, impeding mutual understanding and collaboration. We argued that timely communication, which translates to a quick feedback flow, is a determinant for engagement and fostering a culture of collaboration.

At the same time, organizations face the tension between creating scalable solutions while catering to client-specific needs, necessitating flexible operational frameworks (FC.22 - Balancing Scalability and Customization). Additionally, concerns over ownership and control can complicate the collaborative dynamic (FC.23 - Managing Control and Ownership), as can an overreliance on prescriptive practices that limit innovation (FC.24 - Overreliance on Prescriptive Practices). The ultimate challenge lies in translating the abstract goals of co-creation into tangible, actionable outcomes that resonate with client expectations and deliver measurable value (FC.25 - Materializing Value Co-creation). The following extracts are samples of the corresponding challenges.

> **[P11]:** Our focus was based on the client; it wasn't about conducting research or doing a benchmark. In the end, what mattered were the client's demands at that moment. If the client wants to hire, we'll see later if there are others with similar needs.

The extract above discusses the need to tailor a customized value proposition that aligns with specific types of customers but that still allows for sustainable development that can scale (Sheth, 2020). On the other hand, the concerns of ownership and control may be understood as the institutionalized practices and governance that promote governance, compliance, and transparency (Participant 6 extract below). Then, Participant 9 reiterated that by having participated in another agribusiness software development, he know enough to "guide" his new customer. Finally, Participant 5 talks about materializing the value co-creation to have the buy-in from the customer.

[P6]: Including a tech leader who worked with me at another company always had this disagreement with me, because the technical team tends to think they always have the best solutions. That's not necessarily the case, and it's never the case.

[P9]: Let's use an example: if I've already worked with some agribusiness systems and then another client wants to build an agribusiness system, I can apply my expertise from previous experiences to this system. Because I've already dealt with similar projects, I can add value and provide a business perspective to the client.

[P5]: We develop solutions alongside the client and always make it clear that we can use the same solution for other purposes. This reduces costs for them and makes it feasible for us to commercialize, as there was a risk that the budget could become too high, making it impossible to close the deal.

On the other hand, Client-Driven Challenges in Co-creation center on the behaviors, expectations, and trust dynamics of clients. Many clients focus heavily on end results, often overlooking the collaborative process that enables these outcomes, which can lead to misaligned goals. Avila-Garzon & Bacca-Acosta (2024) suggested that to improve stakeholder engagement, particularly among customers, the organization can implement workshops to enable those stakeholders to participate effectively in co-creation projects.

[P6]: Typically, in e-commerce, clients participate in a very "macro" way, showing interest only in the results. Their attitude is more like, "How are you doing this? It's not really my area, and I don't want to know".

Furthermore, trust deficits and knowledge asymmetries create barriers to effective collaboration. Clients may hesitate to share critical information or misunderstand organizational processes, requiring organizations to actively build credibility and bridge these gaps. Addressing these challenges demands consistent relationship-building efforts and transparent communication to ensure clients feel valued and understood throughout the co-creation journey.

[P7]: A major challenge we face is gaining access to the client's metrics. Usually, we ensure integration, but it's something subjective—the client says it has improved and is integrated. However, it is rare for a client to actually provide metrics in an objective way. For example, "My churn rate dropped from 3.8 to 2.7." They don't share that kind of information.

Navigating the interplay between these categories requires deliberate and coordinated action. Internally, organizations must embrace adaptability, improve feedback loops, and prioritize flexibility to align their processes with client needs using sustainable development goals (Avila-Garzon & Bacca-Acosta, 2024). Externally, they must actively build trust, manage expectations, and reduce knowledge gaps through transparent and iterative engagement (Sheth, 2020). By addressing both dimensions with intentional strategies, organizations can transform these challenges into opportunities, creating a seamless co-creation process that delivers impactful and client-aligned solutions. Table 21 captures the memorandum that consolidates the information corresponding to the Categories associated with Theoretical Code 3.

Table 21
Memo 6.

Navigating the Challenges of Value Co-creation
Related Categories CT.3 - Internal Challenges to Co-creation CT.4 - Client-Driven Challenges in Co-creation
The theme emphasizes significant challenges in collecting timely feedback, aligning product development with client needs, and managing the balance between customization and scalability. Feedback loops are often inefficient, with developers receiving client insights years after implementation, leading to a disconnect between current needs and delivered solutions. Usability improvements are deprioritized in favor of scalability and stability, further widening this gap. Clients' reluctance to share strategic metrics adds another layer of complexity, rooted in trust and negotiation concerns. To address these challenges, organizations have implemented strategies like modular solutions to reduce costs, proactive engagement through professional services, and leveraging existing expertise to guide client decisions. However, these efforts require intentional systems to bridge the gap between development teams and client realities. By fostering transparent communication, accelerating feedback cycles, and prioritizing user-centered innovation, organizations can better navigate the complexities of value co-creation, balancing immediate customization needs with long-term scalability goals.

Source: Research Data (2024).

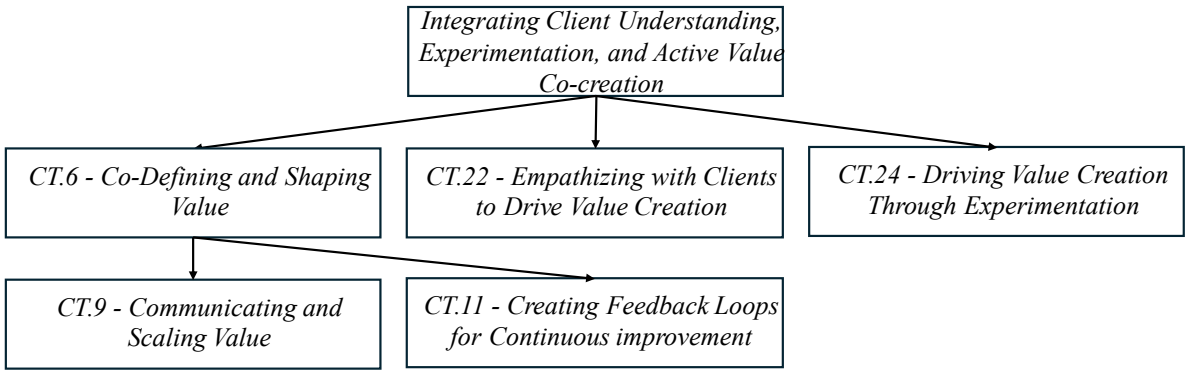
4.4.4 TC.4: Integrating Client Understanding, Experimentation, and Active Value Co-creation

Value co-creation is a multifaceted process that combines deep client understanding, strategic experimentation, and deliberate communication to achieve meaningful outcomes. It requires an iterative approach where organizations and clients work together to define, shape, and scale value. This theoretical code explores the key dimensions of co-creation, highlighting the need for empathy, feedback, and experimentation to drive sustainable innovation. By navigating the inherent challenges in each dimension with intentionality, organizations could create solutions that are not only relevant but scalable and impactful. The theoretical code is

composed of categories, as depicted in Figure 8.

Figure 8

Categories related to TC.4: Integrating Client Understanding, Experimentation, and Active Value Co-creation



Source: Research Data (2024).

Value co-creation can be related to the design thinking process, where before jumping into solving complex problems, it is imperative to frame the problem in the first place. A common understanding in design thinking is that framing the right problem is the only way to create the right solution (Carlgren et al., 2016). Kimbell (2011a) argued that design thinking is a multidisciplinary and participatory process that brings diverse stakeholders to co-create solutions for ill-defined or complex problems using iterative and exploratory methods.

There are benefits of this conceptual exercise of approximation between value co-creation and design thinking. Avila-Garzon & Bacca-Acosta (2024) started their article using co-design and co-creation as different concepts but ended up considering the first one of the steps of the latter. Therefore, they corroborated with the design thinking approximation and ensured that co-creation does not end when value-in-use happens in a service encounter, but it is an organic process where those encounters are intentionally created and often assessed, and feedback flows back to feed new design and adjust the value proposition.

If value co-creation is understood as a process, it begins with co-defining what value means in each context, where clients actively participate in value definition and configuration (Skaržauskaitė, 2013). This is captured by the category “CT.6 - Co-Defining and Shaping Value” which is consistent with service-dominant (S-D) logic (Vargo et al., 2008). In the service context, value is always determined by the customer’s experience (Vargo & Lusch, 2008). Therefore, co-creation is associated with the phenomenological nature of value, where service providers have the chance to collaborate with customers from the beginning, where the

problems are defined, and solutions are ideated, designed, built, and delivered.

Hence, collaboration ensures that both parties share a clear understanding of the goals and priorities. By engaging clients early and deeply in shaping the solution, organizations foster a sense of joint ownership and accountability. This participatory approach builds a foundation for solutions that are closely aligned with the client's unique context (Carlgren et al., 2016). Participant 2 reflects below about what value is, then Participant 5 illustrated how the customer influenced how value should be configured:

> **[P2]:** For me, value […] within what we set out to do, is software that works. To summarize it as much as possible, it is about delivering a system that meets our client's needs and doing so with quality.

> **[P5]:** For example, in our first system, the telemedicine one, the doctors defined a stack structure because they stated it suited the way they issued reports.

Design thinking is not about asking customers what they would like but rather crafting experiments that reveal and provide new data for informed decisions (Carlgren et al., 2016). Design thinking blurs the boundaries between customers and creators, and it is no longer about "us" or "them" but about "us-with-them" (T. Brown & Katz, 2011, p. 382). Empathizing, designing, ideating, prototyping, and testing are steps of the design thinking process that could be mapped to corresponding categories of TC.4. Please observe the mapping proposed in Table 22:

Table 22

Mapping Categories to Design thinking steps

Category	Design Thinking Step
CT.6 - Co-Defining and Shaping Value	Empathize
CT.9 - Communicating and Scaling Value	Design
CT.22 - Empathizing with Clients to Drive Value Creation	Empathize
CT.24 - Driving Value Creation Through Experimentation	Prototype / Test

Source: Research Data (2024).

Once the value is defined, organizations must effectively articulate the value proposition to make its relevance and impact clear to stakeholders, as expressed by the category "CT.9 - Communicating and Scaling Value". At the same time, generalizing the solution involves balancing customization with scalability, ensuring that solutions remain adaptable to various clients and markets. This dimension underscores the importance of communicating and

technical adaptability in extending the reach of co-created solutions without losing their essence. Participant 7 highlights how the solution evolves from a service-oriented design that addresses a custom client requirement.

> **[P7]:** We undertake a thorough design process to understand how to address the issue without being entirely client-specific; we design generic solutions to address the client's pain points. We create an interface that appears platform-based and, behind the scenes, manages the service. Within two months, we deliver this solution to the client.

In the category "CT.11 - Creating Feedback Loops for Continuous improvement", feedback is the lifeblood of value co-creation, and creating feedback flow ensures continuous learning and improvement (Neghina, 2015). By establishing mechanisms to gather, analyze, and integrate feedback from clients, organizations can refine their solutions iteratively. These feedback loops help organizations stay responsive to evolving client needs, fostering a culture of agility and long-term relevance in their offerings.

> **[P9]:** This approach allows the client to provide feedback during the process, not just at the final delivery. However, once the project is delivered, the client performs tests, and subsequently, there may be corrections or improvements—each handled by a different department.

Besides, empathy is the cornerstone of effective co-creation. Category "CT.22 - Empathizing with Clients to Drive Value Creation" suggests that through understanding client needs and finding product-market fit, organizations can design solutions that resonate deeply with users. Therefore, to create meaningful innovation, organizations would need to know their clients and care about their lives, and this is more than a mindset but a process (Rösch et al., 2023).

In fact, teams must go beyond fulfilling orders to act as advisors, not order takers, helping clients navigate challenges and make informed decisions (Carlgren et al., 2016). By thinking like clients, organizations immerse themselves in the client's world, creating solutions that reflect an intimate understanding of their processes, pain points, and aspirations. When questioned about how the developers understand the customers' needs, Participant 8 commented:

> **[P8]:** It is a process without a fixed formula, but its foundation lies in understanding the needs of others. It involves understanding the needs of the person we aim to serve and identifying a value proposition that is meaningful to them. The process can be more

relational or approached in different ways.

Now, picking up on finding the right market fit, Participant 10 highlighted:

[P10]: The practice of being an entrepreneur is always focused on answering "what?" and "for whom?". This was something that wasn't very clear to me at the beginning of my career, but after receiving so much support from Sebrae, mentorships, and consulting, we began to recognize this when developing a new platform or feature. We strive to understand what is going to be developed, what problem it solves, and for whom it solves this problem.

Finally, the category "CT.24 - Driving Value Creation Through Experimentation" underlines that experimentation injects agility and innovation into the co-creation process (Carlgren et al., 2016). By validating hypotheses and creating quick and unembellished value, organizations can respond to client needs with speed and precision. Through managing experimentation with structured governance and embracing a culture of failing fast and learning smart, organizations maintain their adaptability while systematically exploring new opportunities for value creation.

[P8]: Of course, it is important to have hypotheses, but validating those hypotheses is absolutely fundamental to ensure that, by the end of the process, we build something truly relevant and minimize risks. After all, when we talk about product development, everything involves taking risks.

The interplay between these categories is central to successful value co-creation. Co-defining and shaping value sets the stage for meaningful solutions, while communication ensures their relevance can scale to broader contexts. Empathy acts as a guiding principle, ensuring that solutions are rooted in a deep understanding of client needs. Feedback loops provide the mechanism for iterative refinement, and experimentation drives the agility to respond dynamically to new challenges and opportunities. Together, these categories form a cohesive and intentional framework, enabling organizations to navigate the complexities of co-creation while maintaining focus on delivering impactful, client-centered outcomes. Table 23 captures the memorandum that consolidates the information corresponding to the Categories associated with theoretical code 4.

Table 23
Memo 7.

Integrating Client Understanding, Experimentation, and Active Value Co-creation
Related Categories
CT.6 - Co-Defining and Shaping Value CT.9 - Communicating and Scaling Value CT.11 - Creating Feedback Loops for Continuous improvement CT.22 - Empathizing with Clients to Drive Value Creation CT.24 - Driving Value Creation Through Experimentation
The interviews revealed a strong emphasis on understanding client needs and contexts as a foundation for effective value co-creation. Participants frequently noted the importance of tailoring solutions to specific client profiles, such as segmenting shelters by type and size or adjusting workflows in healthcare to match operational realities. This deep understanding not only increased adoption rates but also reinforced the relevance of the solutions provided, especially when clients were directly involved in shaping processes and outcomes. Rapid prototyping and iterative feedback loops emerged as vital practices in reducing risks and aligning solutions with client expectations. Teams prioritized small, functional MVPs that allowed for real-time client input, refining features based on immediate responses. This approach minimized the risk of developing misaligned products and provided a clear path to improvement through continuous feedback. However, participants highlighted that delayed feedback cycles—spanning months or years—often resulted in misalignment and wasted efforts. Experimentation was a recurring theme, with participants balancing innovation with practicality. While the freedom to test new technologies was valued, excessive complexity and overengineering were flagged as counterproductive. Teams found greater success with lean experimentation methods, focusing on quick, measurable outcomes that could be validated and iterated upon. Such governance ensured that innovation remained client-centered and actionable, avoiding "solutions in search of problems." Finally, the shift toward active client collaboration underscored a critical evolution from transactional relationships to partnerships. Teams described educating clients on innovation processes and encouraging their active participation in workshops, discovery phases, and solution validation. This shift not only empowered clients to engage meaningfully but also strengthened the alignment between client expectations and delivered outcomes. By integrating these practices, organizations can create a virtuous cycle of understanding, experimentation, and feedback that drives meaningful and sustained value creation.

Source: Research Data (2024).

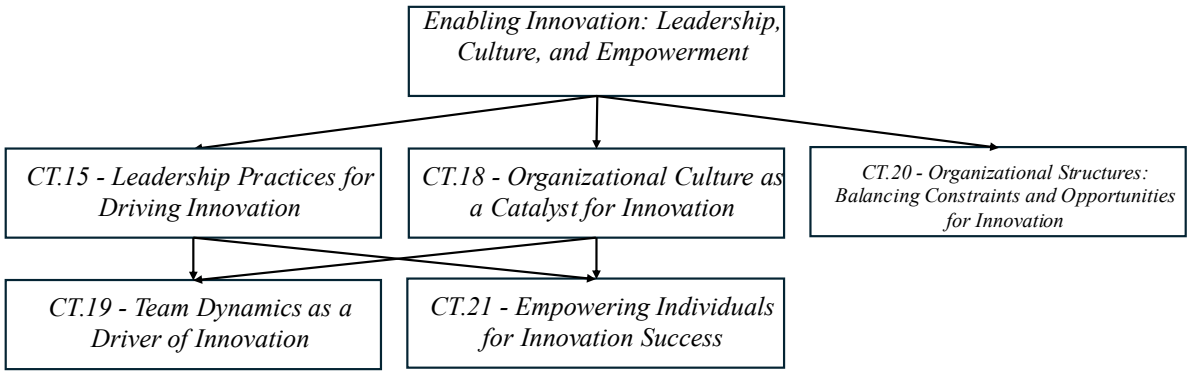
4.4.5 TC.5: Enabling Innovation: Leadership, Culture, and Empowerment

Innovation thrives when organizations create an environment that nurtures creativity, risk-taking, and collaboration (Edmondson, 2018; Martins & Terblanche, 2003). This theoretical code explores the interplay between leadership practices, organizational culture, team dynamics, structural design, and individual empowerment as foundational pillars for fostering innovation. These elements, while distinct, are interdependent, requiring deliberate alignment and thoughtful execution to enable sustained innovation. The theoretical code is composed of categories, as depicted in Figure 9.

.

Figure 9

Categories related to TC.5: Enabling Innovation: Leadership, Culture, and Empowerment



Source: Research Data (2024).

The category "CT.15 - Leadership Practices for Driving Innovation" emphasizes that effective leadership is at the core of enabling innovation, with practices that inspire and guide teams toward creative outcomes. Leaders play a crucial role in creating incentives that reward innovative thinking while promoting diversity to ensure varied perspectives are integrated into problem-solving processes (Martins & Terblanche, 2003). Managing by values establishes a foundation of trust, while governance frameworks provide structure without stifling creativity. Leaders who tolerate honest mistakes create a safe space for experimentation, and by sharing a compelling vision, they align efforts and ignite passion within their teams. Participants 2, 4, and 8 endorsed that it is the responsibility of leaders to create the incentives and promote the values that foster innovation:

> **[P2]:** But it's harder for me because I have a lot of people with different profiles and different mindsets. It's like… different, different people. So it requires a very customized management approach for each one.
>
> **[P4]:** Ah, there's something interesting! The leadership, […] (the team leader), always says that there's no such thing as a silly idea.

> **[P8]:** Something I really enjoy about the innovation process is that it always requires a human perspective on what is happening here and now, as well as a collaborative approach to understand how, as a team, we can find solutions to problems, overcome this phase, and move on to tackle other challenges and discover new solutions.

Conversely, in the category "CT.18 - Organizational Culture as a Catalyst for Innovation", culture is depicted as a bedrock for innovation by fostering an environment where

values are aligned and learning is prioritized (Martins & Terblanche, 2003). Organizations that commit to their values and promote detachment from rigid practices enable agility and adaptability. A culture led by principles rather than rules empowers individuals and teams to focus on continuous learning and purposeful action. Leaders who embed cultural alignment in their organizations help create an innovation-friendly atmosphere where collaboration and creativity flourish. Participants 7 and 8 elaborated on the importance of organizational culture for aligning values and catalyst innovation:

> **[P7]:** Everyone is focused on what's best for the product and the company. There isn't just one person deciding the next step; everyone is collaborating and bringing ideas, willing to engage in discussions about improvements and market insights. It's a highly collaborative environment, which I believe is due to the strong cultural pillars we have in place.

> **[P8]:** In the culture we've established from the beginning, which has been 100% remote, it has always been a goal-oriented culture rather than one focused on time spent sitting at a desk. If we need to reach a certain point, and that is our objective, we simply define what needs to be delivered daily and weekly to ensure we achieve it.

Category "CT.19 - Team Dynamics as a Driver of Innovation" describes that team dynamics significantly impact an organization's ability to innovate. Empowering teams through autonomy and trust enables them to experiment and adapt quickly. Integrating diverse teams fosters inclusion and strengthens collaboration by leveraging varied skill sets and perspectives. Building strong teams requires intentional efforts to nurture cohesion, mutual respect, and shared goals, ensuring that innovation efforts are sustained and impactful. Participants 2 and 4 reflected on the role of empowering teams for innovation:

> **[P2]:** I trust a lot in what they bring to me and what the team considers important for what we do.

> **[P4]:** We have "carte blanche" from the leaders to call whoever is necessary, obviously within limits, since everyone also has other tasks, often "billable," so to speak. Within those limits and the available time, we gather the team here, conduct experiments, and then discuss the results within the forum itself.

Besides, the category "CT.20 - Organizational Structures: Balancing Constraints and

Opportunities for Innovation" is centered on the idea that the design of organizational structures can either enable or constrain innovation (Martins & Terblanche, 2003). Bridging gaps across structures facilitates collaboration and mitigates silos that hinder idea flow. Startups, for instance, often excel in customer orientation and adaptability, while traditional organizations benefit from consistency and stability. Recognizing the strengths and limitations of both models allows organizations to strike a balance that leverages opportunities while managing constraints effectively. Participant 1 explains the need to bridge gaps across the organizational structure to connect resources that otherwise would not collaborate:

> **[P1]:** The engineering team cannot move forward on its own, so we have been building bridges with other teams, whether it's the teams that support sales, the teams that deliver projects, the teams that handle product installation, or even aligning with other product lines.

Reflecting on agility, Participant 12 compares his experience in Startups and traditional organizations using the following metaphor:

> **[P12]:** I make a comparison: a large company is like a Titanic, a massive cruise ship, whereas a startup is like a jet ski. In other words, we can make much quicker maneuvers, turn sideways, and build more innovative things more agilely than a large company. However, the larger company has more financial resources and people to bring these experiments to fruition.

However, innovation begins with individuals who feel empowered to contribute their ideas and grow their skills, as summarized in the category "CT.21 - Empowering Individuals for Innovation Success". Understanding the "whys" behind actions fosters a sense of purpose, while taking ownership of self-development ensures that individuals are continuously evolving. By enabling employees to take control of their growth and encouraging reflective practices, organizations equip them with the confidence and capability to drive innovation forward. Participant 1 explains the need to bridge gaps across the organizational structure to connect resources that otherwise would not collaborate:

> **[P2]:** I believe developers perform much better when they understand the value their software delivers to the client. This is one of the points I always discuss with the team—"Do you know why you're running this? Do you know why you're adding this IF-ELSE statement? In the end, when we compile everything and deploy it, can you explain why we're doing this? Can you explain the purpose behind it?".

> **[P3]:** I need their autonomy, and I believe that if they develop this mindset from the start, it works better. Of course, you need to hire the right person because sometimes you give autonomy to the wrong person, and they just wait for something to happen. In our team, no one waits around. Everyone takes initiative. If there is something to be done, they do it and only ask when they have a doubt.

These categories described above are interconnected, forming a cohesive framework where leadership, culture, team dynamics, structure, and individual empowerment feed into one another. Leaders set the tone and vision, culture provides the enabling environment, teams act as engines of execution, structures determine the pathways for innovation, and individuals bring creativity and commitment. Navigating these elements requires intentionality, as aligning them amplifies their collective impact, creating a virtuous cycle of sustained innovation. Table 24 captures the memorandum that consolidates the information corresponding to the Categories associated with theoretical code 5.

Table 24
Memo 8.

Enabling Innovation: Leadership, Culture, and Empowerment
Related Categories
CT.15 - Leadership Practices for Driving Innovation
CT.18 - Organizational Culture as a Catalyst for Innovation
CT.19 - Team Dynamics as a Driver of Innovation
CT.20 - Organizational Structures: Balancing Constraints and Opportunities for Innovation
CT.21 - Empowering Individuals for Innovation Success
The interviews provided a rich understanding of how leadership, team collaboration, and organizational structures influence innovation. Leadership was consistently highlighted as the driving force behind aligning efforts with a clear vision for the future. Leaders who emphasize the purpose behind tasks and foster an environment where experimentation and learning are encouraged tend to unlock greater creativity and commitment from their teams. These approaches not only ensure technical contributions align with client outcomes but also cultivate a culture where individuals feel invested in the broader organizational mission. A recurring theme was the critical role of collaboration across diverse teams and breaking down silos to promote innovation. Interviewees emphasized that diverse professional backgrounds and perspectives enrich problem-solving but require deliberate management to navigate potential conflicts and maintain focus. By establishing bridges between teams and creating inclusive spaces for dialogue, organizations can harness the strength of their collective insights. At the same time, managing this diversity calls for nuanced leadership capable of tailoring approaches to individual strengths and challenges. The contrast between startups and larger organizations revealed the importance of balancing agility with structure. Startups, with their nimbleness and focus, can quickly adapt and innovate but often struggle with limited resources. Conversely, larger organizations offer stability and resources but are often constrained by rigid processes that slow decision-making. Some interviewees stressed the need to find an equilibrium between governance and operational freedom, allowing innovation to flourish without compromising accountability. Finally, the interviews underscored the power of autonomy and shared ownership within teams. Empowered teams that are trusted to take the initiative and contribute to the decision-making process demonstrate higher

Enabling Innovation: Leadership, Culture, and Empowerment
levels of engagement and creativity. However, sustaining this requires careful attention to motivation, structured incentives, and alignment with both individual aspirations and organizational goals. Together, these elements highlight the interplay of leadership, culture, and team dynamics as essential enablers of meaningful and sustained innovation.

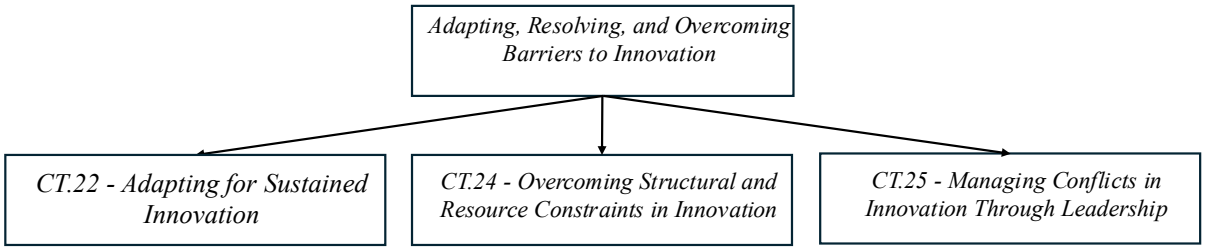
Source: Research Data (2024).

4.4.6 TC.6: Adapting, Resolving, and Overcoming Barriers to Innovation

This theoretical code delves into the dynamic processes and strategies organizations employ to address obstacles that threaten innovation initiatives. The findings reveal a multifaceted approach encompassing adaptation, structural adjustments, and conflict management, underlining the critical role of leadership, cultural alignment, and resource optimization. The theoretical code is composed of categories, as depicted in Figure 10.

Figure 10

Categories related to TC.6: Adapting, Resolving, and Overcoming Barriers to Innovation



Source: Research Data (2024).

In the first category, “CT.22 - Adapting for Sustained Innovation”, the capacity to adapt is a cornerstone of sustained innovation (Manca et al., 2018). A common pitfall occurs when teams lose sight of their customers, focusing more on internal processes than the needs they aim to address. Losing sight of the customer emerged as a key risk, where organizations become inward-focused, leading to misaligned products or services. Organizations often resist service innovation due to entrenched “business-as-usual” mindsets and bureaucratic inertia (Parris et al., 2016). Participant 3, reflecting on his distance to end customers, suggested that the more product-oriented his team is, the more distant it becomes from individual customers:

> **[P3]:** In this case, I have this additional layer with the client. Sometimes, handling tickets or more complex issues that reach L4 allows me to interact directly with the client, but today, this happens much less frequently, and my daily conversations are with the PS (professional services) team.

Resistance to change also emerges as a recurring challenge, as employees and even customers can struggle to let go of established practices or embrace new approaches. To overcome these barriers, organizations must embed adaptability in their culture, ensuring their innovation efforts are not only responsive to immediate challenges but also aligned with long-term goals (Parris et al., 2016). Leaders play a critical role in fostering an environment where experimentation is encouraged yet still guided by a clear vision for the future (Velayati et al., 2020). In the following extract, Participant 5 commented that customers resist upgrading and benefit from new releases, suggesting they prefer to work with the systems as they are instead of trying new things. Next, the same participant underlines the challenge of keeping documentation updated for future support:

> **[P5]:** We have an access link for updates, showing the version the client is using and the latest version they can upgrade to. We always try to get them to migrate to this new version, but it's not always what they want. In fact, they never want to.
>
> […] I think the biggest challenge we've always faced is documentation. Since we operate like a startup, the pace has to be fast, and the documentation doesn't always keep up with it.

Additionally, the category "CT.24 - Overcoming Structural and Resource Constraints in Innovation" suggests that structural and resource constraints present another significant obstacle to innovation (Parris et al., 2016). Limited time and resources require careful prioritization to ensure the most impactful initiatives are advanced. When asked about the main barrier to innovation in his team, Participant 3 points out that available time is the Achilles' heel and, according to Participant 10, the time spent sustaining legacy or existing solutions consumes a good portion of the team's energy:

> **[P3]:** I notice that other teams working here in R&D seem to have a bit more time. I realize they are less overwhelmed than our team, and because of that, they have the availability to study, take a course, or try new practices.
>
> **[P10]:** This is a sincere desire of mine: to listen to the customer more and be more present in their lives. That's a fact. For this, methods and initiatives are necessary. My team is one that really enjoys talking to and serving customers. I always say that at a certain point in our company, we became more specialized in firefighting than in fire

prevention. Preventing the fire would mean listening to the customer first and not letting things happen in the first place.

Additionally, the rise of remote work has introduced complexities in maintaining collaboration and alignment across teams. Organizations must balance the need for agility with the reality of limited capacity, often employing creative solutions to streamline processes and bridge gaps. By leveraging technology and fostering cross-functional collaboration, companies can navigate these constraints while maintaining a focus on their innovation objectives. Participant 9 shared his view about doing innovation with remote personnel:

> **[P9]:** In a remote environment, this same person might feel embarrassed to ask for help in solving the problem, and remotely, it becomes difficult to identify this person's struggles. Depending on their level of seniority, working in person makes sense.

Finally, the category "CT.25 - Managing Conflicts in Innovation Through Leadership" talks about conflict as being an inevitable aspect of innovation, particularly especially when resources are stretched or priorities clash. Resolving conflicts through authority reflects a leadership-driven approach where decisive action is taken to mediate disputes and align teams. Although quick, this approach does not mitigate friction.

Besides, conflicts are inevitable in collaborative service ecosystems due to the diversity of stakeholders, their goals, and expectations. However, managing these conflicts effectively is crucial for fostering innovation (Manca et al., 2018). By establishing clear priorities, fostering open communication, and balancing diverse perspectives, leaders have better options than only addressing immediate tensions with authority. Conflicts open up opportunities to build a culture of trust and accountability, enabling teams to navigate complex innovation challenges collaboratively. The lower groundedness of this category may indicate that this leadership style is less common in companies where innovation is a priority. This is the extract from Participant 3:

> **[P3]:** I wouldn't say there are conflicts because there is always someone who has more authority on certain subjects, and we respect that a lot. Usually, this person ends up making a better decision.

In essence, this theoretical code illustrates how organizations adapt and resolve obstacles to sustain innovation. By addressing customer-centricity, overcoming structural constraints, and managing conflicts with strong leadership, they can foster resilience and ensure

their innovation strategies remain viable in an ever-evolving environment. This interplay between adaptability, resourcefulness, and leadership forms the backbone of overcoming barriers and achieving long-term success in innovation. Table 25 captures the memorandum that consolidates the information corresponding to the Categories associated with theoretical code 6.

Table 25
Memo 9.

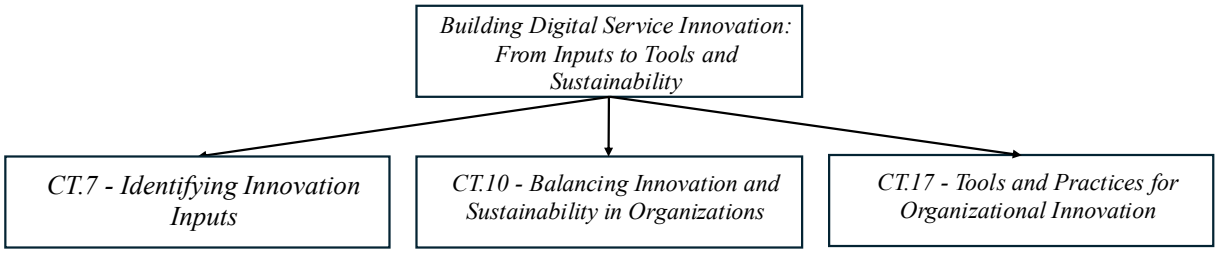
Adapting, Resolving, and Overcoming Barriers to Innovation
Related Categories CT.22 - Adapting for Sustained Innovation CT.24 - Overcoming Structural and Resource Constraints in Innovation CT.25 - Managing Conflicts in Innovation Through Leadership
The interviews highlighted significant challenges in bridging gaps between development teams, customer needs, and organizational processes, all of which impact innovation efforts. A recurring theme is the disconnect between customer insights and technical execution. Ambiguity about client requirements often leads to over-engineering or misaligned solutions, as teams struggle to interpret vague inputs, losing critical context during knowledge transfer. This highlights the need for robust mechanisms to capture and maintain clarity throughout the development lifecycle. Structural inefficiencies and resource constraints emerged as central obstacles. Teams are frequently overburdened with legacy system support and day-to-day operations, leaving little capacity for innovation. For example, many participants lamented the lack of dedicated innovation teams or time, as daily urgencies take precedence. These constraints not only slow progress but also stifle proactive problem-solving, with some organizations focusing more on reactive measures rather than strategic innovation. Cultural and leadership dynamics further influence innovation outcomes. Some interviewees underscored the importance of leadership in fostering clear communication, resolving conflicts, and aligning cross-functional goals. However, others noted organizational silos, where key stakeholders, such as product managers or sales teams, dominate decision-making, isolating engineering from critical customer feedback. This isolation often leads to value misalignment as teams struggle to prioritize features that genuinely benefit end users. Finally, insights drawn from limited or biased data often shaped solutions, resulting in skewed priorities or one-size-fits-all approaches. Several participants stressed the importance of improving data collection and analysis methods to ensure a more representative understanding of customer needs. Moreover, fostering a culture of open dialogue and iterative testing with clients can mitigate these challenges, enabling teams to deliver incremental value while adapting to evolving demands.

Source: Research Data (2024).

4.4.7 TC.7: Building Digital Service Innovation: From Inputs to Tools and Sustainability

Developing innovation within organizations involves identifying critical inputs, integrating sustainable practices, and leveraging appropriate tools and rituals to foster an environment conducive to creativity and progress. This theoretical code captures the dynamic journey from the inception of innovative ideas to their long-term integration into organizational processes. The theoretical code comprises the categories depicted in Figure 11.

Figure 11
Categories related to TC.7: Building Digital Service Innovation: From Inputs to Tools and Sustainability



Source: Research Data (2024).

Category "CT.7 - Identifying Innovation Inputs" key idea is that organizations often begin their innovation journey by identifying inputs derived from market gaps, customer pain points, or reflective analyses of their processes and products. This is the same starting point of Design Thinking, where through empathy, organizations can recognize and understand customers' sensations, emotions, thoughts, motives, and hidden needs (Rösch et al., 2023). Starting from market gaps provides a strategic approach to address unmet needs while focusing on pain points highlighting immediate challenges faced by customers or internal teams. Participants 8 and 12 recounted the context where the ideas to develop a solution came from:

> **[P8]:** We started the company as a social impact startup in 2016, following a frustrating pet adoption experience in the United States, which was carried out by the company's co-founder. We were born out of this context of a frustrating adoption experience, where we realized it took about two months and more than 70 emails between different organizations to adopt a pet due to bureaucracy-related issues.

> **[P12]:** The Pimpus product was aimed at solving issues in child healthcare, specifically the lack of communication or, more precisely, the lack of centralized access to

> information, which did not exist. Three million children are born each year, and a significant portion of them are born to first-time parents who need support from a network and tools to assist them in the process. This was something that didn't exist at the time.

Team reflexivity, on the other hand, allows for a more introspective process of uncovering inefficiencies or opportunities, often leading to innovation that is deeply aligned with the organization's mission and values. According to Hoegl & Parboteeah (2006), team reflexivity is related to the notion there is a need to constantly assess the environment, question, plan, and perform exploratory learning to be more aware of the consequences of its actions. Participant 4 argued that it is easier to innovate through reflection on existing processes and failures than to create something completely new:

> **[P4]:** I see two possible origins: one, through continuous improvement, where we review our process, check what can be improved, or even identify failures in our process.

Once inputs are identified, balancing innovation with sustainability becomes a critical challenge. Category "CT.10 - Balancing Innovation and Sustainability in Organizations" points out that organizations frequently adopt agile methodologies to create flexibility and responsiveness while maintaining operational stability. The category is consistent with the notion of exploration and exploitation (March, 1991). Where exploration includes things such as search, variation, risk-taking, experimentation, flexibility, discovery, and innovation, whereas exploitation refers to things such as refinement, choice, production, efficiency, selection, implementation, and execution (P. P. Li et al., 2023). The extracts below from Participants 7 and 10 illustrate the need to address innovation and sustainability within the same team:

> **[P7]:** We have a very lean team. Most of the time, we can't have a division between a team dedicated to maintenance and another to innovation, so balancing and prioritizing tasks is a significant challenge for the product area.

> **[P10]:** The best metaphor is that of the circus performer balancing plates […] The plate-spinning act lies in the fact that someone is footing the bill — someone is earning revenue and generating money to invest in a new "baby" that is crawling, wobbling, and only incurring costs. Are you paying attention to it? I am paying very little attention,

and we have to balance this plate-spinning act.

P. P. Li et al. (2023) argued that the failure to maintain this balance could lead to stagnation, where teams focus entirely on sustaining current operations without advancing new ideas or adapting to external changes. By contrast, organizations that prioritize continuous learning and experimentation can build more sustainable and adaptive systems, ensuring long-term success.

Sustainable solutions require foresight and adaptability to accommodate inevitable changes in the market or customer preferences and are aligned with exploitation (Y. Li et al., 2008). Segregating team profiles by specialization or innovation focus is another key approach to ensuring that different facets of innovation, from ideation to execution, receive adequate attention. The following extracts illustrate those ideas:

> **[P5]:** Another example: there are doctors who do not have a typist and request to remove that workflow. Other doctors who use voice recording and no longer rely on typing have asked to change the workflow to voice-to-text typing. We make small customizations, as long as they are not too costly so that we can maintain the differentiator of customizing the solution, which many companies do not offer.

> **[P7]:** In this context, knowing how to manage team members is very important. There are people who are very perfectionists, and in such cases, we don't assign them to a project like this because it's not their profile. They would constantly push back, saying things like, "this is a workaround… we need to design the proper structure." They are not the right resource to work on a quick proof of concept.

Tools and practices are indispensable in translating innovative ideas into tangible outcomes, as summarized in the category "CT.17 - Tools and Practices for Organizational Innovation". Here, ritualizing innovation through structured routines or dedicated timeframes fosters a culture where creativity is not an exception but an expectation. The emphasis on implementation practices and resource mobilization is consistent with the findings of Pertuz & Pérez (2021) about innovation management practices. Participant 7 reflects on the importance of setting aside time dedicated to innovation and breaking the routine:

> **[P7]:** If we don't do this, the routine consumes us. If we don't have these moments and rituals where we take some time to focus on new things, we don't grow. There's so much in the daily grind—backlog refinement, evolving solutions, and the drive to keep working on what's already going well—that we need these pauses.

Incremental innovation, often achieved through iterative processes, ensures that new ideas are tested and refined before full-scale implementation. Establishing innovation spaces—both physical and virtual—offers a dedicated environment for brainstorming and experimentation, while incentives for innovation motivate individuals and teams to contribute actively. Additionally, focusing on internal processes streamlines the journey from ideation to execution, reducing friction and enhancing efficiency. About incremental innovation, Participant 6 commented:

> **[P6]:** I understand there is room for innovation, but it should be tied to the existing solution, not something entirely new. What can be done is to deliver what others provide, but at a lower cost and linked to a more modern and affordable technology..

In sum, Building Digital Service Innovation is a multifaceted endeavor that requires organizations to start from meaningful inputs, sustain a delicate balance between creativity and stability, and implement tools and practices that make innovation both systematic and sustainable. This interplay ensures that innovation not only thrives in the short term but also becomes a lasting competitive advantage. Table 26 captures the memorandum that consolidates the information corresponding to the Categories associated with theoretical code 7.

Table 26
Memo 10.

Building Digital Service Innovation: From Inputs to Tools and Sustainability
Related Categories CT.7 - Identifying Innovation Inputs CT.10 - Balancing Innovation and Sustainability in Organizations CT.17 - Tools and Practices for Organizational Innovation
Innovation within organizations emerges from a dynamic interplay between market demands, internal pain points, and reflective practices that foster creative problem-solving. Teams often begin by identifying gaps in the market or addressing inefficiencies within their workflows, adapting solutions to user-specific realities. This user-centric approach, seen in practices like rapid prototyping or incremental customizations, ensures that innovation aligns with end-user needs while remaining feasible and impactful. In startups and agile environments, this adaptability is a core strength, allowing for a closer connection to customer feedback and quicker iteration cycles. However, balancing innovation with sustainability poses significant challenges. Agile frameworks, such as sprints, provide a structure for iterative development, yet the pressure of deadlines often consumes time intended for creative exploration, such as dedicated innovation weeks. Smaller teams face additional constraints, likened to "spinning plates," where limited resources and competing priorities demand careful focus. Leadership plays a pivotal role here, as prioritizing innovation over operational needs or vice versa can define an organization's trajectory. Successful teams mitigate these tensions by aligning roles with individual strengths, ensuring that creative and operational demands are met without overwhelming team members. The practices and tools that drive innovation often depend on fostering a culture of creativity and collaboration. Rituals like hackathons, incentives for ideation, and cross-functional projects encourage diverse perspectives and shared ownership of ideas. These practices not only generate novel solutions but also enhance team cohesion and

engagement. Innovation is frequently incremental, building on existing solutions rather than pursuing disruptive changes, emphasizing practicality and cost-efficiency. Moreover, integrating user feedback into development processes solidifies the connection between innovation and real-world applicability.

Ultimately, innovation requires a balance between structured methodologies and organic collaboration. While formal frameworks provide consistency, informal interactions often spark the creativity needed to address complex challenges. The tension between innovation and sustainment is a persistent reality, yet organizations that embrace adaptability and foster an inclusive, user-focused culture are better equipped to navigate this balance, delivering solutions that are both innovative and sustainable.

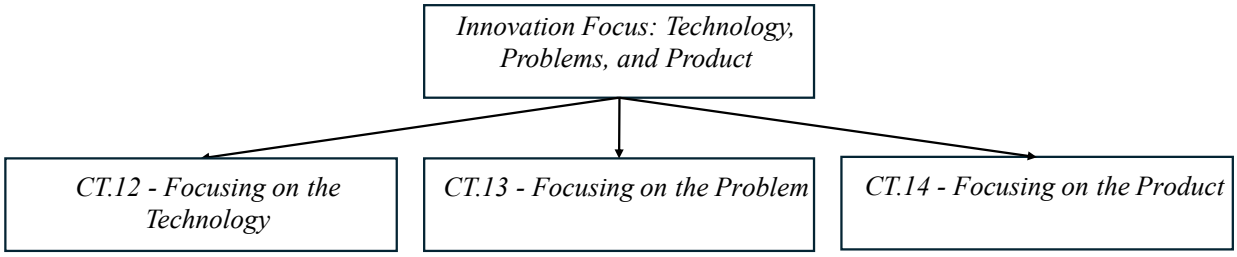
Source: Research Data (2024).

4.4.8 TC.8: Innovation Focus: Technology, Problems, and Product Strategies

The focus of innovation within organizations is shaped by the interplay of three key dimensions: technology, problems, and product strategies. These dimensions serve as lenses through which teams navigate challenges, frame their priorities, and drive innovation. Each focus carries distinct implications for how organizations approach value creation, operational efficiency, and strategic differentiation. The theoretical code is composed of categories, as depicted in Figure 12.

Figure 12

Categories related to TC.8: Innovation Focus: Technology, Problems, and Product Strategies



Source: Research Data (2024).

The category “CT.12 - Focusing on the Technology” indicates that technology emerges as both a catalyst and a constraint in organizational innovation. Many teams view technology as a primary driver of solutions, leveraging it to address complex challenges and unlock new opportunities. However, service innovation is a multidimensional phenomenon and cannot be understood through the lens of new technology alone(Hertog, 2010; Kindström & Kowalkowski, 2014). This perspective underscores the critical role of technical capabilities in shaping innovative practices. Yet, pursuing technological advancement often requires overcoming technical debt, where legacy systems or outdated practices hinder progress.

Addressing these constraints demands a delicate balance between maintaining operational stability and integrating modern, scalable solutions that align with long-term goals.

> **[P1]:** We have been working on removing some technical debts, so to speak, which enable us to move to the next step, such as a SaaS evolution or even allowing existing on-premises clients to have a CI/CD process directly in their environment.

> **[P12]:** We have a certain degree of freedom internally. We can bring in innovations and adopt software (like Jira, for instance) to implement better processes. We can also bring in tools like Red Hat and OpenShift to improve our cloud and infrastructure management. This is how we drive innovation.

On the other hand, the category "CT.13 - Focusing on the Problem" suggests that problem-oriented approaches to innovation emphasize understanding organizational pain points and aligning solutions with core business needs. In this context, teams often act as "organizational therapists," diagnosing inefficiencies, streamlining processes, and prioritizing service improvements over tool-centric fixes. The service-dominant logic invites a focus on improving customer value through co-creation instead of just bringing new offerings to the market (Rubalcaba et al., 2012).

This perspective places the problem – not the product or technology – at the heart of innovation efforts, enabling teams to craft solutions that are both essential to business operations and rooted in service delivery. There is a widely embraced mantra among CEOs and innovators, often attributed to Charles Eames (Kent & Steward, 2008), which emphasizes that the first question in designing an innovation should not be, "Can I make it?" but rather, "Should I make it?" This shift underscores the importance of considering purpose and value before delving into feasibility. By beginning with service rather than product development, organizations build solutions that resonate more deeply with end-users and operational realities.

> **[P7]:** It's like therapy: you go to a therapist to talk about a problem, but over time, you dig deeper until you find the real issue. This doesn't mean the problem doesn't exist, but you lack the clarity to see that it is the actual root issue. In a product team, you have to act like the company's therapist, and you only achieve this through conversation.

> **[P8]:** This gave us a very empirical experience, and we understand that the process of building and developing products needs to be closely connected to reality and the daily pains, not just what we believe to be the pain.

Finally, the category “CT.14 - Focusing on the Product” underlines that product-driven innovations are centered on creating tools that enhance functionality and meet diverse market demands for product-centric organizations (Kindström & Kowalkowski, 2014). However, a shift is observed in prioritizing product support over mere feature expansion, reflecting the importance of sustaining long-term user satisfaction as firms increase service offerings(Kindström et al., 2013). Teams increasingly diversify their offerings to stay competitive, integrating complementary products that enhance overall value propositions. This approach not only drives revenue growth but also enables organizations to adapt to evolving customer expectations and market trends, ensuring their offerings remain relevant and impactful.

> **[P3]:** Sometimes, when you approach with a ready-made system, you have a somewhat distorted view of what the process should be compared to what we impose for it to be.

> **[P11]:** In fact, we intentionally reduced the development team because we wanted to focus more on sales rather than inventing new features. We got caught up in inventing features. It would be great if we could do that. We had three developers available, so we put the resources to work.

By integrating these three focal points—technology, problems, and products/services—organizations construct a nuanced innovation strategy that adapts to varying needs and challenges (Kindström & Kowalkowski, 2014). The interplay between these dimensions underscores the complexity of innovation, requiring teams to balance technological potential, operational realities, and market responsiveness. This multifaceted approach positions organizations to innovate effectively while maintaining alignment with their broader goals and values. Table 27 captures the memorandum that consolidates the information corresponding to the Categories associated with theoretical code 8.

Table 27
Memo 11.

Innovation Focus: Technology, Problems, and Product Strategies
Related Categories
CT.12 - Focusing on the Technology CT.13 - Focusing on the Problem CT.14 - Focusing on the Product
The interview extracts reflect a complex landscape of innovation, process management, and product strategy within diverse organizations. Several themes emerge across the interviews: the need to balance technological

Innovation Focus: Technology, Problems, and Product Strategies
advancements with pragmatic problem-solving, the prioritization of client-focused processes over tools, and the challenges of managing legacy systems while embracing modernization. A recurring insight is the emphasis on focusing on the problem rather than solely the product or technology. Participants liken their roles to organizational therapists, identifying root causes through iterative dialogue and adapting solutions to client-specific needs. This approach underscores the importance of understanding user contexts and operational challenges with empathy, such as in the case of healthcare professionals burdened by time constraints or shelters dealing with inefficiencies in adoption workflows. The value of simplicity in design and avoiding unnecessary complexity, as emphasized by the "Keep It Simple and Stupid" (KISS) philosophy, appears central to fostering scalability and maintaining client trust. Technological innovation, while not dismissed, is viewed as a means rather than an end. Participants describe efforts to integrate methodologies such as SaaS deployment, CI/CD pipelines, and automation tools, highlighting how these approaches reduce technical debt and improve responsiveness. However, innovation is not defined by cutting-edge technologies alone but by the effective alignment of processes, tools, and user needs—such as optimizing appointment confirmations or streamlining financial and operational management in health sectors. Finally, the interviews reveal an ongoing tension between focusing on immediate operational priorities and longer-term strategic innovation. Many organizations struggle with legacy constraints and resource limitations while striving to deliver value incrementally. This tension is further complicated by the need for alignment across teams, stakeholders, and clients to ensure that solutions are not only technically feasible but also meaningful and impactful in their contexts.

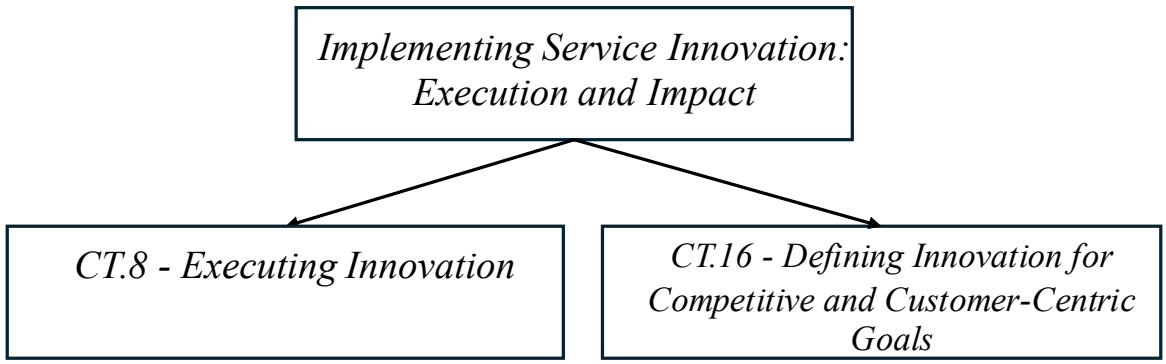
Source: Research Data (2024).

4.4.9 TC.9: Implementing Service Innovation: Execution and Impact

Implementing service innovation involves a fine balance between ideation and practical execution. Organizations must navigate operational complexities while focusing on outcomes that matter most to customers and the business. This requires a disciplined approach emphasizing scalability, adaptability, and clear alignment with strategic goals. The theoretical code is composed of categories, as depicted in Figure 13.

Figure 13

Categories related to TC.9: Implementing Service Innovation: Execution and Impact



Source: Research Data (2024).

A critical aspect of execution is the ability to innovate within existing systems and processes. This is captured in the category “CT.8 - Executing Innovation”. Within that category, legacy structures often pose challenges, yet they can also serve as a foundation for transformation when approached creatively. Organizations must remain focused on scalable solutions that deliver practical benefits, ensuring that innovations not only meet current needs but also pave the way for future growth. The idea of maintaining focus amid complexity is, in some sense, aligned with the paradoxical link between exploitation and exploration, where too much exploration (too many options) can starve resources, and too much exploitation can favor learning myopia (P. P. Li et al., 2023).

> **[P7]:** One of them is focus, or rather, the lack of it. Even with an excess of willingness to innovate, in a small company with a small team looking at eight different market pain points, we end up bringing several small things and falling into the analogy of the duck—it flies, swims, and walks, but doesn’t do any of them well.
>
> […] To innovate with quality and reach a point where we truly solve a customer’s pain point, the focus is essential. We’ve experienced both scenarios: at one point, we looked at too many things, and that was when we made the least progress in terms of results.
>
> **[P9]:** The speed of a startup’s development comes from the team being focused on just one problem. In a software house, we have multiple projects happening at the same time.

This pragmatic mindset helps prioritize actions that drive tangible results, aligning innovation efforts with clear objectives (Pertuz & Pérez, 2021). This is captured in the following extract from Participant 1:

> **[P1]:** Much of what we encounter daily is that clients have requirements specific to their needs, but what we need to do is extrapolate so that these can be configurable enough to serve the hundreds of clients we have.

Equally important is the broader vision of innovation as a tool for competitive advantage and customer-centric progress as described by the category “CT.16 - Defining Innovation for Competitive and Customer-Centric Goals”. By continuously enhancing the customer experience, organizations create deeper connections with their users, fostering loyalty and differentiation in crowded markets (Prahalad & Ramaswamy, 2004). Innovation also drives efficiency, enabling teams to optimize internal operations while delivering greater value to

clients (Agazu & Kero, 2024).

> **[P7]:** We are a small company competing in markets dominated by large players, so innovation is essential. If we do what is already available, there are much larger companies with broader portfolios and greater market reputations, so we would easily be swallowed up.
>
> […] With our market experience, specialization, and deep understanding of what we are working with, we have the advantage of being agile. This is where large companies, when they seek out other large companies, struggle a bit. We can quickly grasp a context, configure a solution, and adapt to something we identify on the spot.

In sum, the process of implementing service innovation is as much about disciplined execution as it is about creative ideation. By focusing on meaningful impact and aligning efforts with strategic goals, organizations can turn bold ideas into transformative solutions that drive growth, efficiency, and long-term relevance. Table 28 captures the memorandum that consolidates the Categories associated with theoretical code 9.

Table 28
Memo 12.

Implementing Service Innovation: Execution and Impact
Related Categories CT.8 - Executing Innovation CT.16 - Defining Innovation for Competitive and Customer-Centric Goals
The interviews reveal a complex interplay between innovation, operational constraints, and market realities. Organizations serving diverse client bases face a recurring challenge: balancing scalability with the need for tailored solutions. This tension is exacerbated by the limitations of legacy systems, which, while stable, often hinder transformative changes. Incremental innovation becomes the preferred strategy in these contexts, as it ensures stability while introducing improvements that align with client expectations. However, this cautious approach often clashes with the ambitious aspirations of teams eager to experiment and disrupt traditional workflows. Resource constraints further complicate the picture. Teams often lack the diversity of expertise needed to drive innovation, with heavy reliance on a few overburdened individuals. In startups, these challenges are magnified by funding limitations, which restrict the ability to scale or explore innovative solutions. Even within established organizations, market preferences for stability and brand recognition frequently overshadow the appetite for groundbreaking technologies, creating a dynamic where smaller players struggle to gain traction despite offering superior cost-to-value propositions. At the heart of effective innovation lies the ability to deeply understand and address client pain points. Successful teams focus not on flashy technologies but on pragmatic solutions that deliver measurable value, such as cost savings or process simplifications. Agile methodologies and iterative deliveries, including MVPs, have emerged as effective frameworks for bridging client needs and organizational priorities. Yet, fostering focus and alignment remains critical. Organizations that attempt to tackle too many problems simultaneously dilute their impact, often falling short of meaningful outcomes.

Implementing Service Innovation: Execution and Impact
Culturally, resistance to change and communication silos pose significant hurdles. Long-standing workflows and entrenched mindsets, particularly among senior or traditional stakeholders, can stifle progress. Addressing these barriers requires leadership that not only facilitates collaboration across specialized teams but also demonstrates the tangible benefits of innovation through incremental successes. Ultimately, the findings emphasize that innovation is not merely about creating something new; it is about delivering solutions that resonate with client needs and evolve alongside a rapidly changing market landscape.

Source: Research Data (2024).

4.5 Theoretical Saturation

The concept of theoretical saturation is pivotal in grounded theory research. It refers to the point in the data collection process where no new themes, categories, or theoretical insights emerge, and additional data collection yields diminishing returns (Charmaz, 2014). At this stage, the researcher reaches confidence that the developed categories comprehensively represent the phenomenon under investigation (Corbin & Strauss, 2008).

In the present study, efforts were made to approach theoretical saturation by conducting interviews and analyzing data iteratively. The process allowed initial themes and categories to emerge and evolve over time. However, it became evident that some aspects of the investigated phenomenon still required further exploration to fully develop theoretical categories and validate their relationships. This limitation stems from the study's constrained timeframe, which necessitated halting data collection before theoretical saturation could be conclusively achieved.

It is worth noting that while achieving theoretical saturation is ideal, grounded theory studies acknowledge practical constraints such as time and resources (Bowen, 2008). In such cases, researchers must strike a balance between methodological rigor and practical feasibility, ensuring that emerging insights still provide valuable contributions. Although the study did not fully achieve saturation, the analysis nonetheless provides a meaningful foundation for understanding the phenomena and highlights areas for future research.

Future research should aim to revisit and expand the scope of this investigation, particularly by conducting additional interviews to further test and refine the categories identified here. The iterative nature of grounded theory offers flexibility, allowing subsequent studies to build upon the work presented in this thesis.

5. DISCUSSION

The discussion section summarizes the findings of this research, situating them within the broader theoretical and practical contexts while addressing their implications for organizational learning, service innovation, and value co-creation. Using grounded theory as the primary methodology (Charmaz, 2014), this research systematically advanced through levels of abstraction, from initial coding to developing high-level theoretical codes.

The integration of theoretical codes, as outlined in the findings, provides a framework for understanding the interconnected processes that drive innovation, learning, and value co-creation. Despite the limitations of not reaching theoretical saturation due to time constraints, the depth of the data allowed for the development of meaningful insights. These insights reflect the dynamic interplay between organizational practices, client interactions, and innovation strategies. This section will contextualize the theoretical codes, compare emergent patterns with extant literature, and explore their implications for academia and industry.

Not only that, but the alignment between the theoretical codes developed in this study and the specific research objectives also provides a clear roadmap for understanding how the findings address the core aims of the research. Table 29 highlights the connections between the theoretical constructs emerging from the data and the predefined objectives, illustrating the coherence and relevance of the analytical process. By mapping these relationships, the table emphasizes how each theoretical code contributes to answering the research objectives while offering a structured perspective on integrating empirical insights with the study's goals. This approach underscores the study's methodological rigor and theoretical depth, offering a comprehensive framework to guide scholarly exploration and practical application.

Table 29
Theoretical Codes and Research Objectives

Theoretical Code	Specific Objectives
TC.1 - Organizational Learning Across Teams and Client Partnerships	Objective 1: Understand the key elements of the phenomenon of organizational learning and value co-creation in software development teams.
TC.2 - Building Collaborative Relationships and Trust for Co-creation	Objective 1: Understand the key elements of the phenomenon of organizational learning and value co-creation in software development teams.
TC.3 - Navigating the Challenges of Value Co-creation	Objective 2: Understand the relationships of these elements in the context of innovation in digital services.
TC.4 - Integrating Client Understanding, Experimentation, and Active Value Co-creation	Objective 2: Understand the relationships of these elements in the context of innovation in digital services.
TC.5 - Enabling Innovation: Leadership, Culture, and Empowerment	Objective 3: Offer explanations for the behavior of individuals in development teams working in innovation processes with co-creation of value.

Theoretical Code	Specific Objectives
TC.6 - Adapting, Resolving, and Overcoming Barriers to Innovation	Objective 3: Offer explanations for the behavior of individuals in development teams working in innovation processes with co-creation of value.
TC.7 - Building Digital Service Innovation: From Inputs to Tools and Sustainability	Objective 4: Develop a conceptual scheme capable of relating organizational learning, value co-creation, and digital service innovation.
TC.8 - Innovation Focus: Technology, Problems, and Product Strategies	Objective 4: Develop a conceptual scheme capable of relating organizational learning, value co-creation, and digital service innovation.
TC.9 - Implementing Service Innovation: Execution and Impact	Objective 4: Develop a conceptual scheme capable of relating organizational learning, value co-creation, and digital service innovation.

Source: Research Data (2024)

5.1 Integration of Findings with Extant Literature

This section bridges the findings of this study with the existing body of literature, illustrating how the emergent theoretical codes align with, extend, or challenge established theoretical frameworks. By integrating findings with extant theories, we aim to contextualize the contributions of this research and provide a robust foundation for understanding the dynamics of organizational learning, service innovation, and value co-creation.

Table 30 examines each theoretical code against relevant scholarly works to highlight convergences and divergences, fostering a deeper appreciation of the insights gained and their broader implications. The integration underscores the relevance of adopting multifaceted approaches, such as leveraging trust-building, exploring design thinking principles, and balancing innovation with sustainability, to drive impactful and sustainable organizational practices.

Table 30
Theoretical Codes and Extant Literature

Theoretical Code	Extant Literature	Comments
TC.1 - Organizational Learning Across Teams and Client Partnerships	Team Learning (Watzek et al., 2019; Wilhelm et al., 2019); Organizational Learning (Bido et al., 2010; Bontis et al., 2002; Pérez López et al., 2005); Organizational Learning from Errors (Edmondson, 2012; Leoncini, 2016; Tahirsylaj, 2012); Value Co-creation (Desmarchelier et al., 2020)	The authors argued that as an interdependent process, organizational learning requires coordination of the underlined stages, a shared purpose that gives sense and direction to activities. Furthermore, as a social endeavor, learning takes place in the presence of social relations, where emotions and perceptions can foster or stifle learning
TC.2 - Building Collaborative Relationships and Trust for Co-creation	Value co-creation antecedents (Neghina, 2015); Service Experience and innovation (Rubalcaba, 2018)	Relationships and trust are driving forces for effective value co-creation. Embracing vulnerability can foster trust and demonstrate commitment.

Theoretical Code	Extant Literature	Comments
TC.3 - Navigating the Challenges of Value Co-creation	Value co-creation(Prahalad & Ramaswamy, 2003); Managing Value co-creation (Payne et al., 2008)	Internal and external challenges were identified as obstacles to value co-creation. Nevertheless, getting timely and quality feedback, as well as balancing scalability (product) and customization (service) are the prominent themes.
TC.4 - Integrating Client Understanding, Experimentation, and Active Value Co-creation	Design Thinking (T. Brown, 2008; T. Brown & Katz, 2011); Service-Dominant Logic (Lusch & Vargo, 2014; Vargo et al., 2008; Vargo & Lusch, 2008)	The design thinking process is a learning process where you start from understanding the problem to understand how and why the solution (prototype) adds value to the customer you're solving the problem.
TC.5 - Enabling Innovation: Leadership, Culture, and Empowerment	Innovation (Baregheh et al., 2009); Innovation as a Learning process (Lundvall, 2010)	Innovation is a multi-stage process involving distinct social actors and requires leadership and strategic intentions to mobilize individuals, teams, and the organization.
TC.6 - Adapting, Resolving, and Overcoming Barriers to Innovation	Overcoming Barriers to Service Innovation (Manca et al., 2018; Parris et al., 2016; Velayati et al., 2020)	The risk of drifting away from the customer needs during the innovation efforts is a well-known obstacle. However, it is potentially harder when considering that organizations need to sustain and evolve existing offerings.
TC.7 - Building Digital Service Innovation: From Inputs to Tools and Sustainability	Exploration and Exploitation (A. K. Gupta et al., 2013; March, 1991; Popadiuk, 2012); Design Thinking (T. Brown, 2008; T. Brown & Katz, 2011); Team Reflexivity in Innovation (Hoegl & Parboteeah, 2006)	Exploration and exploitation are central concepts in the innovation literature. These concepts describe two distinct approaches to innovation and resource allocation, each crucial in driving a balance between sustaining existing services and innovating. Design Thinking and Team Reflexivity (learning) are key concepts used to reflect on how organizations get inputs for innovation.
TC.8 - Innovation Focus: Technology, Problems, and Product Strategies	Service Innovation dimensions (Hertog, 2010; Kindström & Kowalkowski, 2014); Enabling Service Innovation and value co-creation (Kindström et al., 2013; Rubalcaba, 2018)	Service innovation is a multidimensional phenomenon and cannot be understood through the lens of new technology alone. Framing the "why", "to whom", and better, "should we ever make it?" need to be addressed before gravitating toward a solution/product.
TC.9 - Implementing Service Innovation: Execution and Impact	Exploration and Exploitation (A. K. Gupta et al., 2013; March, 1991; Popadiuk, 2012); Innovation Management (Pertuz & Pérez, 2021)	Implementing service innovation requires a delicate balance of focus, scalability, resources, and prioritization.

Source: Research Data (2024)

The theoretical code "TC.1 - Organizational Learning Across Teams and Client Partnerships" underlines the interplay between team-based learning (internal component) and learning with/from clients (external component) as precursors to organizational learning.

Desmarchelier et al. (2020) highlighted the proximity between concepts of organizational learning and value co-creation. According to these authors, both are models of adaptation to a changing environment, whether motivated by competition or instigated by prospective customers. Besides, services can be seen as the application of competencies between parties through integration and exchange of resources, and, in this context, value is inherently experiential and determined as value-in-use (Vargo et al., 2008).

Additionally, teams are seen as the vehicle for the type of learning that fosters innovation and creates the capabilities necessary for value co-creation (Hoegl & Parboteeah, 2006; Watzek et al., 2019). The theoretical code 1 (TC.1) suggests the type of learning involved by its forming categories: a) Learning from Mistakes (Wilhelm et al., 2019), b) Learning through Experimentation (Argote & Miron-Spektor, 2011), c) Learning from Problems (Argyris & Schon, 1978), and d) Combining Knowledge (Dyer & Singh, 1998; Sambamurthy et al., 2003).

On the other hand, TC.1 points out the learning that happens with/from clients where external knowledge about the business gaps, opportunities, and problems needs to be internalized, integrated, and mutually understood. Learning from external sources aligns with the concept of absorptive capacity, where prior knowledge is a prerequisite for assessing and incorporating new information and transforming ideas into new offers (Zahra & George, 2002).

The theoretical code “TC.2 - Building Collaborative Relationships and Trust for Co-creation” emphasizes trust as a driver for value co-creation. As previously discussed, trust is a well-known antecedent to value co-creation (Neghina, 2015; Rubalcaba, 2018). Nevertheless, the referred authors do not indicate the mechanisms organizations can rely on to build trust. The empirical data, however, revealed that organizational vulnerability can be an important mechanism that builds trust with transparency and consistent commitment. This finding is consistent with Mayer et al. (1995), who defines trust as a willingness to be vulnerable to the actions of others based on perceptions of their ability, benevolence, and integrity.

In the context of service innovation and value co-creation, vulnerability represents a bold and deliberate openness to sharing work-in-progress, acknowledging that the path to excellence often begins with imperfection. It is about fostering an environment where incomplete ideas and early prototypes are not hidden away but are shared transparently with customers to invite genuine, constructive feedback. This willingness to expose the “messy middle” of the creative process establishes trust and encourages collaboration, positioning customers as recipients and co-creators of value.

Drawing on a metaphor, it is like inviting customers into a kitchen, and this approach

goes beyond merely serving the final dish; it offers them a seat at the table where recipes are conceptualized, ingredients are combined, and new flavors are discovered. By engaging them in this process, organizations create better offerings and deeper, more meaningful relationships.

The theoretical code "TC.3 - Navigating the Challenges of Value Co-creation" is related to the work of Payne et al. (2008) and Sheth (2020) on managing internal and external challenges, scaling solutions, and dealing with customization. Nevertheless, the empirical data reinforces the need to integrate client feedback into workflows and equip teams with tools, processes, and values to speed up the reception and treatment of the feedback.

The theoretical code "TC.4: Integrating Client Understanding, Experimentation, and Active Value Co-creation" emphasizes iterative learning processes that are aligned with design thinking and service innovation concepts. Design Thinking (DT) promotes collaborative learning processes where developers and clients start by understanding how and why a solution would add value to stakeholders interested in solving the problem (T. Brown, 2008). Distinct participants highlighted empathy as a key theme. Instead of asking what participants would like, DT helps developers craft experiments (prototypes) that provide rich data and reveal people's emotions and experiences when using the solution (Carlgren et al., 2016).

Besides, the theoretical code also aligns with the Service-Dominant Logic (S-D) that asserts that use-value is co-created by definition and always phenomenologically determined by the customer during the service experience (Vargo & Lusch, 2008). Additionally, the experience should be understood as an empathetic, emotional, and memorable interaction with intrinsic value; therefore, the sooner developers promote and collect data about the experience, the sooner they learn what is important to customers.

On the other hand, "TC.5: Enabling Innovation: Leadership, Culture, and Empowerment" resonance with leadership and cultural dimensions of innovation (Baregheh et al., 2009; Lundvall, 2010). The theoretical code points out the role of leadership in promoting alignment and psychological safety and creating incentives for empowering individuals to experiment, be vulnerable to tolerate honest errors and be committed to organizational values (Martins & Terblanche, 2003). Innovation relies on leadership practices and organizational structures for balancing constraints, time and resources, and leverage collaboration within teams and with customers.

Both "TC.6: Adapting, Resolving, and Overcoming Barriers to Innovation" and "TC.7: Building Digital Service Innovation: From Inputs to Tools and Sustainability" converge on the need for balancing sustainability and innovation. Here, sustainability shall be understood regarding the capacity to maintain and evolve existing services/solutions, ensuring they are

relevant for service providers and customers. The theoretical codes TC.6 and TC.7 reflect the tension between exploration and exploitation, as discussed by March (1991) and A. K. Gupta et al. (2013).

Exploration and exploitation are central concepts in the innovation literature (Andriopoulos & Lewis, 2009; P. P. Li et al., 2023), particularly in studies of organizational learning (March, 1991), innovation strategies (Y. Li et al., 2008), and ambidexterity (Duc et al., 2020). These concepts describe two approaches to innovation and resource allocation, each crucial to organizational success.

According to March (1991), Exploration refers to activities to discover new knowledge, technologies, or processes. It involves experimentation, risk-taking, and the pursuit of novel opportunities. The focus is on developing radical or disruptive innovations, entering new markets, or creating new product categories. It requires investing in research and development (R&D), experimentation, and prototyping.

On the other hand, Exploitation involves refining, improving, and optimizing existing knowledge, processes, or products. It focuses on efficiency, reliability, and incremental improvements (A. K. Gupta et al., 2013; Popadiuk, 2012).

Balancing sustainability and innovation reflects the tension organizations face about resource allocation, as exploration and exploitation compete for limited resources like time, money, and personnel. Excessive focus on exploration can lead to a lack of immediate returns and inefficiency, while overemphasizing exploitation can result in stagnation and vulnerability to disruption (P. P. Li et al., 2023). Organizational Ambidexterity is the capability of balancing exploration and exploitation and achieving superior long-term performance and innovation outcomes.

Finally, “TC.8: Innovation Focus: Technology, Problems, and Product Strategies” and “TC.9: Implementing Service Innovation: Execution and Impact” describe, respectively, the dimensions of service innovation, the priorities and strategies for translating that innovation into efficiency, and competitive advantage.

In TC.8, service innovation is seen as a multidimensional concept (Hertog, 2010; Kindström & Kowalkowski, 2014) where the focus should shift from technology and products as service vehicles to experiences and business needs (Rubalcaba, 2018; Vargo & Lusch, 2008). Drawing from a metaphor, Participant 7 suggested that software development teams should play the role of organizational therapists in diagnosing root causes and underlined inefficiencies and pain points. Additionally, TC.9 surfaces the need for focus and a pragmatic mindset to align actions with tangible results and keep customer-centric goals.

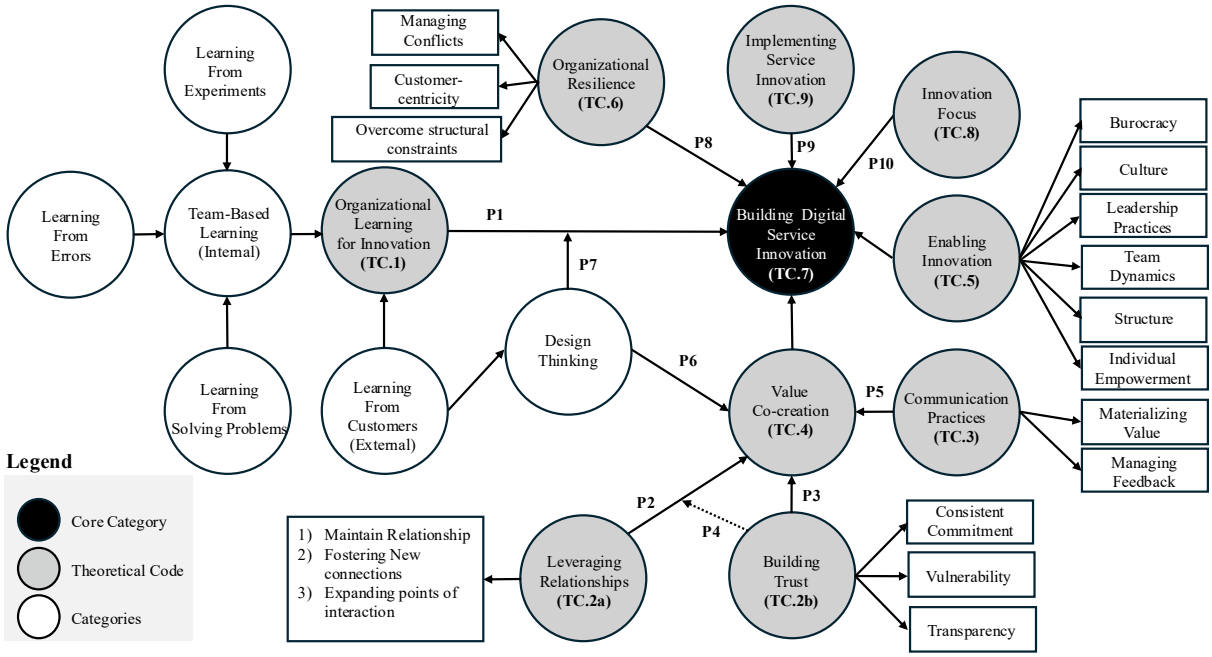
5.2 Theoretical Contributions

This section outlines the theoretical contributions of the study, focusing on how the findings advance existing knowledge and offer novel insights into the examined phenomena. Theoretical contributions stem from integrating empirical data with grounded theory methodology, leading to the development of theoretical codes that reflect the nuanced dynamics of the research context.

These contributions bridge gaps in the literature by addressing underexplored areas, synthesizing relationships between concepts, and providing a coherent framework for understanding innovation processes. The insights generated through this research offer a foundation for academic exploration and practical application, paving the way for future investigations and organizational strategies. Figure 14 shows the conceptual model derived from the empirical data.

Figure 14

Theoretical Model.



Source: From the Author.

5.2.1 Theoretical Propositions

The theoretical code "TC.1 - Organizational Learning Across Teams and Client Partnerships" is renamed to align with the overarching concept of Organizational Learning with the addition of the qualifier "for Innovation" to indicate this is a specific kind of learning. One where social relations, emotions, and perceptions may foster or stifle innovation. Besides, learning from errors, according to the literature, is bound to the emotional context where employees would be more inclined to learn from their failure experiences when medium to high levels of psychological safety are present in the working group (Wilhelm et al., 2019).

Additionally, Vargo et al. (2008) emphasized that value is always co-creation in service systems and is driven by adaptability and knowledge integration facilitated by collaborative exchanges. Learning from errors and experimentation (Edmondson, 1999) fuels iterative cycles of building-measuring-learning so common in startups (Ries, 2011) and, while problem-solving and knowledge combination (Argyris & Schon, 1978) support teams in addressing systemic challenges and co-creating value. These types of learning enhance the knowledge intensity of service systems, fostering innovations that deliver value-in-use.

Also, learning from errors or failures (McGrath, 2011) and collaborative learning with clients facilitate knowledge-intensive exchanges (Ley, 2020), enabling service systems to internalize and apply knowledge to improve adaptability. According to Vargo et al. (2008), value co-creation depends on resource integration and the dynamic application of knowledge in service systems, making these types of learning essential for fostering innovations that align with value-in-use principles. From the above elaboration, we derive:

> ***Proposition 1****: Building Digital Service innovation (TC.7) emerges from organizational learning processes (TC.1) that facilitate knowledge-sharing, collaborative problem-solving, and experimentation, allowing service systems to co-create value by improving their ability to adapt and integrate resources in response to environmental changes.*

The theoretical code "TC.2 - Building Collaborative Relationships and Trust for Co-creation" captures the intersection of value co-creation, relationships, and trust by exploring how organizations leverage collaborative relationships and establish trust as foundational enablers of value co-creation. Nevertheless, to enhance clarity and analytical precision, the code was divided into two distinct codes, "TC.2a" - Building Collaborative Relationships and "TC.2b" - Building Trust for Co-creation. While closely related, these concepts represent

distinct dimensions of the co-creation process. Collaborative relationships focus on the interactions, partnerships, and shared efforts that enable effective teamwork ("CT.5: Leveraging Relationships for Effective Co-creation"), whereas trust for co-creation emphasizes the foundational confidence and mutual understanding required to foster openness and innovation ("CT.23: Building Trust in Collaborative Relationships"). Separating these elements allows a more nuanced exploration of their individual contributions and interplay in enabling service innovation.

Effective value co-creation begins with cultivating and sustaining relationships that foster collaboration and innovation. Neghina (2015) describes value co-creation as planned coordination involving clear goals, defined roles, and an understanding that value creation is interdependent. This description aligns with the study's findings, emphasizing the need for maintaining and strengthening existing partnerships, fostering new connections, and diversifying interaction points as essential practices for effective co-creation. These activities enhance engagement by integrating diverse perspectives, enabling knowledge-sharing, and creating opportunities for resource alignment.

***Proposition 2**: Maintaining and strengthening relationships, fostering new connections, and diversifying interaction points (TC.2a) enable organizations to create collaborative platforms that enhance value co-creation (TC.4) through shared knowledge and resource alignment.*

Besides, trust is the cornerstone of co-creation, built through transparency, vulnerability, and demonstrated commitment. While Neghina (2015) highlights trust as central to value co-creation, the study extends this understanding by identifying specific mechanisms. Transforming vulnerability into trust emerges as a key strategy, where openly sharing limitations or uncertainties fosters authenticity and mutual respect. For instance, presenting "imperfect" products or prototypes accelerates feedback, creating a sense of shared purpose and empowering collaborators to contribute meaningfully.

***Proposition 3**: Building trust (TC.2b) through vulnerability enables organizations to deepen collaborative relationships, fostering a culture that drives effective co-creation and innovative outcomes (TC.4).*

The research data aligns with Rubalcaba (2018) by suggesting trust is essential for

service co-production. However, the study extends this by emphasizing mechanisms such as vulnerability and commitment, which transform relationships into resilient partnerships capable of driving innovation. Balancing transparency with protection also plays a critical role, ensuring that openness does not compromise strategic interests while maintaining credibility. Demonstrating commitment reinforces trust through consistent follow-through on feedback, proving that vulnerability is paired with action. The synergy between relationships and trust ensures that co-creation transcends transactional exchanges, becoming a transformative partnership (Neghina, 2015; Vargo & Lusch, 2008). Vulnerability acts as a bridge, strengthening both relational and trust dimensions.

***Proposition 4**: The interplay between Leveraging Relationships (TC.2a) and the effectiveness of Value Co-creation (TC.4) is moderated by Building Trust (TC.2b).*

Indeed, value co-creation is a dynamic process that requires collaboration between organizations and their clients to generate shared value (Payne et al., 2008). However, this process is fraught with challenges that arise from both internal organizational barriers and client-driven complexities. TC.3, "Navigating the Challenges of Value Co-creation," explores these dual dimensions and highlights the importance of communication, alignment, and adaptability in addressing them.

Internally, organizations face difficulties in integrating feedback, balancing scalability with customization, and managing control over co-creation initiatives. Externally, client behaviors such as an outcome-driven focus, trust deficits, and knowledge asymmetries introduce additional layers of complexity. Together, these challenges require deliberate strategies to foster transparent communication, accelerate feedback cycles, and build trust to create a seamless co-creation process that delivers impactful and innovative solutions.

The analysis of research data emphasizes that effective communication, particularly in managing feedback and aligning expectations, is central to overcoming internal and client-driven co-creation challenges. Feedback delays, as highlighted by Participant 1, hinder the iterative refinement of solutions and disconnect developers from current client needs. Timely feedback loops foster engagement and promote transparent and iterative stakeholder communication, which is aligned with the findings of Avila-Garzon & Bacca-Acosta (2024). Consequently, addressing knowledge asymmetries, through open dialogue and accessible metrics builds trust and credibility, may reduce barriers to collaboration. From the above elaboration, we derive the following proposition:

***Proposition 5**: Communication Practices (TC.3) are critical to overcoming co-creation challenges (TC.4), enabling organizations to build service innovations that align with client needs and foster long-term scalability*

The research data describes the archetype of effective communication practices. They are timely and help to accelerate the feedback cycle; besides, they are transparent in bridging knowledge asymmetries between the software development team and the customer.

Building on the internal and client-driven challenges outlined in TC.3, TC.4 focuses on how organizations actively collaborate with clients to co-define, shape, and scale value. By integrating empathy, feedback, and experimentation into the co-creation process, organizations can foster deeper client engagement and create solutions that are both relevant and adaptable. This approach ensures that co-creation becomes a continuous, iterative process, driving innovation and long-term impact.

Design thinking's multidisciplinary and participatory approach (Kimbell, 2011b) aligns closely with the iterative and collaborative nature of value co-creation. The empathize phase corresponds to understanding client needs (CT.22), while prototyping and testing map to driving value creation through experimentation (CT.24). This framework allows for deeper engagement by shifting the dynamic from "us and them" to "us-with-them" (T. Brown & Katz, 2011), ensuring solutions are client-centered, co-defined, and responsive to feedback. By fostering collaboration early and iteratively refining value propositions, design thinking enhances the co-creation process. Based on that rationale, the following proposition was derived:

***Proposition 6**: Design thinking facilitates value co-creation (TC.4) by providing a structured framework that integrates empathy, feedback loops, and experimentation, enabling organizations to collaboratively define, shape, and scale value in alignment with client needs and contexts.*

Besides, Organizational learning emphasizes acquiring and internalizing knowledge from client interactions, experimentation, and feedback (Argyris & Schon, 1978; Edmondson, 1999). Therefore, Design thinking can act as a bridge between this learning and innovation by operationalizing insights through its iterative phases – particularly prototyping and testing (CT.24). For instance, rapid prototyping allows teams to validate hypotheses (FC.17) and

minimize risks, ensuring that learning feeds directly into the development of innovative, market-relevant services. This approach reduces misalignment and enhances adaptability. Based on that rationale, the following proposition was derived:

> ***Proposition 7****: Design thinking moderates the effect of organizational learning (TC.1) on Building Digital Service innovation (TC.7) by transforming insights gained through learning processes into actionable prototypes and validated solutions, thereby accelerating the cycle from knowledge acquisition to innovation delivery.*

Building on the foundations of co-creation and client engagement, "TC.5: Enabling Innovation: Leadership, Culture, and Empowerment" delves into the internal organizational drivers that enable innovation. It highlights the interplay between leadership practices, organizational culture, team dynamics, structural design, and individual empowerment as critical components for fostering creativity and delivering impactful solutions.

Innovation is known to be deeply influenced by leadership practices that inspire trust, reward creativity, and provide a clear vision (Edmondson, 2018; Martins & Terblanche, 2003). Leaders who prioritize values-based management, tolerate honest mistakes, and promote diversity (CT.15) establish a foundation where teams feel supported to take calculated risks. This aligns with the emphasis on team dynamics (CT.19) as execution engines, where collaboration and inclusion foster the creative synergy needed for innovation.

Moreover, organizational culture (CT.18) acts as a catalyst, embedding a learning mindset and agility within teams, while structures (CT.20) enable the balance of stability and flexibility required for service innovation. Empowered individuals (CT.21) who understand the purpose behind their tasks contribute with creativity and initiative, bridging the gap between strategic intent and practical execution.

Hertog (2010) had already identified culture as a key dimension in their framework for service innovation, while Vargo & Lusch (2004) had already emphasized the role of knowledge and technology in support of service-oriented innovation. Hence, customer involvement is critical and intertwined with the definition of services (Prahalad & Ramaswamy, 2004; Vargo & Lusch, 2008), as well as a human-centric design approach (T. Brown, 2008).

This study, however, brings a new perspective by examining how organizational structure, bureaucracy, team dynamics, and individual empowerment collectively act as critical enables of service innovation, highlighting their interdependencies and uncovering mechanisms through which they foster or hinder innovation within service contexts.

By aligning these elements, leadership not only drives innovation internally but also enhances the organization's ability to respond dynamically to client needs and market opportunities, translating innovative practices into impactful and scalable service solutions. Based on that rationale, the service innovation enablers (TC.5) are bureaucracy, culture, leadership practices, team's dynamics, organizational structure, and individual empowerment.

Next, "TC.6: Adapting, Resolving, and Overcoming Barriers to Innovation" examines the critical role of organizational resilience in navigating barriers to sustained innovation. Resilience is characterized by an organization's capacity to adapt to challenges, overcome resource constraints, and resolve conflicts while focusing on long-term goals (Duchek, 2020). By fostering adaptability, managing structural inefficiencies, and leveraging strong leadership, resilient organizations create the conditions necessary for continuous service innovation. This theoretical code emphasizes the interplay of customer-centricity, structural optimization, and conflict resolution as essential components for achieving transformative and sustainable outcomes in dynamic environments.

Organizational resilience is a critical capability that allows firms to navigate uncertainty and complexity while maintaining a focus on innovation (Madani & Parast, 2023). It involves adapting to change (CT.22), overcoming resource constraints (CT.24), and resolving conflicts through strong leadership (CT.25). This multidimensional concept reflects an organization's ability to anticipate disruptions, reconfigure processes, and maintain alignment with customer needs and strategic objectives (Duchek, 2020).

In the context of service innovation, resilience enables organizations to sustain creativity and collaboration even under pressure. For example, overcoming resistance to change (FC.56) and managing time and resource constraints (FC.57) ensures teams remain agile and focused on delivering value. Accordingly, fostering open communication and collaborative conflict resolution creates an environment where diverse perspectives contribute to innovation rather than becoming sources of friction. Based on that rationale, the following proposition was derived:

***Proposition 8**: Organizational resilience (TC.6), characterized by adaptability, resource optimization, and effective conflict management, fosters building digital service innovation (TC.7) by enabling organizations to sustain creativity and responsiveness in the face of constraints and challenges.*

Innovation begins with identifying meaningful inputs and evolves into sustainable

practices that integrate creativity into the organizational fabric (Rösch et al., 2023). “CT.7 - Identifying Innovation Inputs” explores the process of Building Digital Service Innovation, focusing on how organizations source ideas from market gaps, pain points, and reflective practices. It emphasizes balancing innovation with operational sustainability and leveraging tools and rituals to make creativity systematic and impactful. By rooting innovation in user needs and fostering iterative development, organizations ensure their solutions remain relevant and adaptable over time.

Identifying gaps in the market (FC.61) ensures that efforts are strategically directed at unmet needs while addressing customer pain points (FC.62) highlights immediate challenges requiring resolution. Reflective practices within teams (FC.63) further refine innovation efforts by uncovering inefficiencies and aligning solutions with organizational values (Hoegl & Parboteeah, 2006). Based on that rationale, the following proposition was derived:

Proposition 9*: The search for innovation inputs, driven by market gaps, customer pain points, and reflective team practices (TC.9), lays the foundation for building digital service innovation (TC.7) by aligning creative efforts with user-centric needs and organizational goals.*

By rooting innovation in inputs that reflect both external opportunities and internal insights, organizations set the stage for creative processes that yield sustainable, user-centered service innovations.

Finally, Innovation strategies are shaped by three interrelated dimensions: technology, problems, and product strategies. “TC.8: Innovation Focus: Technology, Problems, and Product Strategies” explores how organizations leverage technological capabilities, address pressing problems, and develop product-centric solutions to align with user needs and market demands. This multifaceted approach underscores the importance of balancing these dimensions to create sustainable, impactful innovations.

Service innovation thrives when organizations focus on identifying and solving core problems rather than starting with technology or products. Acting as “organizational therapists” (FC.75), teams uncover root causes through empathy and iterative dialogue, ensuring solutions are deeply connected to user needs (Rubalcaba, 2018). This approach prioritizes value co-creation and aligns with service-dominant logic by emphasizing user-centric problem-solving over tool-centric fixes.

Technological advancements (FC.73) provide the necessary infrastructure for delivering

these solutions, enabling scalability, automation, and operational efficiency. However, as Kindström & Kowalkowski (2014) argued, technology should serve as an enabler rather than the focal point of innovation. Similarly, product diversification (FC.81) ensures that organizations can adapt to changing market demands, enhancing their service offerings without losing sight of core problem-solving objectives. Based on that rationale, the following proposition was derived:

> ***Proposition 10****: Focusing solely on technology, problems, or product strategies in isolation can hinder service innovation (TC.8); building digital service innovation (TC.7) requires a balanced integration of these dimensions to align solutions with user needs, operational realities, and market opportunities.*

Integrating all three dimensions (technology, product, problems) allows organizations to harness each other's strengths. Problem-oriented approaches ensure user-centricity and value co-creation(Rubalcaba, 2018)while technological innovation enhances scalability and efficiency (Hertog, 2010). Conversely, product strategies adapt solutions to evolving market demands and ensure long-term relevance (Kindström & Kowalkowski, 2014). By balancing these dimensions, organizations create service innovations that are not only impactful and efficient but also aligned with user expectations and strategic goals.

5.2.2 Substantive Theory

The process **"Building Digital Service Innovation"** emerges as the central phenomenon through which software development teams integrate organizational learning, trust-building, value co-creation, and adaptive resilience to create, refine, and sustain digital service innovations. This core category encapsulates the iterative and collaborative activities that transform knowledge, relationships, and experimentation into impactful and sustainable solutions. The grounded theory demonstrates how the identified categories interact dynamically to enable and sustain Building Digital Service Innovation in software development teams.

At its core, Building Digital Service Innovation represents the culmination of efforts to transform diverse inputs – such as trust, learning, and customer insights – into meaningful outcomes. It is a dynamic and iterative process that balances competing demands (exploitation x exploration), integrates resources and adapts to constraints. Through this process, teams may generate impactful and user-centered solutions while maintaining alignment with organizational goals and client expectations.

Organizational Learning Across Teams and Client Partnerships emerges as a key enabler of Building Digital Service Innovation by providing the foundation for adaptability and creativity. Learning processes like experimentation, problem-solving, and collaboration allow teams to acquire new insights, refine practices, and address systemic challenges. These learning mechanisms enhance responsiveness to client needs while fostering the innovative capabilities required to sustain competitive advantage. Additionally, learning from errors, particularly when supported by psychological safety, empowers teams to iterate on feedback and generate more effective solutions.

Trust-building mechanisms also play a pivotal role in enabling Building Digital Service Innovation. The findings underscore the importance of Building Collaborative Relationships and Trust for Co-Creation, where trust acts as a bridge between internal teams and external stakeholders. Transparency, vulnerability, and demonstrated commitment are critical for fostering authenticity and mutual respect. For instance, openly sharing early-stage prototypes or acknowledging uncertainties encourages feedback and accelerates the iterative refinement of ideas. Trust transforms transactional relationships into resilient partnerships, creating the psychological safety needed for experimentation and collaboration to flourish.

The process of Value Co-Creation further amplifies Building Digital Service Innovation by aligning innovation efforts with client-centered goals. Engaging stakeholders as active participants ensures that solutions resonate with real-world needs and contexts. The co-creation

process, supported by empathy and iterative feedback, bridges the gap between ideation and execution, allowing teams to deliver solutions that are both relevant and impactful. Value co-creation shifts the innovation dynamic from a unilateral effort to a collaborative journey, enhancing the depth and applicability of innovation outcomes.

Building Digital Service Innovation is sustained through Adaptation, Resilience, and Overcoming Barriers, which equip teams to navigate resource constraints, operational pressures, and structural inefficiencies. Organizational resilience, characterized by flexibility, resource optimization, and conflict management, ensures that innovation processes remain focused even in the face of uncertainty. The ability to adapt to challenges and balance operational stability with the need for exploration is essential for maintaining creative momentum. Resilience enables teams to sustain Building Digital Service Innovation while addressing the complexities of dynamic environments.

The strategic focus on Building Digital Service Innovation is further reinforced by the integration of diverse priorities. The findings reveal that innovation strategies are shaped by three interrelated dimensions: technology, problems, and product strategies. This multidimensional focus ensures that teams do not rely solely on technology as the primary driver of innovation but instead prioritize problem-solving and client needs. By uncovering root causes and addressing user pain points, teams align their efforts with meaningful outcomes. At the same time, diversifying product strategies ensures that innovation efforts remain scalable and adaptable to evolving market demands.

The iterative nature of Building Digital Service Innovation is central to its effectiveness. Feedback loops, cultivated through trust-based collaborations and co-creation efforts, inform future learning and adaptation. For example, learning from user interactions enhances subsequent experimentation and ideation, while trust-building deepens collaborative relationships, enabling more effective co-creation. These reinforcing mechanisms create a continuous cycle of learning, adaptation, and value generation, ensuring that Building Digital Service Innovation remains responsive and sustainable.

In sum, Building Digital Service Innovation emerges as the central phenomenon in service innovation, integrating organizational learning, trust-building, value co-creation, and resilience into a dynamic, iterative process. It balances the interplay of strategic focus, adaptive capabilities, and collaborative relationships to create sustainable, user-centered solutions that align with organizational and market needs.

This substantive theory positions Building Digital Service Innovation as the primary framework for understanding how software development teams navigate the complexities of

digital service innovation. Emphasizing the integration of diverse enabling factors provides a robust and comprehensive lens through which organizations can approach innovation processes, offering both, theoretical insights and practical guidance, for fostering sustainable and impactful outcomes.

Therefore, with this study, we provided a comprehensive framework that deepens the understanding of service innovation by integrating diverse theoretical perspectives with grounded empirical insights. The theoretical contributions span organizational learning, trust-building, collaborative relationships, communication practices, leadership, and organizational resilience. These dimensions converge to highlight how innovation is dynamically shaped by both internal enablers and external client-driven complexities.

By connecting these dimensions into a cohesive framework, the study bridges theoretical gaps and provides a robust foundation for mic exploration and practical application, equipping organizations with strategies to navigate the complexities of service innovation in dynamic environments.

5.3 Practical Implications

This section focuses on the practical implications of the research findings, emphasizing their relevance to organizational leaders, practitioners, and stakeholders. By bridging the gap between theory and practice, the study offers actionable insights that can inform decision-making processes, drive innovation, and enhance organizational performance.

A central recommendation is the adoption of **design thinking** as a structured, user-centered approach to innovation. By embedding empathy, iterative feedback, and experimentation into their processes, organizations can co-create solutions that align closely with client needs and operational realities. This participatory approach shifts the relationship between stakeholders from transactional to collaborative, ensuring innovations are relevant and sustainable. Design thinking also offers a systematic way to bridge gaps between theoretical learning and practical execution, enabling organizations to operationalize insights effectively.

Another critical implication lies in the role of **building trust through vulnerability**. Trust is foundational to successful collaboration and value co-creation. Organizations can foster deeper relationships with clients and partners by embracing openness, such as sharing early prototypes, acknowledging uncertainties, and inviting constructive feedback. This practice accelerates iterative refinement while empowering collaborators to engage authentically. By

transforming vulnerability into a strength, organizations build long-lasting partnerships that support sustained innovation.

The study also highlights the importance of **experimentation as a driver of innovation**. Agile methodologies, such as rapid prototyping and iterative testing, allow organizations to validate hypotheses, refine solutions, and reduce risks. This approach encourages a culture of adaptability, where teams are motivated to "fail fast and learn smart." By systematically integrating experimentation into their processes, organizations can balance creativity with practicality, ensuring that innovations are both impactful and feasible.

Additionally, **organizational resilience** emerges as a crucial enabler of sustained innovation. Organizations must foster adaptability, optimize resource allocation, and address conflicts proactively to maintain creative momentum. This resilience equips teams to navigate resource constraints, structural challenges, and shifting market dynamics without losing focus on innovation goals. By embedding resilience into their practices, organizations can sustain responsiveness and creativity even in the face of uncertainty.

Finally, this study underscores the need for organizations to **align innovation strategies with client-centric goals**. Rather than overemphasizing technology or products, organizations should prioritize problem-first approaches that address customer-specific pain points and market gaps. By rooting innovation in user needs and reflective practices, organizations ensure that their solutions remain relevant and adaptable to evolving market demands. This balance allows organizations to sustain their competitive edge while fostering long-term value co-creation.

These implications collectively provide organizations with a robust framework to navigate the complexities of service innovation, leveraging design thinking, trust-building, experimentation, and resilience to deliver solutions that are adaptive, impactful, and aligned with user-centric goals.

5.4 Critical Reflections

This study provides valuable insights into the dynamics of service innovation, offering a robust framework that integrates organizational learning, trust-building, and co-creation processes in software development teams. However, like any research endeavor, it is not without its limitations. A critical reflection on the methodology and scope highlights areas where the study's breadth and execution may have constrained its depth and theoretical saturation.

One significant limitation of the study is that **theoretical saturation was not fully achieved**. Despite rigorous data collection and analysis, the breadth of the research theme likely contributed to the emergence of new ideas and contexts with each additional interview. The diversity of insights underscores the richness of the phenomenon under study but also suggests that the broad scope may have hindered the ability to exhaustively capture recurring patterns or narrow the focus on specific dimensions of service innovation. Future research could address this limitation by refining the scope of inquiry to focus on narrower aspects of the phenomenon, enabling a more in-depth exploration.

Another limitation stems from the **diversity of organizations represented in the sample**, including startups, traditional companies, and software houses. While this variety adds richness and comparative value to the findings, it also means that the perspectives offered by participants may reflect unique organizational contexts that are not fully generalizable. These different organizations approach innovation from distinct angles, shaped by their resources, cultures, and operational constraints. Although this diversity broadens the applicability of the findings, it also complicates efforts to distill universal insights. Future research could consider stratifying the sample to focus on specific organizational types, offering a more tailored analysis of service innovation dynamics within each context.

Additionally, this study acknowledges the **potential for biases in data collection and interpretation**. Participants' responses may reflect personal experiences, organizational priorities, or contextual constraints that influence their views on service innovation. Researcher bias, while mitigated through a rigorous application of grounded theory methods, may also shape how data was interpreted and theoretical codes were constructed. Triangulating these findings with additional data sources or involving multiple researchers in coding and analysis could reduce these biases and enhance the validity of the conclusions.

Furthermore, the **context and conditions under which the theory applies** warrant careful consideration. The findings are primarily derived from organizations operating in different environments, ranging from startups to software houses and more traditional or resource-constrained settings. While the theoretical framework provides a broad understanding of service innovation processes, its relevance may vary depending on factors such as industry, organizational size, and market maturity. Future studies could explore how these variables influence the theory's applicability, offering a more nuanced understanding of its boundaries.

Finally, an additional opportunity lies in **engaging participants in a reflective feedback loop**. Conducting follow-up interviews or workshops to share and discuss findings with participants could enhance the study's rigor and practical relevance. This approach would

not only validate the interpretations derived from the data but also provide participants with an opportunity to refine and contextualize the insights further. Collaborative reflection with participants could also uncover nuances or contradictions that deepen the understanding of the phenomenon.

In conclusion, while this study advances the understanding of service innovation and its enabling factors, these limitations highlight avenues for future exploration. Narrowing the thematic scope, stratifying organizational contexts, addressing potential biases, and engaging participants in reflective dialogue would strengthen the findings' robustness and applicability, ensuring that the complexities of service innovation are comprehensively addressed and understood.

5.5 Future Research Directions

This section highlights potential directions for future research based on the findings and limitations of this study. By identifying gaps and underexplored areas, it provides a roadmap for advancing knowledge in the field and deepening our understanding of the processes and phenomena uncovered. First, a more focused investigation into specific aspects of service innovation—such as trust-building mechanisms or the interplay of organizational learning and design thinking—could provide deeper insights and achieve greater theoretical saturation. Narrowing the scope would allow for a more detailed exploration of recurring patterns and dynamics.

Second, stratified research focusing on specific organizational contexts, such as startups, traditional firms, or software houses, could uncover how different organizational types approach service innovation. This would enable a comparative analysis to identify context-specific strategies and challenges, enhancing the generalizability of findings.

Third, future studies could adopt a longitudinal design to track the evolution of service innovation practices over time. Such an approach would provide richer insights into how organizations adapt to changing environments and sustain innovation efforts, particularly in response to external pressures and resource constraints.

Additionally, engaging participants in reflective feedback loops—through follow-up interviews or collaborative workshops—would enhance the validity and practical relevance of findings. This iterative process could refine theoretical frameworks and uncover nuanced perspectives that were not initially apparent.

Lastly, examining the boundary conditions of the proposed framework by testing its

applicability across industries, cultural contexts, and organizational sizes could provide a clearer understanding of its scope. This would help to delineate the factors that influence the effectiveness of service innovation strategies and expand the framework's relevance to diverse organizational settings. These directions offer a roadmap for advancing the study of service innovation, bridging theoretical gaps, and enhancing the practical application of its insights.

5.6 Revisiting the Research Question

This study set out to explore the research problem: *How do software development teams experience organizational learning and value co-creation in the context of digital service innovation?* Guided by this inquiry, the study sought to understand the nuanced dynamics of organizational learning and value co-creation within the innovation processes of software development teams. Through the application of grounded theory methodology, this research has generated a conceptual framework that illuminates the key elements, relationships, and mechanisms underlying these phenomena, offering theoretical contributions and practical implications.

Grounded theory provided a robust methodological foundation for addressing the research problem, enabling the iterative development of theoretical codes directly derived from empirical data. By engaging with participants from diverse organizational contexts—ranging from startups to traditional companies—this study captured a broad spectrum of perspectives and experiences. The resulting theoretical framework reveals the interplay between organizational learning, value co-creation, and digital service innovation, highlighting key processes such as building trust through vulnerability, leveraging design thinking, and integrating experimentation into innovation practices.

The findings advance our understanding of how software development teams learn and co-create value. For instance, organizational learning was found to be deeply intertwined with processes such as problem-solving, knowledge-sharing, and iterative experimentation, all of which are critical for fostering adaptability and responsiveness in service systems. Similarly, value co-creation emerged as a collaborative process, and it is facilitated by mechanisms like trust-building, transparent communication, and empathy-driven engagement. These insights offer explanations for the behaviors of development team members in innovation contexts, shedding light on how they navigate challenges and leverage resources to deliver impactful digital services.

The practical implications underscore the study's significance, emphasizing actionable

strategies for organizations. For example, adopting design thinking enables teams to align innovation efforts with client needs while fostering a culture of trust and experimentation, accelerating learning and collaboration. Additionally, the research highlights the importance of balancing operational stability with innovation agility, ensuring that teams can sustain creativity while meeting ongoing demands.

In revisiting the research problem, it becomes clear that the study has achieved its stated objectives. It has identified the key elements of organizational learning and value co-creation, explained their relationships within digital service innovation, and provided a conceptual scheme to guide further research and practice. By addressing a complex and multifaceted problem, this work contributes to the growing body of knowledge on digital service innovation, offering both academic insights and practical guidance for software development teams and the organizations that support them. The findings underscore the importance of fostering collaborative, adaptive, and user-centered approaches to innovation, ensuring that development teams are equipped to thrive in dynamic and evolving environments.

6. FINAL REMARKS

With this study, we focused on exploring how software development teams experience organizational learning and value co-creation within the context of digital service innovation. By employing grounded theory methodology, the research uncovered a substantive theory of Building Digital Service Innovation as a central process through which organizations navigate the complexities of modern service ecosystems. This conceptualization integrates the dynamics of organizational learning, value co-creation, and the role of enabling factors such as trust, experimentation, and resilience in fostering innovation.

Our findings revealed that Building Digital Service Innovation is a dynamic and multifaceted process driven by a convergence of organizational learning practices, collaborative relationships, and problem-solving. Organizational learning emerged as a cornerstone of innovation, characterized by experimentation, knowledge-sharing, and reflection. These processes enable teams to adapt to shifting conditions, address systemic challenges, and align their efforts with the needs of clients and stakeholders. Within this framework, trust – particularly when built through openness and vulnerability – was found to be a critical enabler. Transforming vulnerability into trust allowed teams to foster collaboration, accelerate feedback loops, and establish resilient partnerships that drive innovation forward.

The study also illuminated the challenges faced by software development teams in balancing the demands of operational stability with the need for innovation. Teams often operate under constraints such as limited resources, misaligned communication, and the pressure to meet client expectations. Despite these challenges, key enabling factors—strong leadership, adaptive cultures, and empowered teams—emerged as essential drivers for overcoming these barriers. These enablers not only mitigate the impact of constraints but also provide the foundation for Building Digital Service Innovation as an iterative, sustainable process. The adoption of strategies such as leveraging design thinking, embedding experimentation into workflows, and fostering organizational resilience were shown to be critical in supporting this process.

By positioning Building Digital Service Innovation as the core category, we advance a cohesive theoretical framework that integrates disparate elements of service innovation. This core process bridges the gaps between learning, trust, and collaborative engagement, highlighting how organizations iteratively develop solutions that are both user-centric and adaptive to market demands. The substantive theory emphasizes that innovation is not merely a product of technological advancements or problem-solving; rather, it is a holistic and iterative

process that synthesizes insights, aligns strategies and engages multiple actors in co-creation.

Grounded theory methodology proved invaluable in capturing the complexity of this phenomenon. Its inductive approach allowed the study to move beyond preconceived notions and instead develop theory directly from the rich, lived experiences of participants. By emphasizing iterative analysis and constant comparison, the study revealed the interconnectedness of theoretical codes, culminating in the identification of Building Digital Service Innovation as the core process underpinning successful service innovation. This methodology also facilitated the articulation of practical insights that resonate with real-world challenges faced by software development teams.

The theoretical contributions of this study are significant. By proposing Building Digital Service Innovation as a substantive theory, the research extends existing knowledge on service innovation, organizational learning, and value co-creation. It offers a novel lens through which to understand how teams navigate the complexities of innovation in dynamic environments, synthesizing insights from trust-building, design thinking, and resilience into a coherent framework. The study also bridges gaps in the literature by emphasizing the iterative and human-centered nature of innovation processes, positioning software development teams as pivotal agents of digital transformation.

For practitioners, the findings provide actionable strategies to foster innovation. These include embedding design thinking as a structured, user-centered approach, leveraging trust as a foundation for collaboration, and institutionalizing experimentation to drive adaptability. The concept of Building Digital Service Innovation underscores the importance of aligning organizational goals with client needs, ensuring that innovation remains relevant, sustainable, and impactful. By adopting these practices, organizations can navigate constraints while fostering a culture of resilience and creativity.

Reflecting on the research process, it is evident that while the study achieved significant depth, the breadth of the theme prevented full theoretical saturation. Each additional interview added new perspectives, underscoring the richness of the phenomenon but also suggesting that narrower thematic scopes might yield more focused insights in future studies. Moreover, the diversity of organizations in the sample enriched the findings but also introduced variability, limiting generalizability. Future research could stratify organizational contexts or employ longitudinal designs to explore how Building Digital Service Innovation evolves over time and across different environments.

In conclusion, this study highlights that fostering innovation in software development teams requires more than technical expertise – it demands a holistic approach that integrates

learning, trust, and co-creation into the fabric of organizational practices. The substantive theory of Building Digital Service Innovation provides a robust framework for understanding how organizations navigate service innovation's complexities, offering theoretical insights and practical strategies. As the field evolves, future research will undoubtedly refine and expand upon these findings, further illuminating the pathways to meaningful and sustainable innovation.

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APPENDIX A – ANCHORAGE MATRIX

Research Question: How do software development teams experience organizational learning and value co-creation in the context of digital service innovation?

Research Objectives		Theoretical References	Research Topics	Data Collection technique	Data Analysis technique
General	**Specifics**				
Understand the perceptions and experiences of members of software development teams when they participate in innovation processes with value co-creation.	• Understand the key elements of the phenomenon of organizational learning and value co-creation in software development teams.	**Service Innovation** Hipp and Grupp (2005); Lusch and Nambisan (2015); Vargo et al. (2015); Kowalkowski and Witell (2020); Helkkula et al. (2018). **Value Co-creation** Karpen et al. (2012); Prahalad and Ramaswamy (2003); Akbar and Hoffmann, (2020); Patrício et al. (2018); Desmarchelier et al. (2020). Organizational Learning Weiss (1990); Takahashi (2015); Putz et al. (2013)	**A) Identify key categories** • What is your view about service innovation in your organization and how does it affect your work? • How do you perceive the participation of end customers in the process of value creation? • Could you describe a typical day for you when you and your team had to focus on innovation? How different it is from regular working days? • What is your experience when dealing with conflicting opinions during teamwork?	Intensive Interview (Charmaz, 2014)	Constructivist Grounded Theory (Charmaz, 2014).
	• Understand the relationships of these elements in the context of innovation in digital services.	Desmarchelier et al. (2020); Hackman (2002); Decuyper et al. (2010); Patrício et al. (2018); Wang (2021); Helkkula et al. (2018).	**B) Relation of categories** • How would you describe the role learning plays in the way you work? • Are you concerned with how your customers perceive the work you do/create?	Intensive Interview (Charmaz, 2014)	Constructivist Grounded Theory (Charmaz, 2014).

	• Offer explanations for the behavior of individuals in development teams working in innovation processes with co-creation of value.	S. G. Cohen and Bailey (1997); Hoegl and Parboteeah, 2006); Decuyper et al. (2010); Koc (2007); Dixon (2017); Sicotte and Langley (2000).	**C) Behaviors in Software Development Teams** • Tell me about your experience as a software developer. • What helps you to deal with the challenge of innovation? What problems do you encounter? • Could you tell me what you think is the most important aspect of your work as a team when you consider the company's objective?	Intensive Interview (Charmaz, 2014)	Constructivist Grounded Theory (Charmaz, 2014).
	• Develop a conceptual scheme capable of relating organizational learning, value co-creation, and digital service innovation.	Desmarchelier et al. (2020); Hackman (2002); Decuyper et al. (2010);	**D) Substantive Theory** • Could I ask you to describe the most important lessons you learned through the experience of creating value for your organization or customers?	Intensive Interview (Charmaz, 2014)	Constructivist Grounded Theory (Charmaz, 2014).

APPENDIX B – INTERVIEW GUIDE

Following the initial interview, the guide was revised to foster a more open and exploratory dialogue. The researcher recognized that adhering strictly to pre-established concepts guiding the study risked introducing bias by steering responses toward existing knowledge. Instead, the updated approach aimed to create space for the emergence of novel themes and topics of interest from the participant's perspective. The revised interview guide is presented below:

TITLE: Organizational learning and value co-creation in the context of digital service innovation: a study in software development teams

The present interview questionnaire is part of research for the Graduate Program in Business Administration at Mackenzie Presbyterian University and led by the student José Antônio Afonso Nonato, under the supervision of professor Dr. Gilberto Perez. Your contribution to this study is appreciated, and please feel free to respond to the questions in your own words and to ask for clarification at any point in time.

Study Objective

This research aims to understand the perceptions and experiences of members of software development teams when participating in innovation processes with value co-creation.

Participant Information

Organization

1) Name:
2) Sector/Industry:
3) Annual Revenue:
4) Employees (Total including third parties):

Participant

5) Name:
6) Department:
7) Role:
8) Education Level:

9) The number of members on the team:

The following driving questions will be used to collect initial information about the phenomena. They aim to uncover: 1) conditions (reasons why); 2) actions and interactions (responses to); 3) consequences or outcomes (anticipated or actual outcomes).

Open Questions

1) Can you briefly describe your company and the main product/market you serve?
2) How do you perceive the involvement of end customers in the value-creation process? Tell me about your experience. What are the existing barriers and benefits?
3) How does your group deal with the need to innovate and sustain what exists?
4) Who is usually involved in the conception of ideas and definitions?
5) How do you obtain customer feedback on the work you do/create?
6) What are the challenges for innovation in your work group?
7) What were the most important lessons you learned about creating value for your organization or customers?
8) What advice would you give to other software development teams that aim to improve their innovation and value co-creation processes?

Follow-up questions:

10) Can you tell me more about this?
11) Can you provide an example?
12) How did you feel about this experience?
13) Could you tell me how you define it, so I can have it in your words?
14) What was the result of this approach?

APPENDIX C – E-MAIL INVITATION FOR INTERVIEW

Hello, everyone! 🌟

We are looking for amazing companies to participate in a study! We want to talk to developers, leaders, and managers of software teams about their experiences in innovation and value co-creation in digital services.

With just a 30-minute interview, your insights can make a difference in my thesis project. As a way of thanking you, you will receive personalized feedback with the result that can offer new insights for your company.

If you are interested in being part of this enriching exchange, please contact me or schedule your interview directly at https://calendly.com/jose-nonato/entrevista-tese-inovacao-e-co-criacao-de-valor !

I thank you and I look forward to hearing your stories and experiences. 🚀

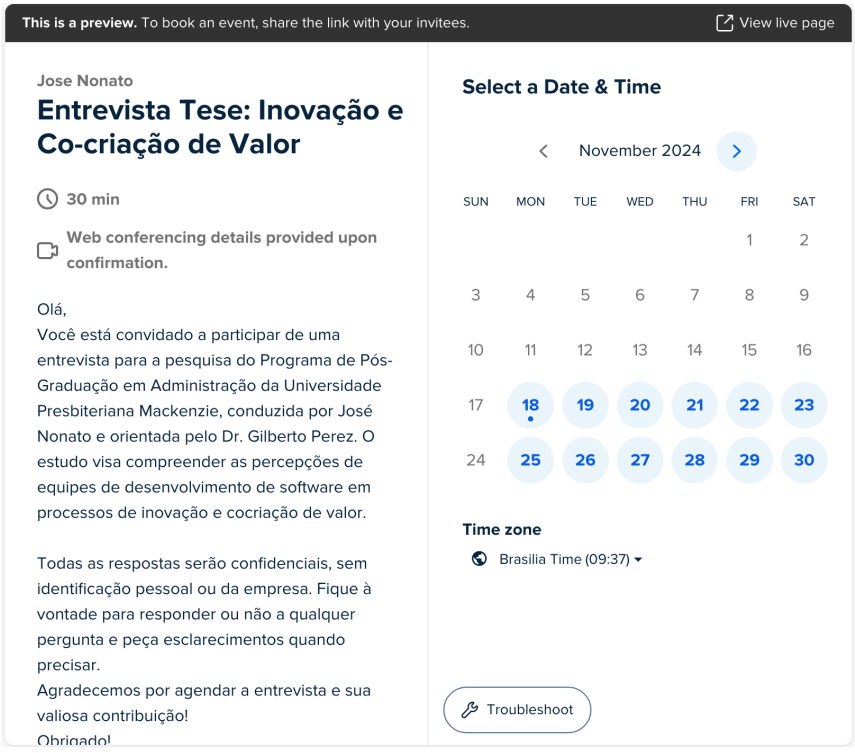
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APPENDIX D – ONLINE APPOINTMENT SCHEDULE TOOL

The process of scheduling interviews often proves cumbersome, requiring numerous interactions between the researcher and the participants to identify a mutually convenient time. To streamline this process, an online scheduling platform was employed. Calendly, a well-known scheduling tool offering a 14-day free trial was selected for its ease of use and functionality.

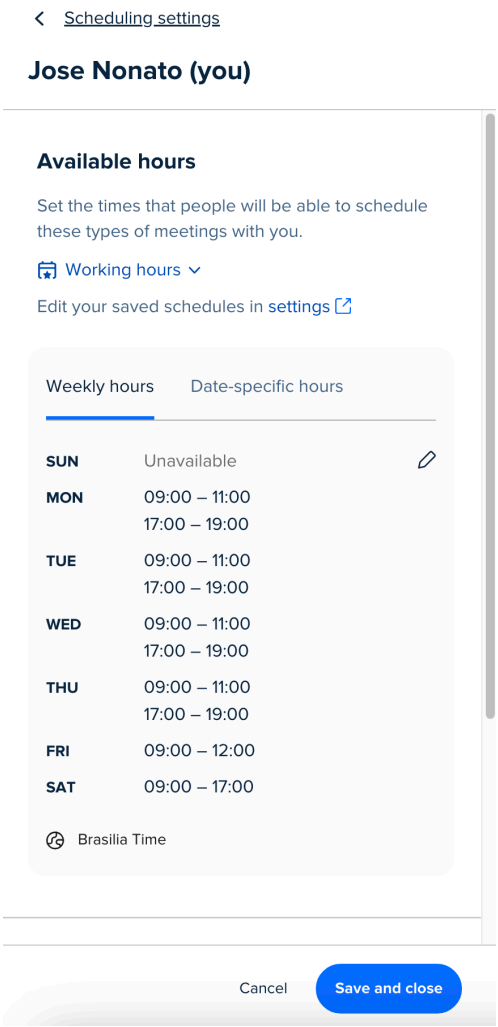
Before adopting this tool, the scheduling process was fraught with challenges, including frequent rescheduling, no-shows, and prolonged lead times between initial contact and the actual interview. Over six months, only six interviews were successfully conducted. However, with the integration of personalized emails generated using ChatGPT and the implementation of Calendly, the remaining six interviews were scheduled and conducted within just 14 days.

The accompanying image provides an overview of the scheduling webpage presented to participants. On the left side, a concise description outlines the purpose of the interview. On the right, participants can choose among the available dates and times. Before sending the invitations, the researcher predefined the time slots and dates during which he was available, allowing participants to select an option that best suited their schedule.

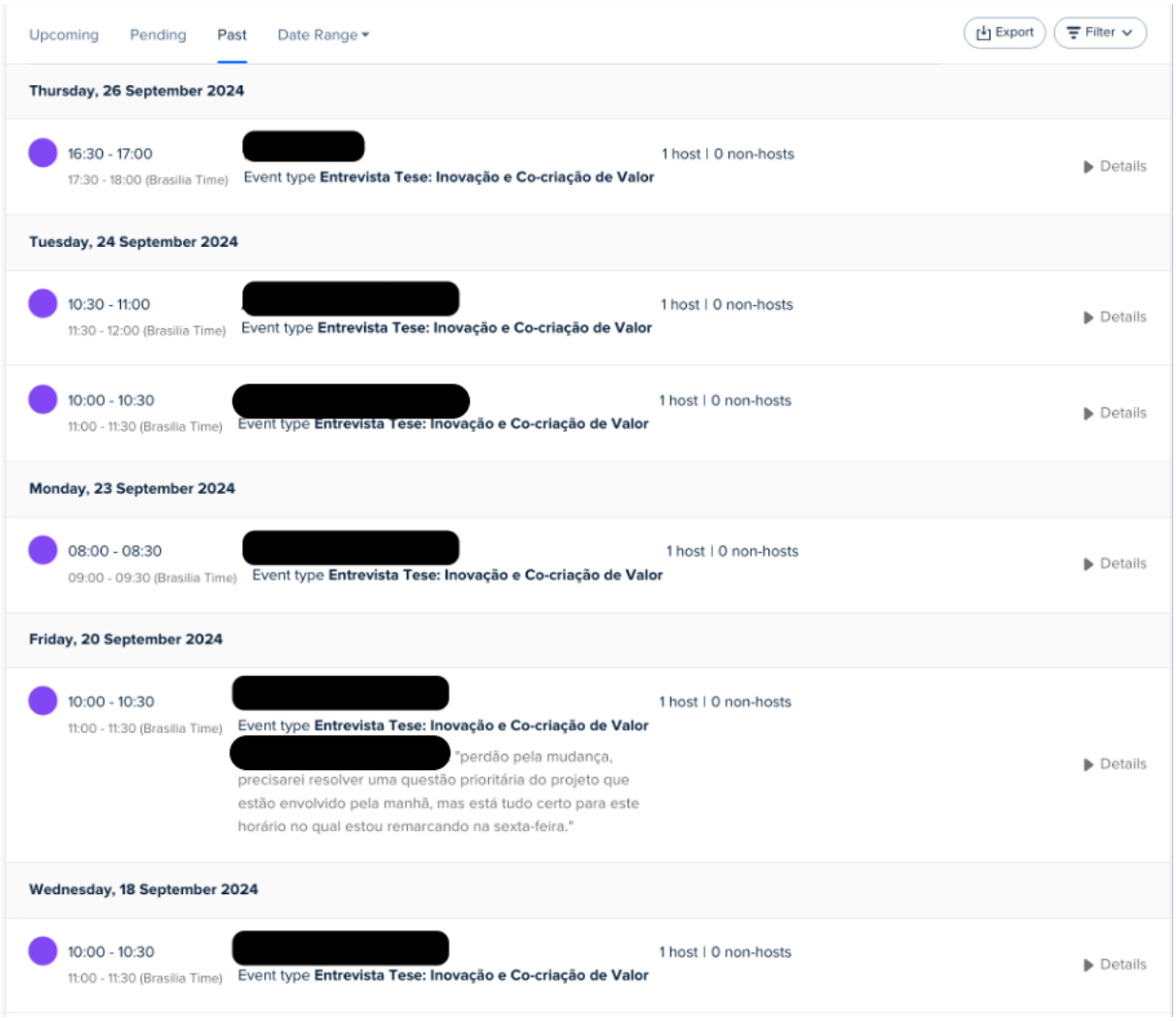


The screen below shows the configuration of available timeslots for booking interviews.

Whenever a new scheduling was done, I was notified, and a meeting appointment was automatically created on the configured calendar.



Rescheduling was simplified, as participants could modify their bookings directly via the scheduling webpage without needing apologies or additional interactions through email or WhatsApp. As the researcher and organizer of the interviews, I could view upcoming appointments and manage them effortlessly. Additionally, participant information, such as email addresses and phone numbers, was collected during the scheduling process through Calendly's customizable settings, ensuring easy follow-up if needed. The screenshot below illustrates a completed booking.


Upcoming
Pending
Past
Date Range
Export
Filter
Thursday, 26 September 2024
16:30 - 17:00
17:30 - 18:00 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
Details
Tuesday, 24 September 2024
10:30 - 11:00
11:30 - 12:00 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
Details
10:00 - 10:30
11:00 - 11:30 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
Details
Monday, 23 September 2024
08:00 - 08:30
09:00 - 09:30 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
Details
Friday, 20 September 2024
10:00 - 10:30
11:00 - 11:30 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
[redacted]
"perdão pela mudança, precisarei resolver uma questão prioritária do projeto que estão envolvido pela manhã, mas está tudo certo para este horário no qual estou remarcando na sexta-feira."
Details
Wednesday, 18 September 2024
10:00 - 10:30
11:00 - 11:30 (Brasilia Time)
[redacted]
Event type Entrevista Tese: Inovação e Co-criação de Valor
1 host | 0 non-hosts
Details

APPENDIX E – DETAILED CODING EXAMPLES

This appendix provides detailed examples of the coding process used in this study, demonstrating how raw data were analyzed and transformed into codes, categories, and themes. These examples illustrate the systematic application of grounded theory principles, including initial coding, focused coding, categorization, and theoretical coding.

Raw Data Segment (Interview Excerpt):

> **[P7]:** If you are not willing to show yourself, be vulnerable, and take a certain risk in this relationship, it will never be as fruitful as it could be in terms of creation. If you close yourself off to all the pain and only present a perfect world to the client, they will have little to contribute.

Initial Code:

1) **Cod.20 -** Transforming Vulnerability into Trust**.**

Focused Code:

2) **FC.12 -** Transforming Vulnerability into Trust

Category:

3) **CT.23 -** Building Trust in Collaborative Relationships

Theoretical Code:

4) **TC.2 -** Building Collaborative Relationships and Trust for Co-creation

Description captures as a memo:

The initial code explores how openness about limitations or uncertainties can foster trust. Besides, the theoretical code emphasizes the interconnectedness of leveraging relationships and fostering trust as complementary processes in successful co-creation.

Summarizing the coding process:

Table 31 illustrates the progression from raw data to codes, categories, and core concepts informed by Theoretical codes using the above example and two new excerpts.

Table 31

Samples of the coding process.

Raw Data Excerpt	Initial Code	Focused Code	Category	Theoretical Code
If you are not willing to show yourself, be vulnerable, and take a certain risk in this relationship, it will never be as fruitful as it could be in terms of creation. If you close yourself off to all the pain and only present a perfect world to the client, they will have little to contribute	Cod.20 - Transforming Vulnerability into Trust.	FC.12 - Transforming Vulnerability into Trust	CT.23 - Building Trust in Collaborative Relationships	TC.2 - Building Collaborative Relationships and Trust for Co-creation
The work of building relationships never ends, because as long as the product remains alive, there will always be more to do and new opportunities to explore.	Cod.40 – Demonstrating Transparency	FC.82 - Balancing Transparency and Protection	CT.23 - Building Trust in Collaborative Relationships	TC.2 - Building Collaborative Relationships and Trust for Co-creation
We also realize what we didn't even see during the design phase and identify the blind spots. What do I ultimately want from this? The goal here is learning. Learning means understanding that I could make mistakes, that I could fail, but in the next project, we aim for new mistakes because these ones, we already know how to solve	Cod.3 – Learning From Mistakes	FC.1 - Learning from Mistakes	CT.1 - Team-Based Learning	TC.1 - Organizational Learning Across Teams and Client Partnerships

Source: Research Data (2024).

APPENDIX F – CHATGPT PROMPTS

The analysis process in this study was enriched by leveraging ChatGPT as a thinking partner, offering a novel and dynamic approach to qualitative research. As an advanced language model, ChatGPT facilitated various stages of data interpretation and synthesis, acting as a supportive tool for reflection, organization, and validation. Its contributions were particularly valuable in revisiting initial codes, summarizing interview impressions, synthesizing newly reviewed literature, validating focused codes, and refining text readability. By integrating ChatGPT into the workflow, the analysis process benefited from increased efficiency, critical engagement with the data, and a structured approach to refining theoretical insights. Table 32 details how prompts were designed and utilized across these activities to support rigorous and thoughtful analysis.

Table 32
ChatGPT Prompts used during the Analysis Process

Analysis Stage / Process	Prompt
Refining Text Readability	
Naming Categories	In grounded theory, according to Charmaz, how could we name a category where several interviews mentioned the challenges associated with value co-creation. For example: a) difficulty in obtaining feedback; b) giving emphasis in scalability; c) fearing losing the control; d) accustomed to prescribing solution; e) difficulties to materialize the cocreation
Search for additional literature	Hey ChatGPT, let us do some role-play again. Imagine you're an experience researcher in social science and this time you're an expert in the field of value co-creation. You're well versed on most recent literature and will propose a few articles that describe the challenges of co-creation. Can you suggest those articles?
Summarizing Articles	Hey! You're a talented research assistant with vast experience synthesizing complex articles and studies into actionable summaries. You have an ability to tease out key points, synthesize relevant insights, underline controversies, and suggest takeaways for ruminating insights. Please give me the TLDR ("too long, didn't read") on the following article by following these three steps: 1) Please read the following / attached, 2) then give me a brief TLDR report, which includes the fundamental thesis, three main points (with a key quote for each point). 3) Please also make at least 3 takeaways I can leverage this knowledge, as well as your advice for three areas I might want to explore more deeply: (hit "shift + enter" and then paste text of article)
Supporting the Analysis	The literature says that value is always experiential. In the service context, value is always determined by the customer's experience (Vargo & Lusch, 2008), therefore, co-creation is associated with the phenomenological nature of value where service providers have the chance to collaborate with customers from the beginning, where the problems are defined, solutions are ideated, designed, built, and delivered. Please, considering the themes and categories we have discussed, what are the implications of this mesmo?
Using it as Thinking Partner	The following memo, written in Portuguese, talks about the initiation of new employees in the company. It is important to understand the innovation does not happen in a vacuum and that processes, rituals, documentation, and training is necessary to onboard new collaborators and ensure that fresh insights can still happen. Please, not only translate the memo, but looking

Analysis Stage / Process	Prompt
	back to all themes that you translate, where can we fit the following memo: Colaboradores ao serem iniciados na empresa passam por um processo inicial de normalização de conhecimento, onde conhecem os ritos, práticas e produtos oferecidos. Ferramentas e processos são também apresentados nessa fase. Em seguida, por se tratar de uma empresa de nicho, a forma de trabalhar é ensinada por meio de resolução de problemas. Isto é, tickets de clientes são direcionados aos novos colaboradores para que possam tratar das desconformidades. Isso pode sugerir que esses colaboradores adquirem uma visão parcial da solução, sempre sob a ótica do software, e quase nunca sob a ótica do negócio ou do cliente
Validating Initial Coding	Hello ChatGPT, I would like you to assume the role of a social scientist specializing in the domains of organizational learning, value co-creation, and digital service innovation. Additionally, you have extensive experience conducting qualitative studies using grounded theory. I would like you to leverage your vast expertise and knowledge of Charmaz's coding methods to help us analyze interview data together. Please ask me 2 or 3 contextualization questions before I submit the interview data. What I expect from you is to help me identify codes and categories that group these codes.
Summarizing Interview Impressions	Getting summaries using Voice Hey ChatGPT! I'd like to unload a couple of thoughts following the interview I have done with Participant X. When I'm done with it, please help me summarize all that I have shared in the form of a quick debriefing. Please pay special attention for definitions, processes, counterintuitive comments and metaphors. (Shared my thoughts using Voice)

Source: From the Author.